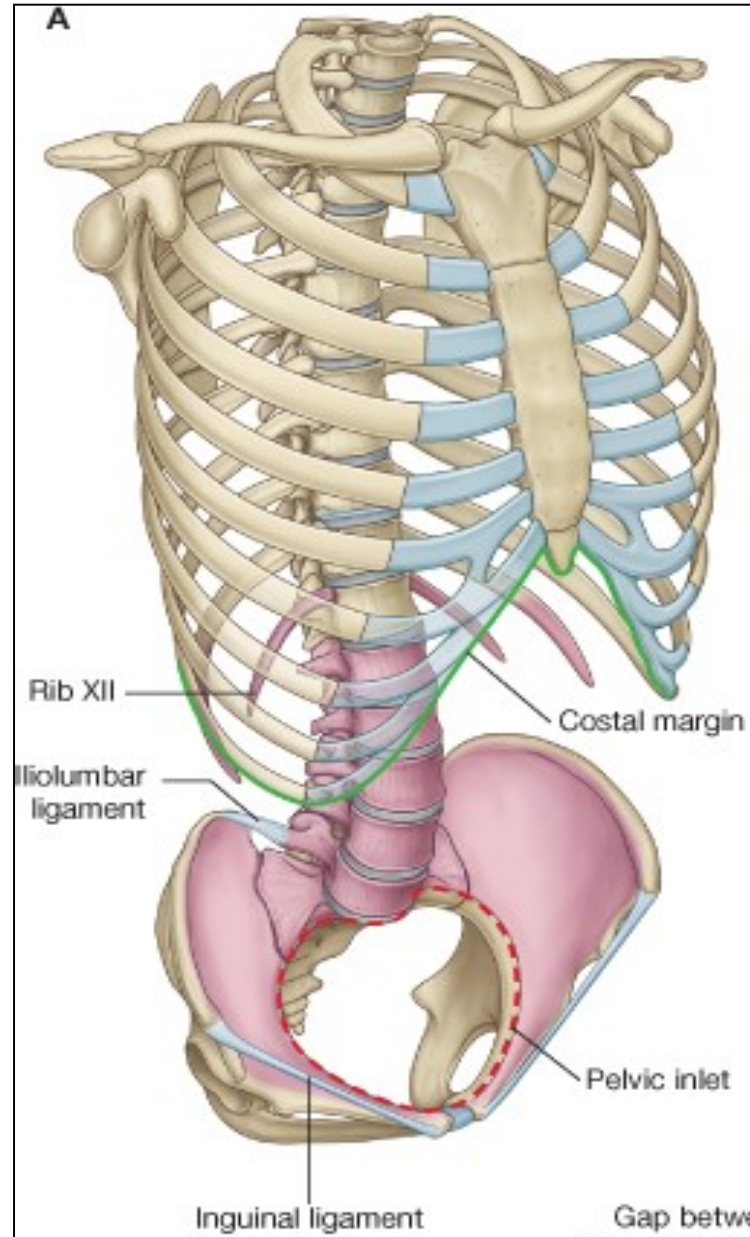
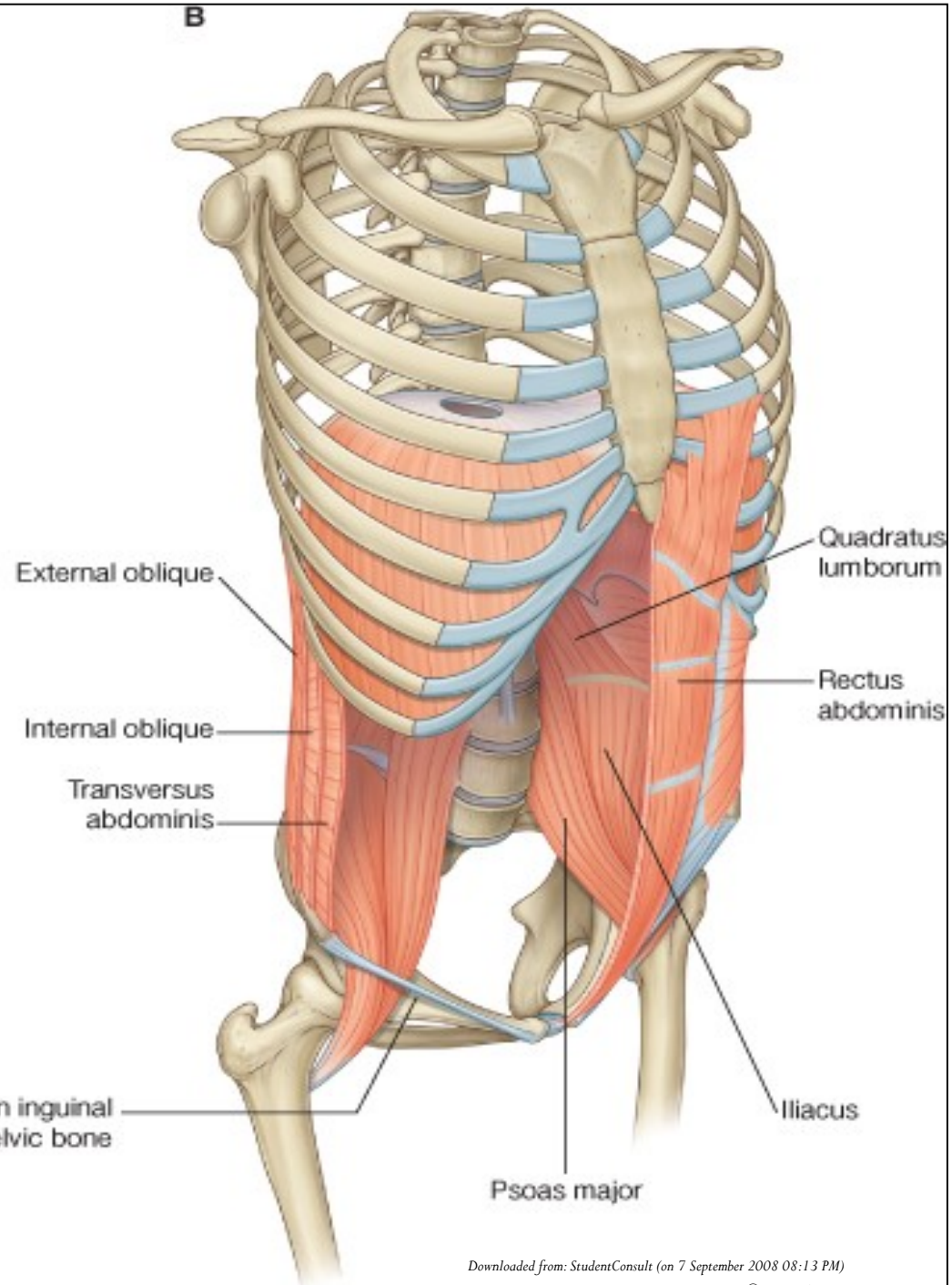


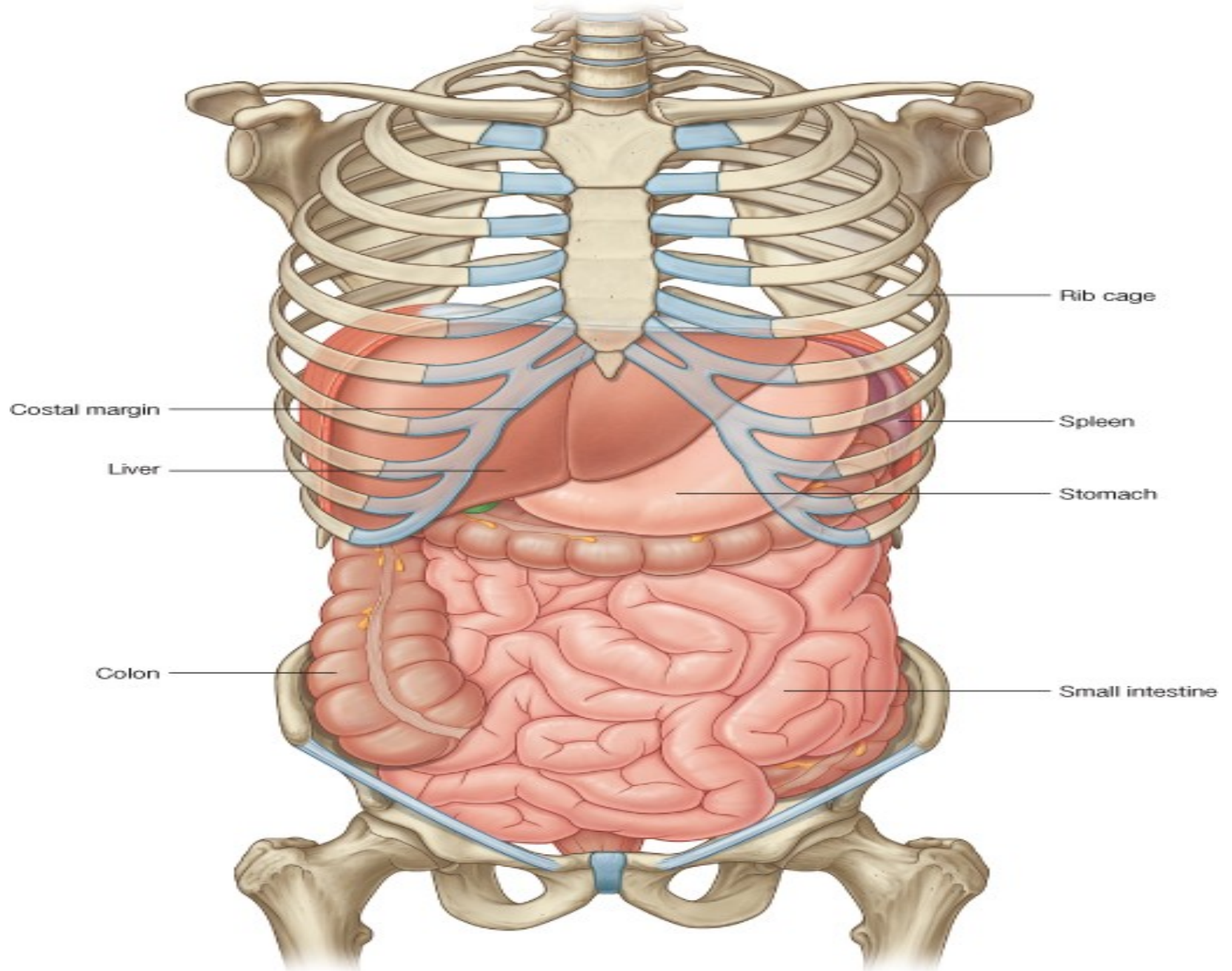
**ANATOMY
OF THE
ABDOMINAL REGION**

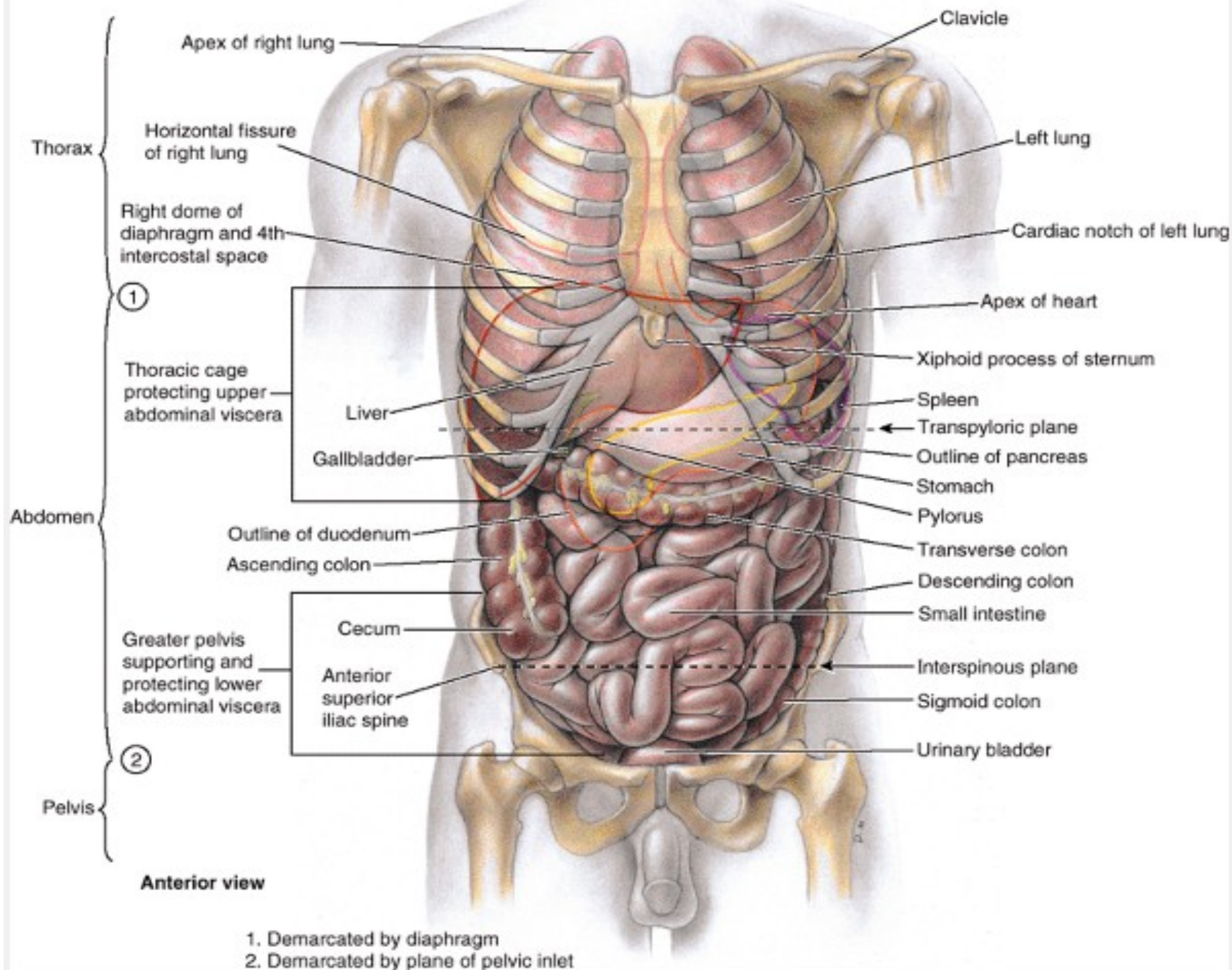
Bacha Deleta

Introduction

- Abdomen is part of the trunk between the thorax and the pelvis
- It is a flexible, dynamic container, housing most of the organs of the digestive system and part of the urogenital system
- The abdomen is able to enclose and protect its contents while allowing the flexibility between the more rigid thorax and pelvis required by respiration, posture, and locomotion

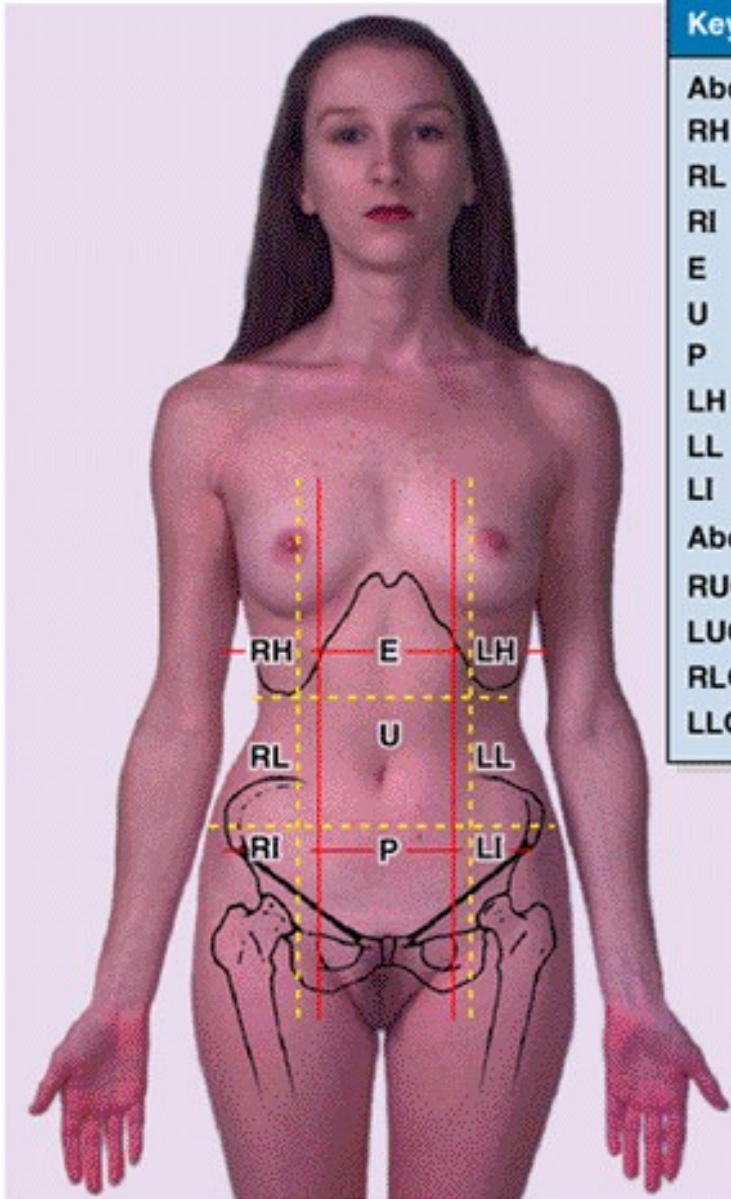
A**B**





Regions of abdomen

- Clinicians subdivide abdomen into 9 regions to locate abdominal organs, pain sites, swelling or incision
- Delineated by 4 planes
- Two horizontal
 - Subcostal plane: passing through inferior border of 10th costal cartilage
 - Transtubercular plane: passing through iliac tubercles and body of L5 vertebra
- Two vertical
 - Midclavicular planes: passing from midpoints of clavicles to midinguinal points



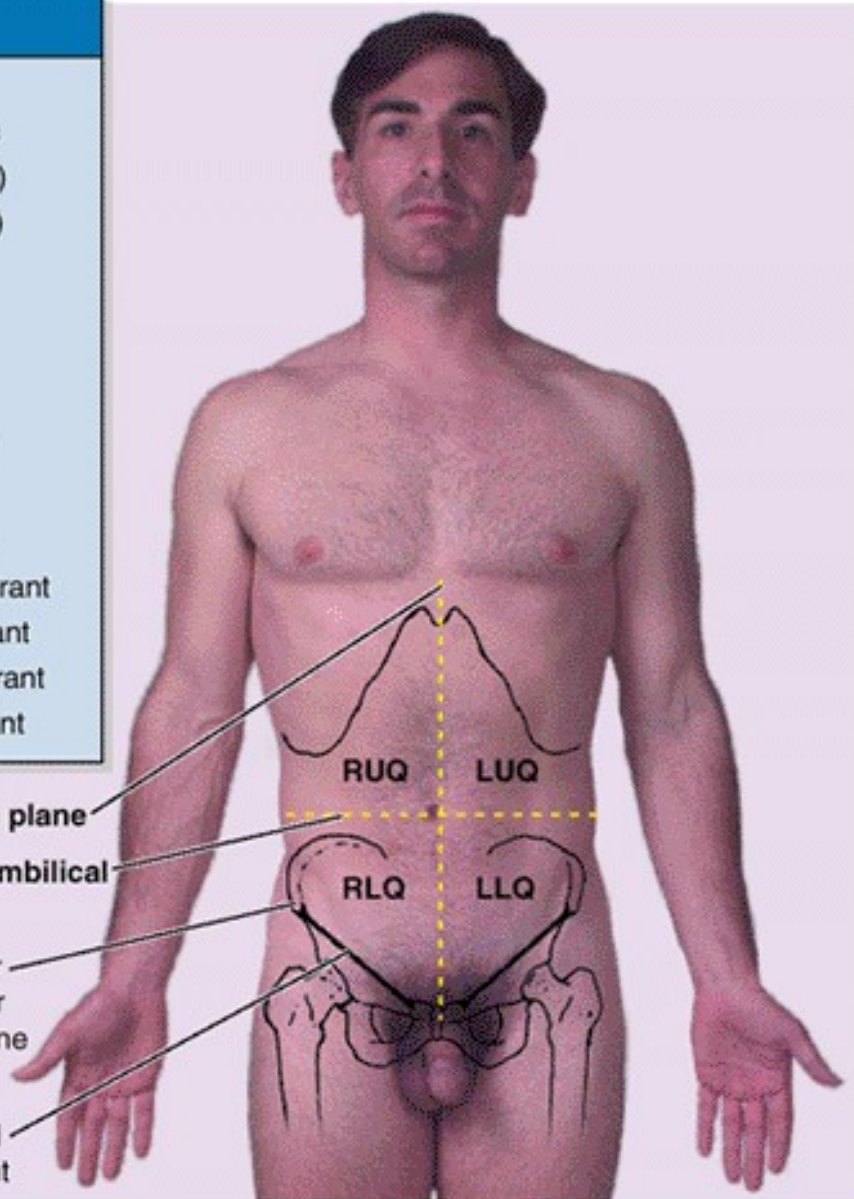
Key

Abdominal Regions:

- RH Right hypochondriac
- RL Right lateral (lumbar)
- RI Right inguinal (groin)
- E Epigastric
- U Umbilical
- P Pubic (hypogastric)
- LH Left hypochondriac
- LL Left lateral (lumbar)
- LI Left inguinal (groin)

Abdominal Quadrants:

- RUQ Right upper quadrant
- LUQ Left upper quadrant
- RLQ Right lower quadrant
- LLQ Left lower quadrant

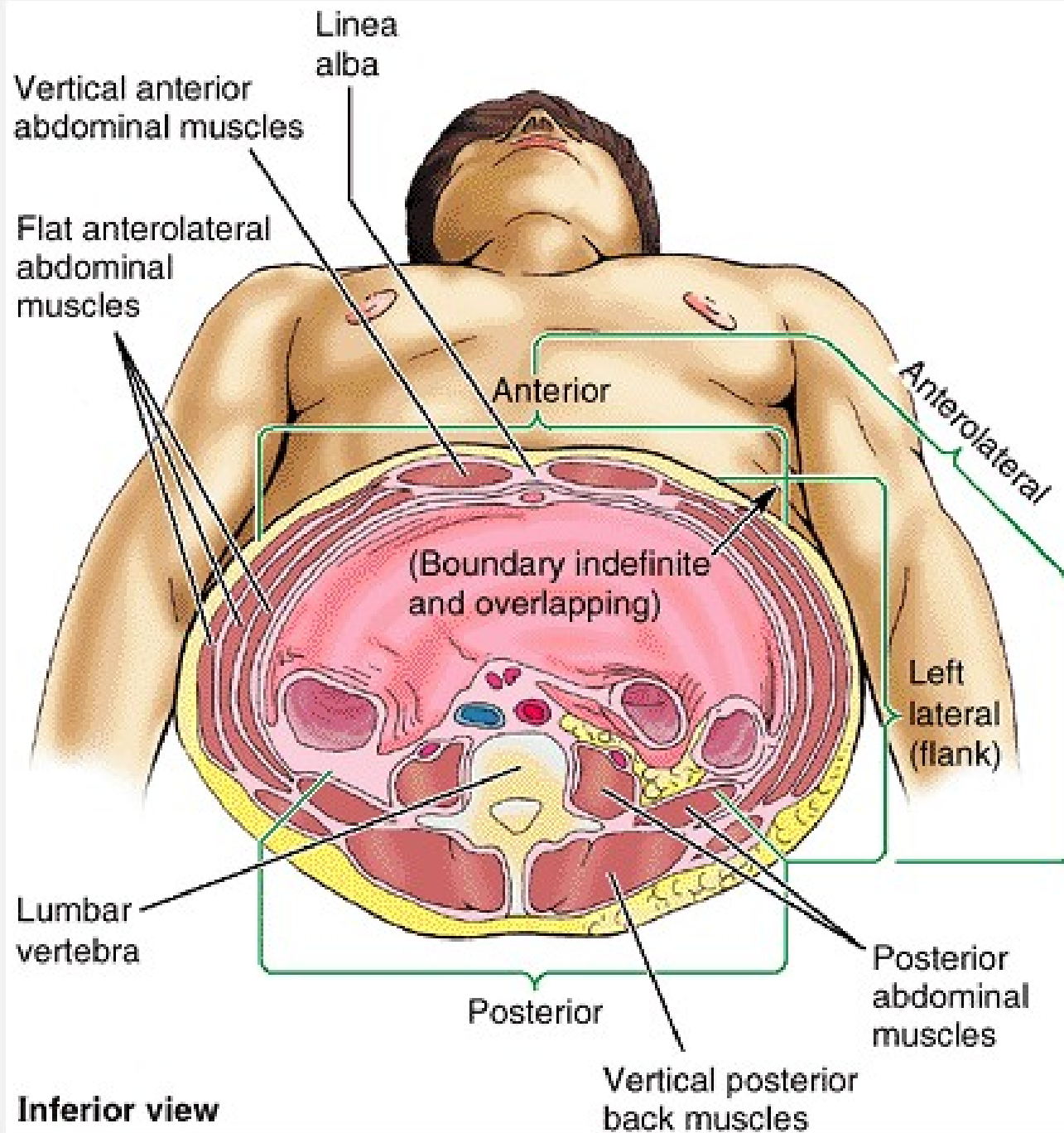


(A) Abdominal regions

(C) Abdominal quadrants

Abdominal walls

- ⊙ Subdivided into
 - Anterior wall
 - Lateral walls
 - Posterior wall
- ⊙ Boundary between anterior and lateral walls is indefinite thus called anterolateral wall



Anterolateral abdominal wall

- ⦿ Extends from thoracic cage to pelvis
- ⦿ Bounded
 - superiorly by cartilages of 7th -10th ribs and xiphoid process
 - Inferiorly by inguinal ligament and pelvic bones
- ⦿ During physical examination, the anterolateral wall is inspected, palpated, percussed, and auscultated
- ⦿ Surgeons usually incise this wall during abdominal surgery
- ⦿ The wall consists of
 - skin
 - subcutaneous tissue
 - muscles and their aponeuroses
 - deep fascia
 - extraperitoneal fat
 - parietal peritoneum

Fascia of anterolateral abdominal wall

- ⊙ The fascial layers from superficial to deep include
 - Subcutaneous tissue (superficial fascia)
 - Inferior to the umbilicus, the deepest part of the subcutaneous tissue is reinforced by many elastic and collagen fibers, so the subcutaneous tissue here has two layers:
 - superficial fatty layer (Camper fascia)
 - deep membranous layer (Scarpa fascia)
 - The membranous layer continues inferiorly into the perineal region as the superficial perineal fascia (Colles fascia), but not into the thighs
 - Investing fascia
 - Cover external aspects of the three muscle layers
 - Endoabdominal fascia
 - Lines internal aspect of abdominal wall

External oblique
Internal oblique
Transverse
abdominal
muscle

Abdominal wall
Iliac crest
(pelvic bone)

(A) Anterior view

Rectus abdominis
Inguinal ligament
Pubic crest and symphysis

Superficial ← → Deep

Skin (cut edge)

Superficial fatty layer of
subcutaneous tissue
(Camper fascia)

Deep membranous layer
of subcutaneous tissue
(Scarpa fascia)

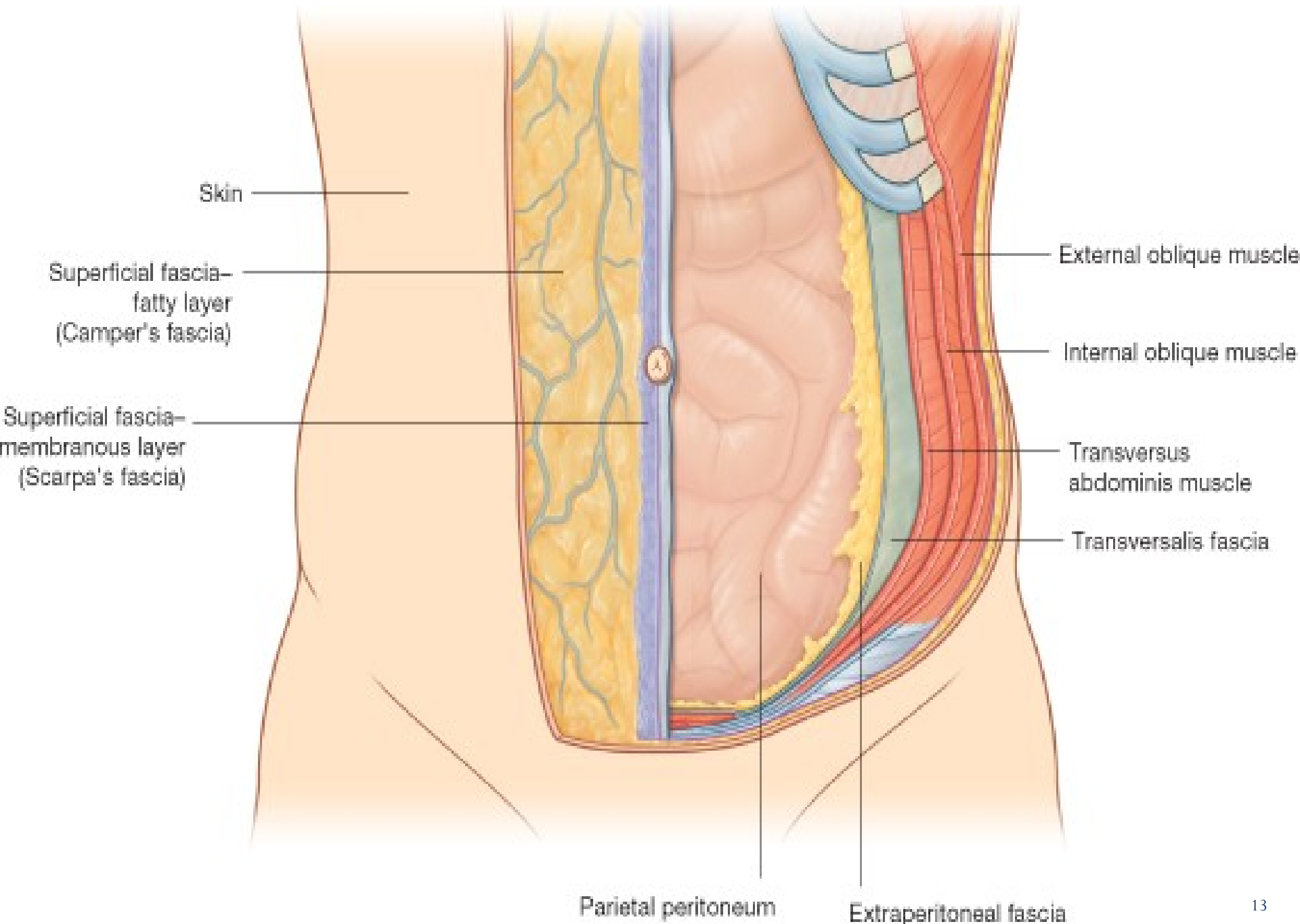
Investing (deep)
fascia—superficial,
intermediate, & deep

External oblique muscle
Internal oblique muscle
Transverse
abdominal muscle

Extraperitoneal fat

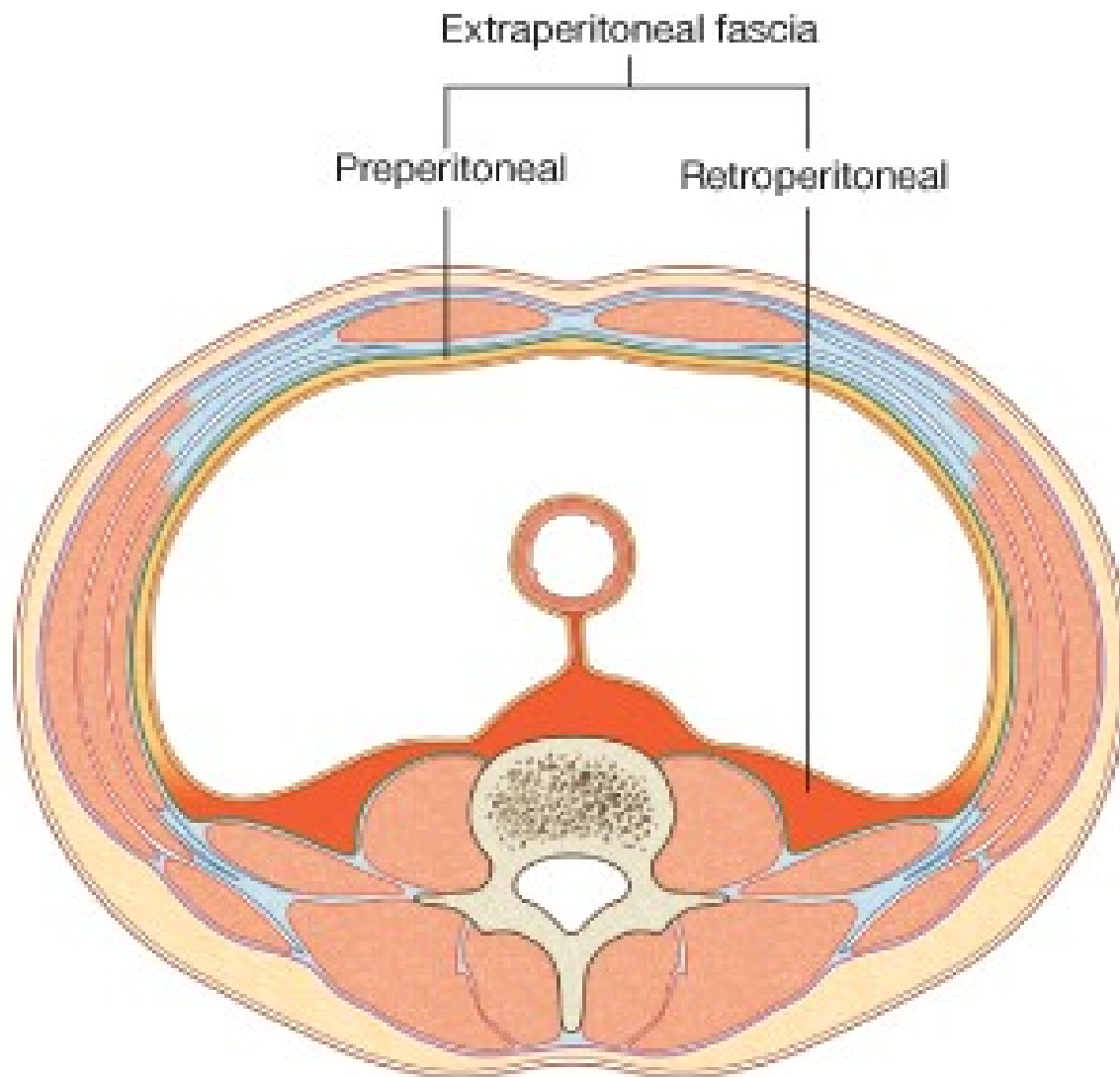
Endoabdominal
(transversalis) fascia
Parietal
peritoneum

(B) Longitudinal section



Extraperitoneal fascia

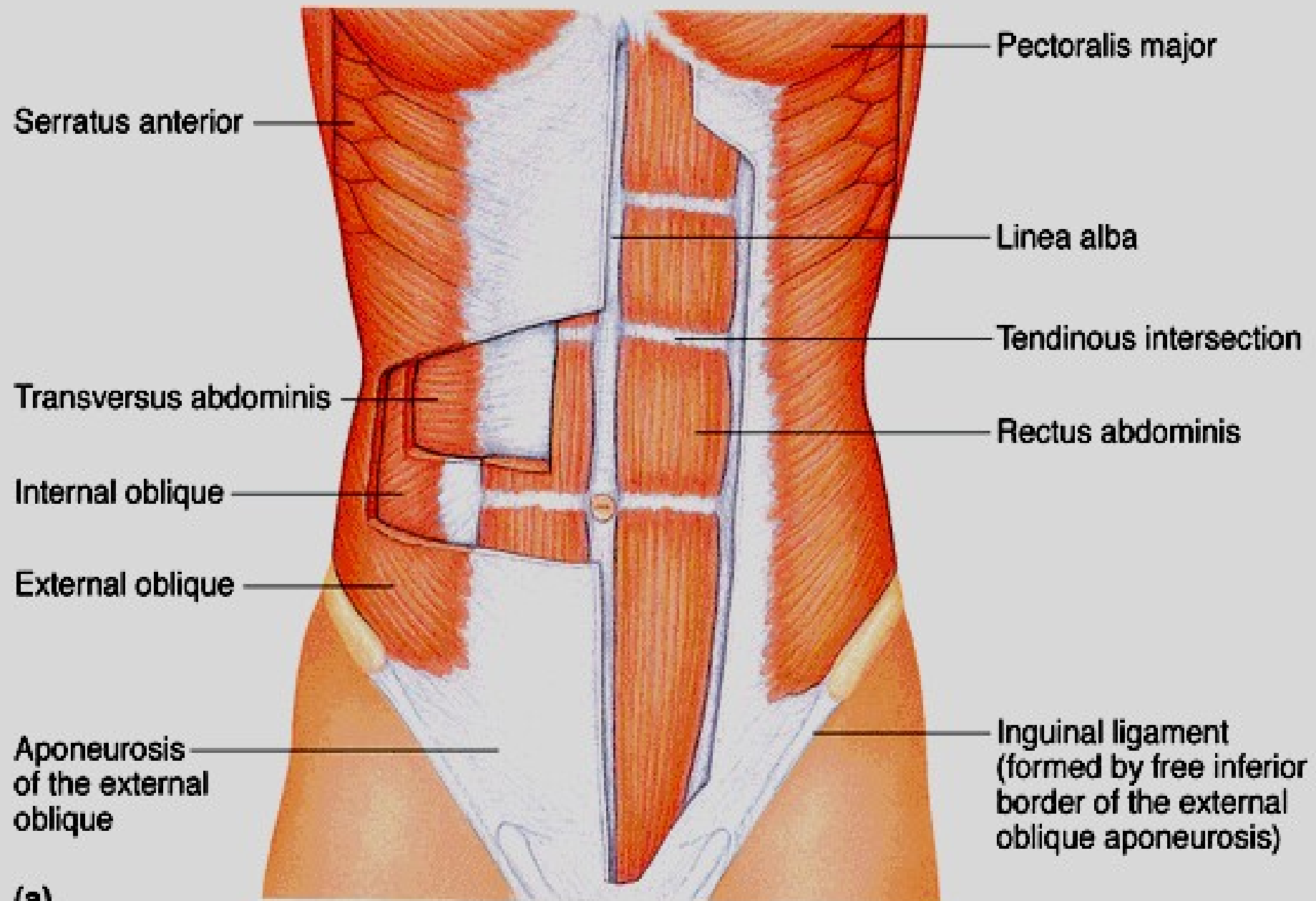
- ⦿ Lies deep to the muscles, it separates the muscles from the peritoneum
- ⦿ Contains
 - fat
 - blood vessels
 - organs and viscera such as kidneys and pancreas which are retroperitoneal
- ⦿ Extends into the mesentery with blood vessels, nerves and lymphatics



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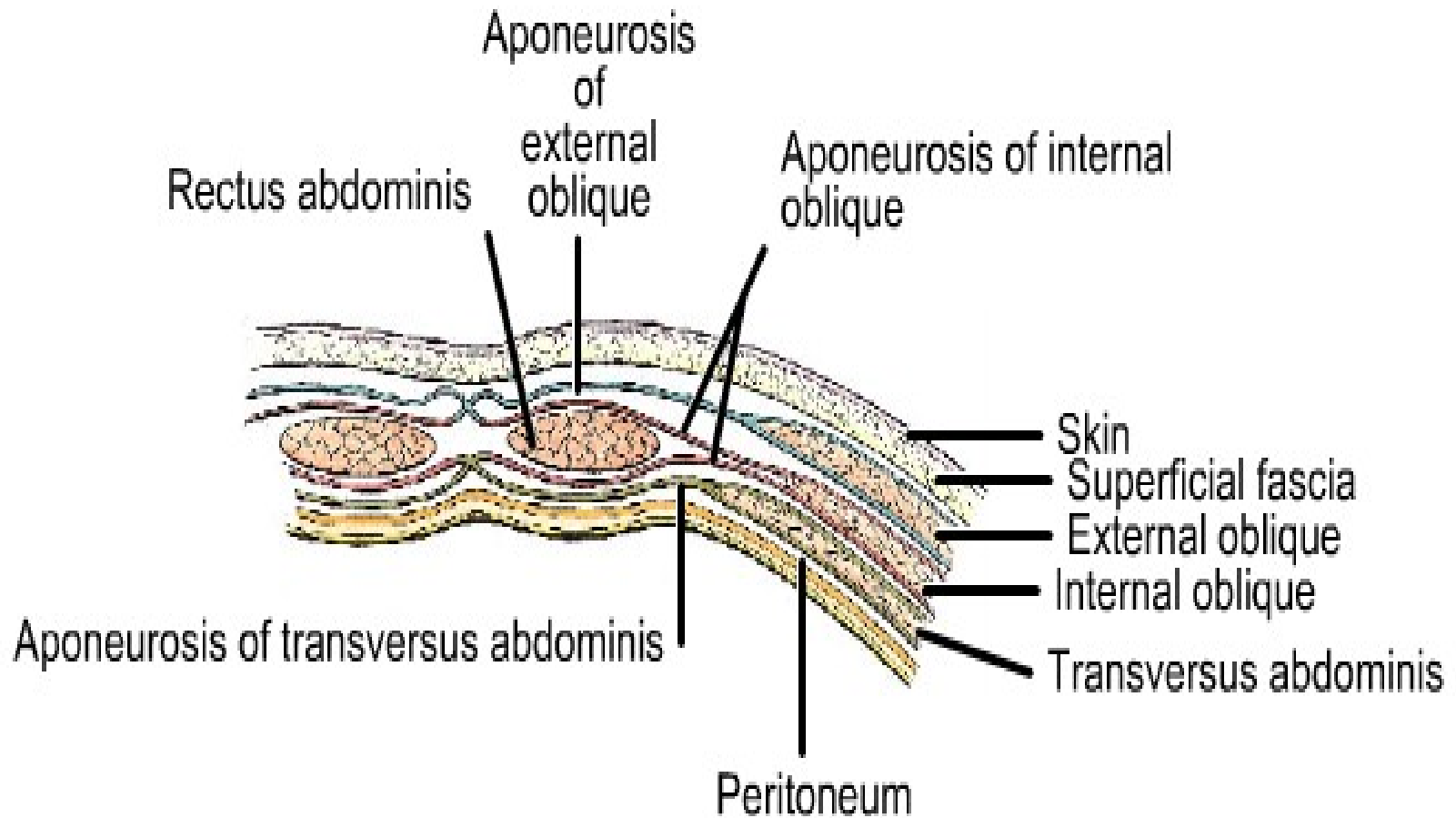
Muscles of anterolateral abdominal wall

- ⦿ 5 muscles: 3 flat and 2 vertical
- ⦿ Flat muscles
 - External oblique
 - Internal oblique
 - Transverse abdominal
- ⦿ Vertical muscles
 - Rectus abdominis
 - Pyramidalis



(a)

Transverse section superior to the umbilicus



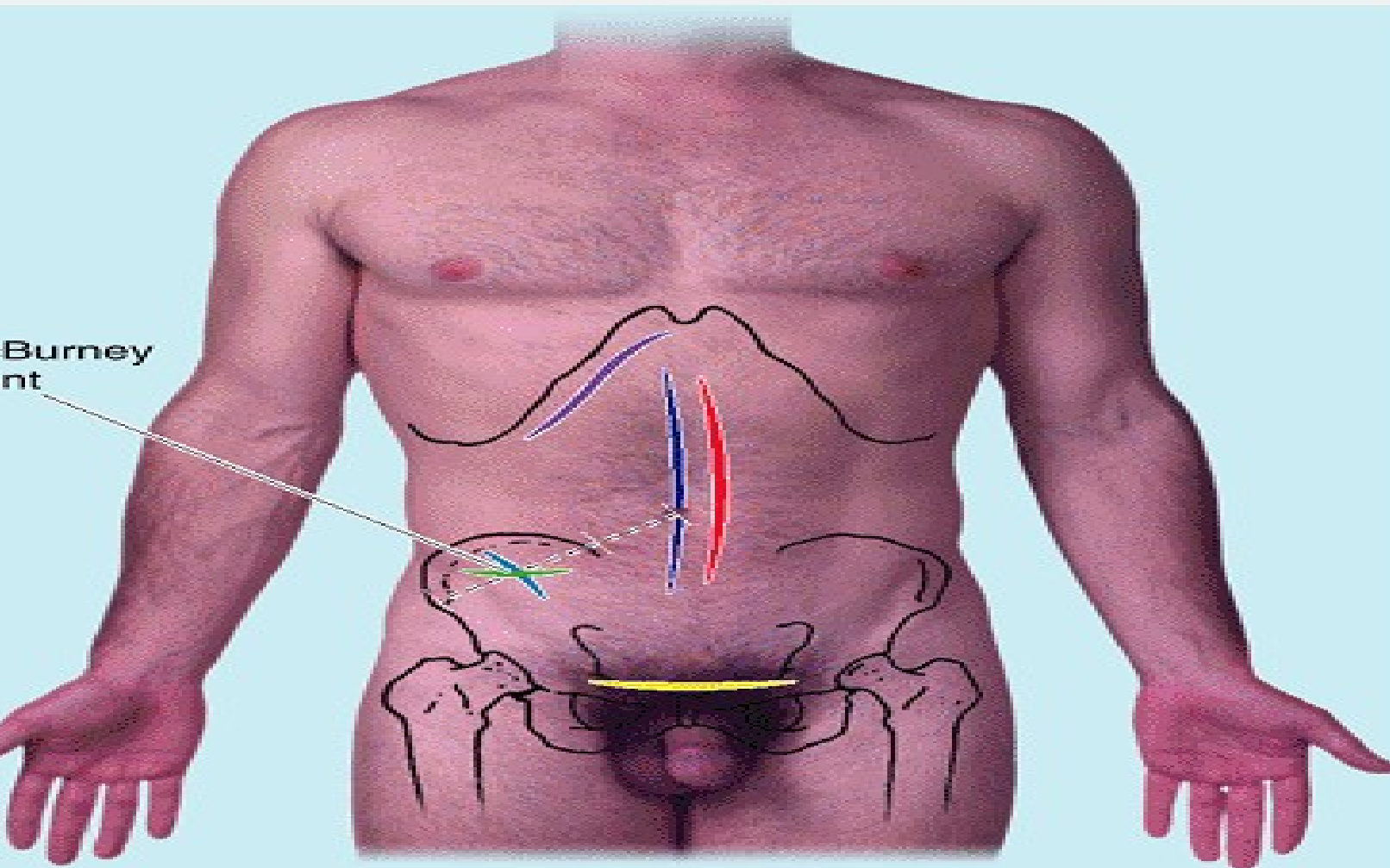
Nerves of anterolateral abdominal wall

- ◎ The skin and muscles of the anterolateral abdominal wall are supplied by the following nerves:
 - Thoracoabdominal nerves: the distal, abdominal parts of the anterior rami of the inferior six thoracic spinal nerves (T7 - T11)
 - Lateral (thoracic) cutaneous branches: of the thoracic spinal nerves T7 - T11
 - Subcostal nerve: anterior ramus of spinal nerve T12
 - Iliohypogastric and ilioinguinal nerves: terminal branches of the anterior ramus of spinal nerve L1

Abdominal Surgical Incisions

- Surgeons use various incisions to gain access to the abdominal cavity
- When possible, the incisions follow the cleavage lines (Langer lines) in the skin
- The incision that allows adequate exposure and, secondarily, the best possible cosmetic effect, is chosen
- The location of the incision also depends on
 - the type of operation
 - the location of the organ(s) the surgeon wants to reach
 - bony or cartilaginous boundaries
 - avoidance of nerves
 - maintenance of blood supply
 - minimizing injury to muscles and fascia of the wall while aiming for favorable healing

Burney
nt

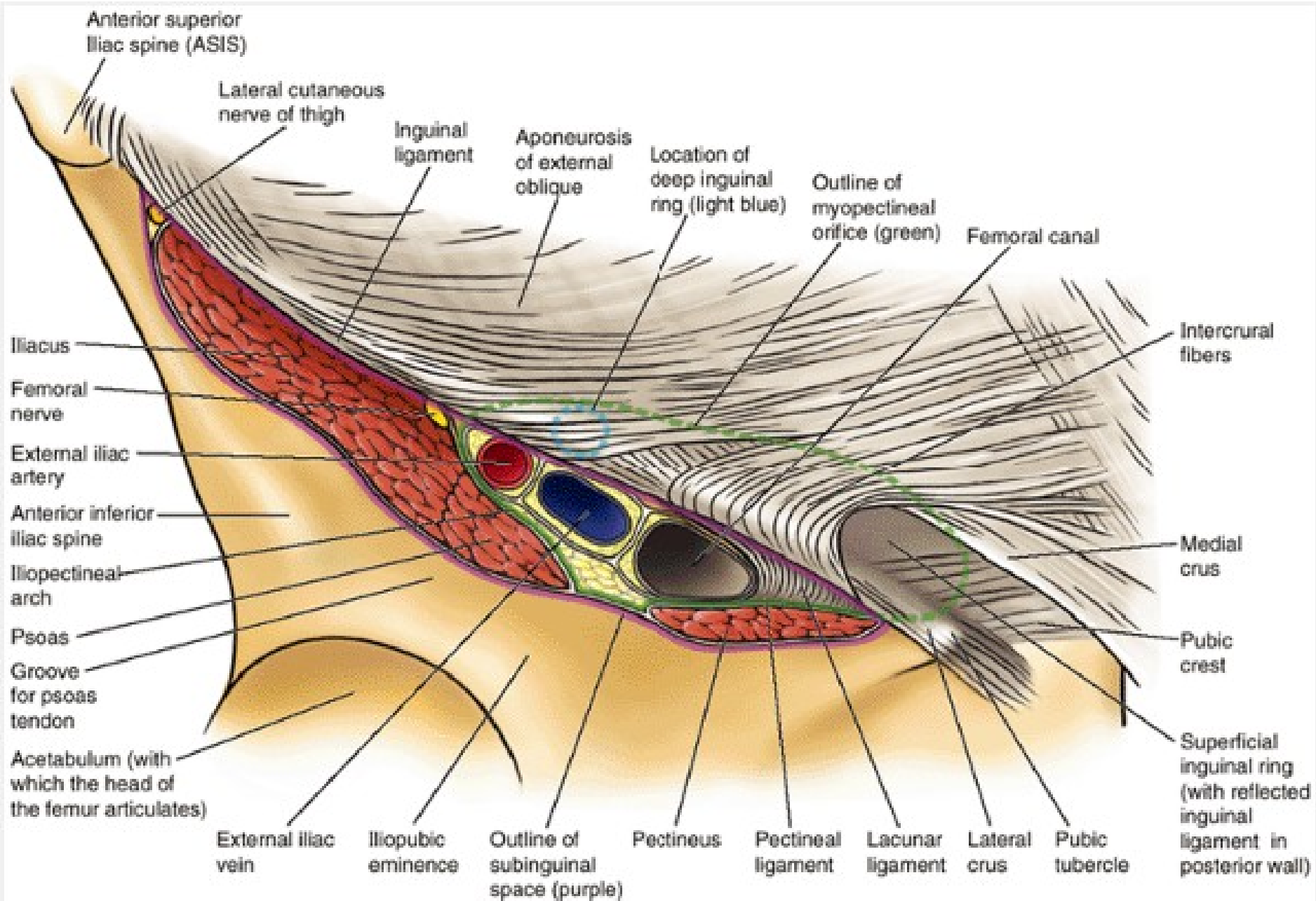


Key

-  Median or midline incision
-  Left paramedian incision
-  Gridiron (muscle-splitting) incision
(traditional McBurney incision
and horizontal incision)
-  Suprapubic (Pfannenstiel) incision
-  Transverse (abdominal) incision
-  Subcostal incision

Inguinal Region

- ⦿ The area between anterior superior iliac spine and pubic tubercle
- ⦿ A region where structures enter and exit abdominal cavity
- ⦿ Weak area in the abdominal wall
- ⦿ Potential sites of herniation



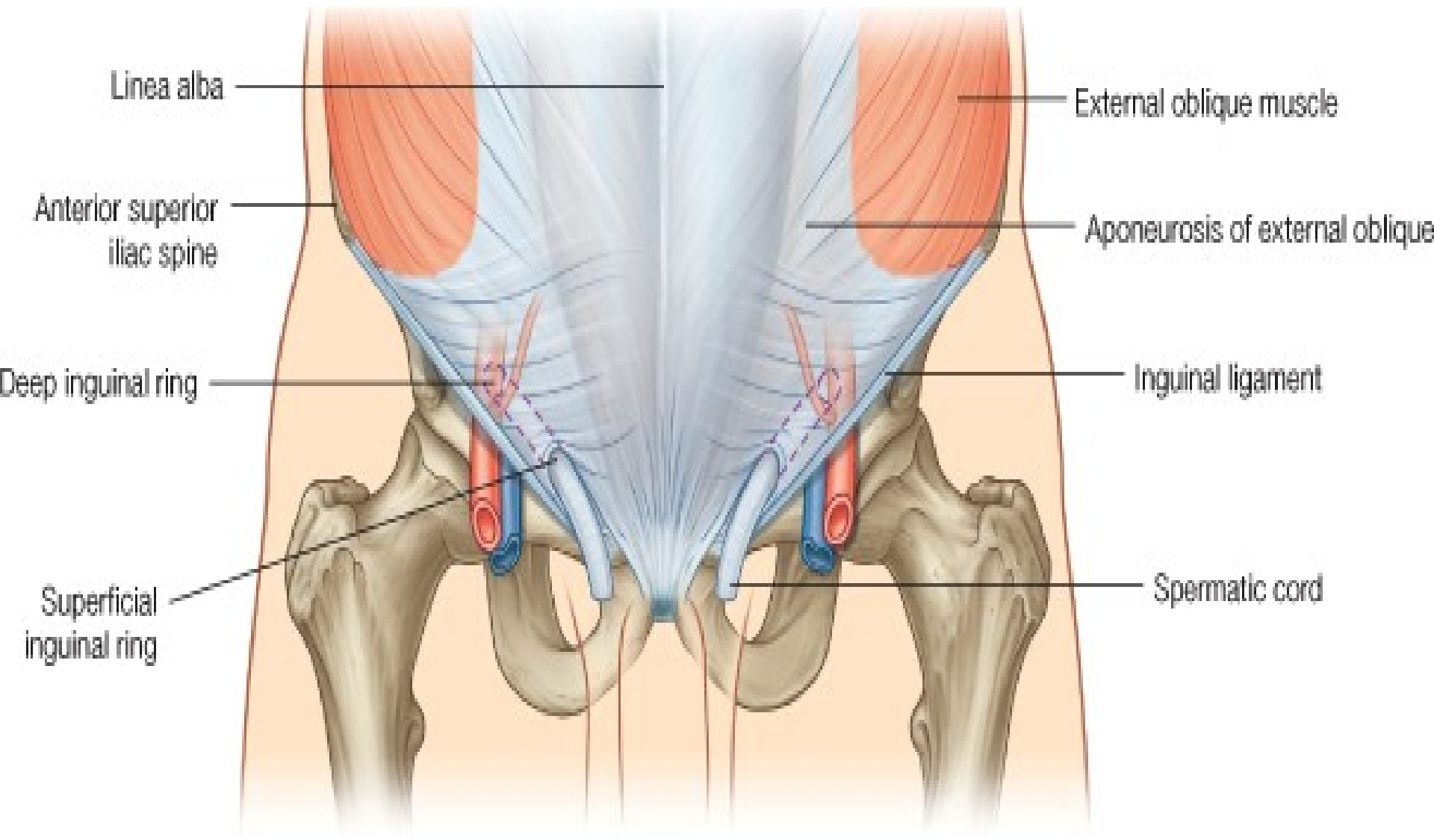
Anteroinferior view

Inguinal canal

- Oblique passage (4 cm long) inferomedially directed at inferior margin of anterior abdominal wall parallel and above to the lower half of the inguinal ligament
- The path taken by the testis during its descent
- It is bounded by the deep and superficial inguinal rings
- Contents
 - spermatic cord in males and round ligament of uterus in females
 - blood and lymphatic vessels and the ilioinguinal nerve in both sexes

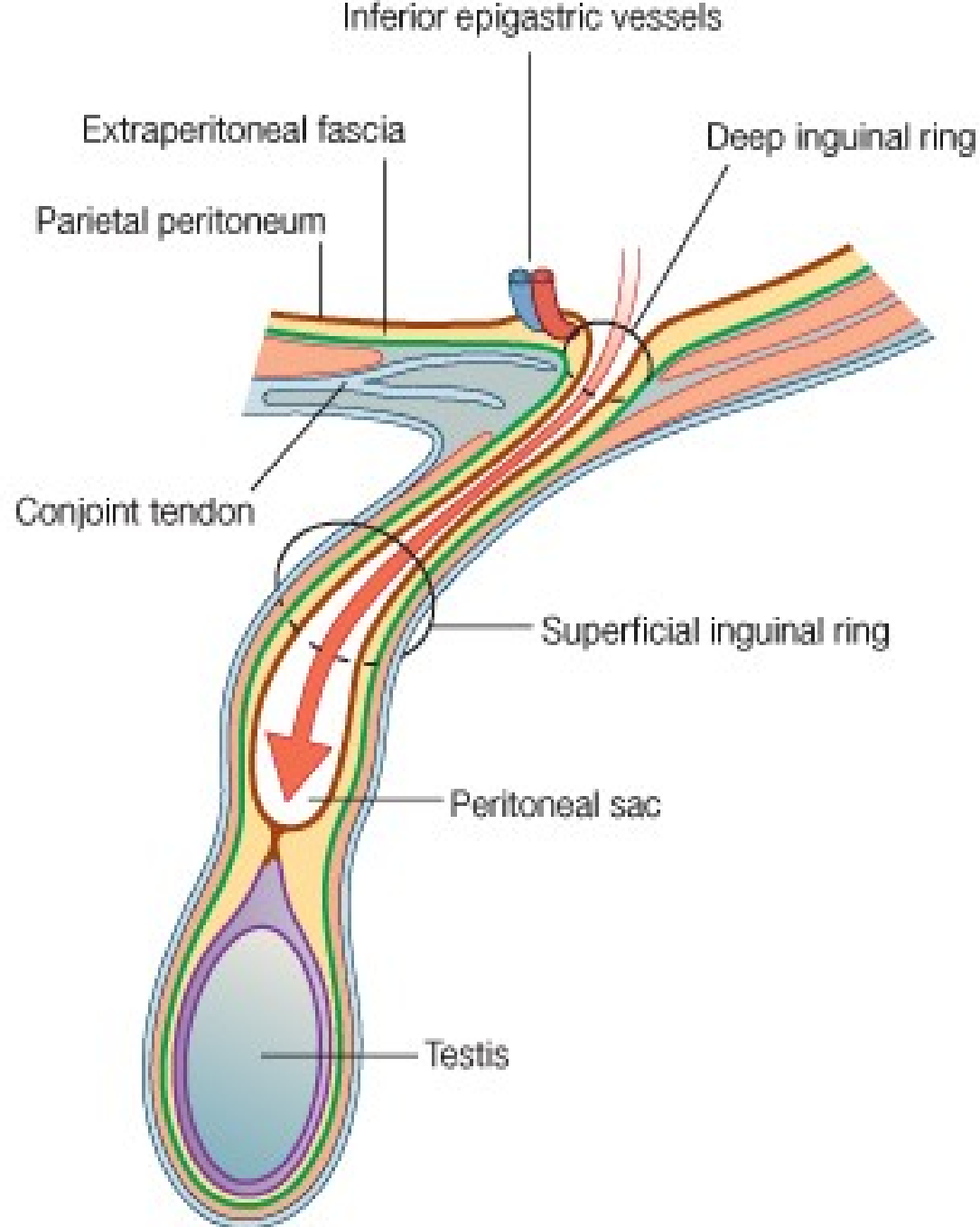
Inguinal canal: openings

- The inguinal canal has an opening at each end
- Deep (internal) inguinal ring
 - internal entrance to the inguinal canal
 - located superior to the middle of the inguinal ligament and lateral to the inferior epigastric artery
 - Through this opening, the extraperitoneal ductus deferens (vas deferens) and testicular vessels in males (or round ligament of the uterus in females) pass to enter the inguinal canal
- Superficial (external) inguinal ring
 - The exit by which the spermatic cord in males, or the round ligament in females, emerges from the inguinal canal
 - Slit like opening between diagonal fibers of external oblique aponeurosis just superolateral to the pubic tubercle



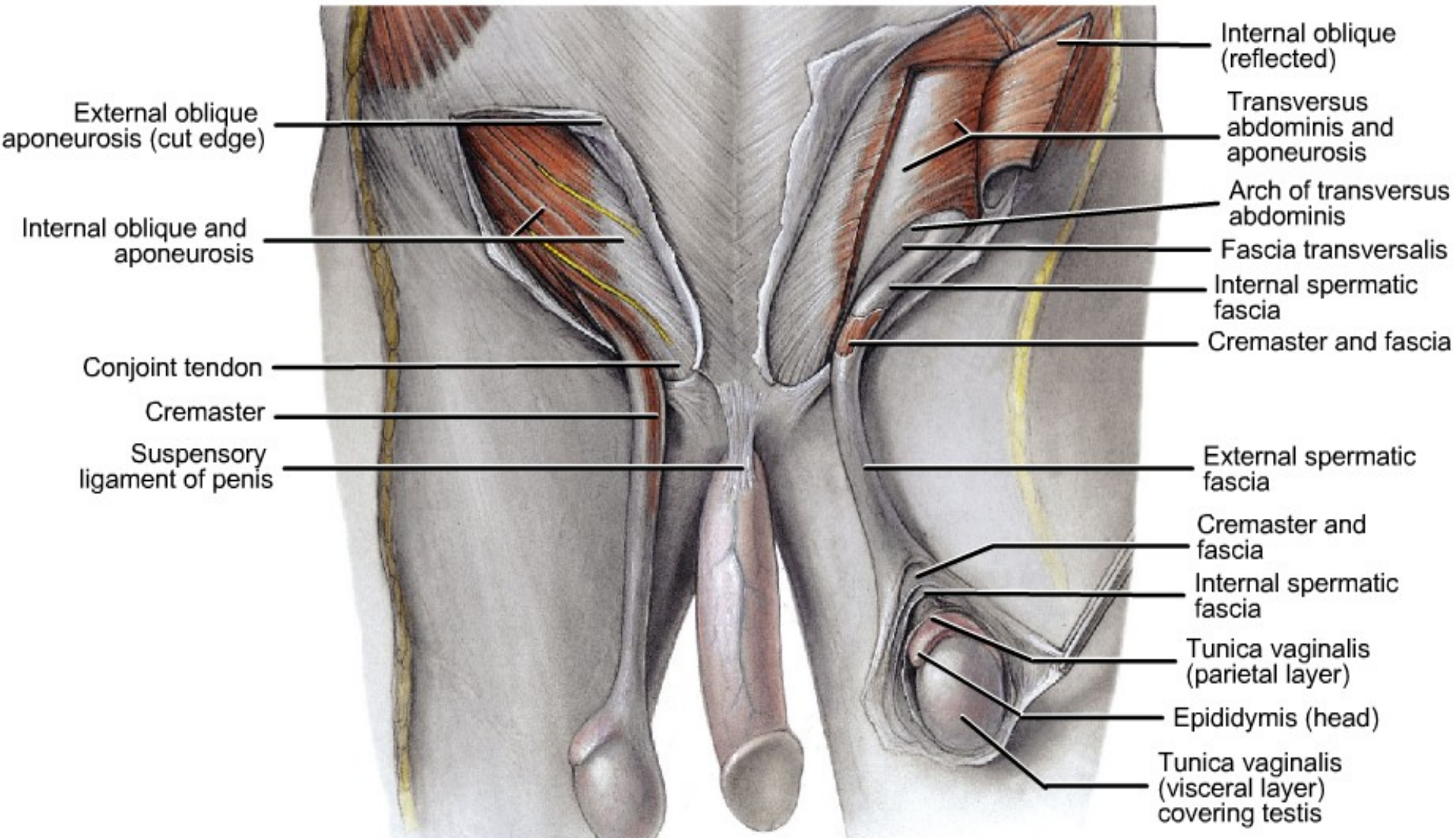
Inguinal hernias

- ⦿ Inguinal hernia is the protrusion of a part of the peritoneal sac, with or without abdominal contents, into the groin
- ⦿ A hernia may enter the inguinal canal either: indirectly, through the deep inguinal ring or directly, through the posterior wall of the inguinal canal
- ⦿ An inguinal hernia is therefore classified as either an indirect inguinal hernia or direct inguinal hernia



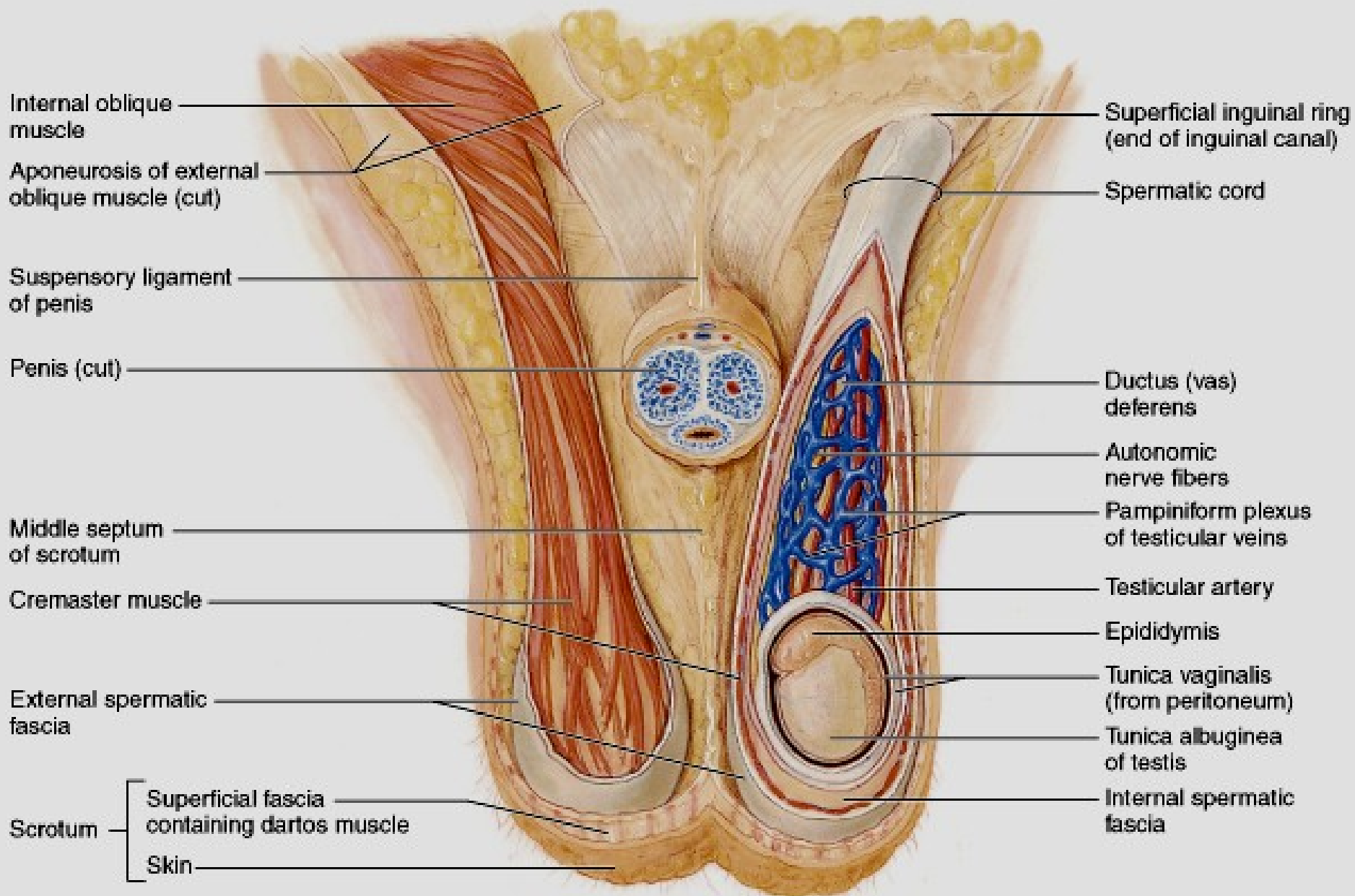
Spermatic cord

- Contains structures running to and from the testis and suspends the testis in the scrotum
- Begins at the deep inguinal ring lateral to the inferior epigastric vessels, passes through the inguinal canal, exits at the superficial inguinal ring, and ends in the scrotum at the posterior border of the testis
- Fascial coverings
 - derived from the anterolateral abdominal wall during prenatal development
 - Internal spermatic fascia: derived from the transversalis fascia
 - Cremasteric fascia: derived from the fascia of both the superficial and the deep surfaces of the internal oblique muscle
 - External spermatic fascia: derived from the external oblique aponeurosis and its investing fascia



Scrotum

- ⊙ Cutaneous sac consisting of two layers
 - Skin
 - Dartos fascia
- ⊙ Skin is heavily pigmented and covered with sparse hairs
- ⊙ Fat-free fascial layer including smooth muscle fibers (dartos muscle) responsible for the rugose (wrinkled) appearance of the scrotum
- ⊙ Because the dartos muscle attaches to the skin, its contraction causes the scrotum to wrinkle when cold, thickening the integumentary layer while reducing scrotal surface area and assisting the cremaster muscles in holding the testes (testicles) closer to the body, all of which reduces heat loss

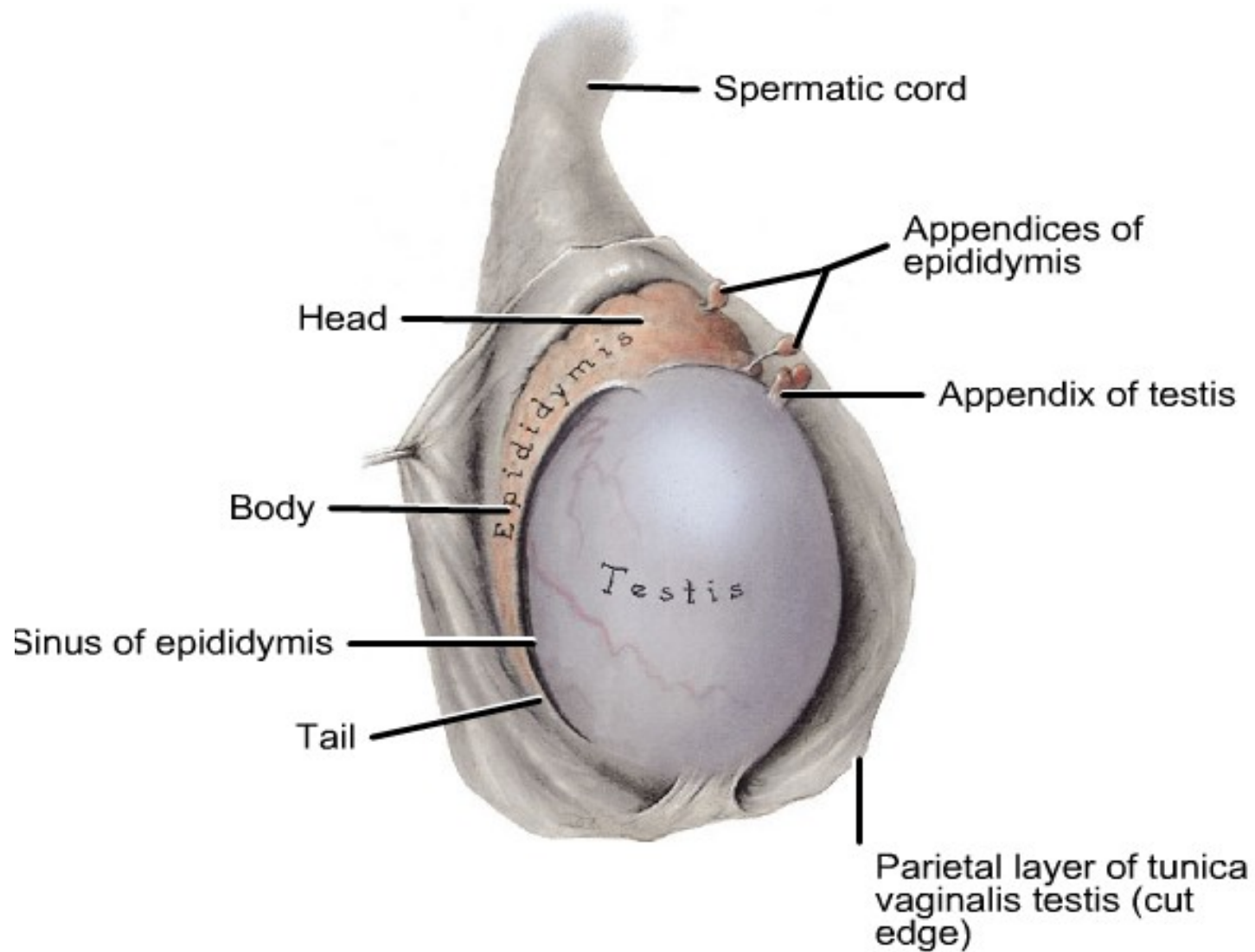


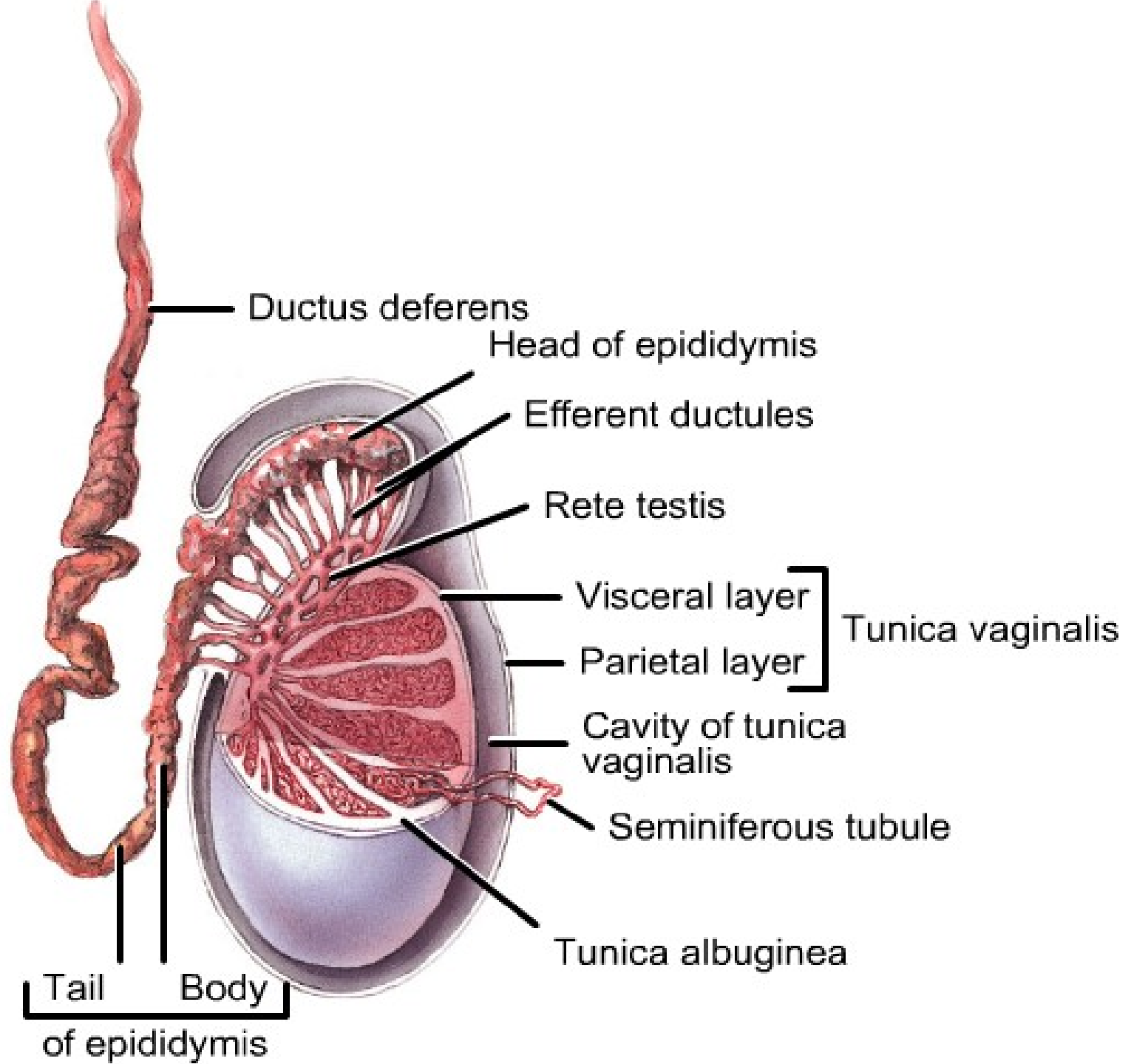
Scrotum: temperature regulation

- ◉ The location of testes appears to make them vulnerable to injury
- ◉ However, viable sperm cannot be produced at core body temperature
- ◉ The superficial location of the scrotum provides a temperature which is about 1 - 2 °C cooler
- ◉ The scrotum also responds to temperature changes
 - When it is cold, the testes are drawn closer to the warmth of the body and the scrotum becomes shorter and heavily wrinkled to reduce heat loss
 - When it is warm, the scrotal skin is flaccid and loose to increase cooling, and the testes hang lower
- ◉ These changes reflect the activity of the two sets of muscles
 - Dartos
 - Cremaster

Testes

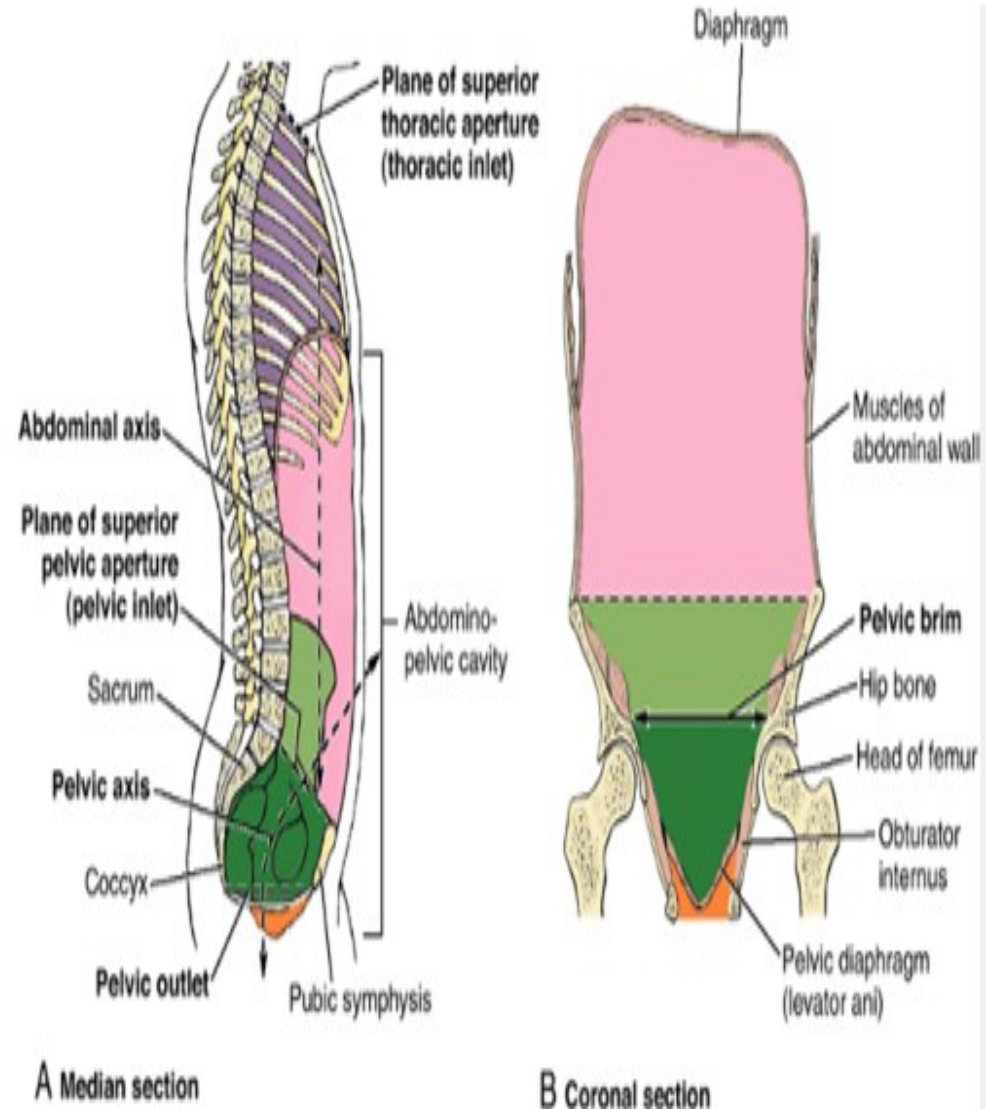
- The testes are the male gonads that produce the male germ cells (sperms, or spermatozoa) and male hormones, primarily testosterone
- The testes are suspended in the scrotum by the spermatic cord, with the left testis (testicle) usually suspended (hanging) more inferiorly than the right testis
- Each testes is approximately 4 cm long and 2.5 cm in diameter
- It is surrounded by two tunics
 - Tunica vaginalis: outer; derived from the peritoneum
 - Tunica albuginea: deep; fibrous capsule; thickens into a ridge on its internal, posterior aspect as the mediastinum of the testis





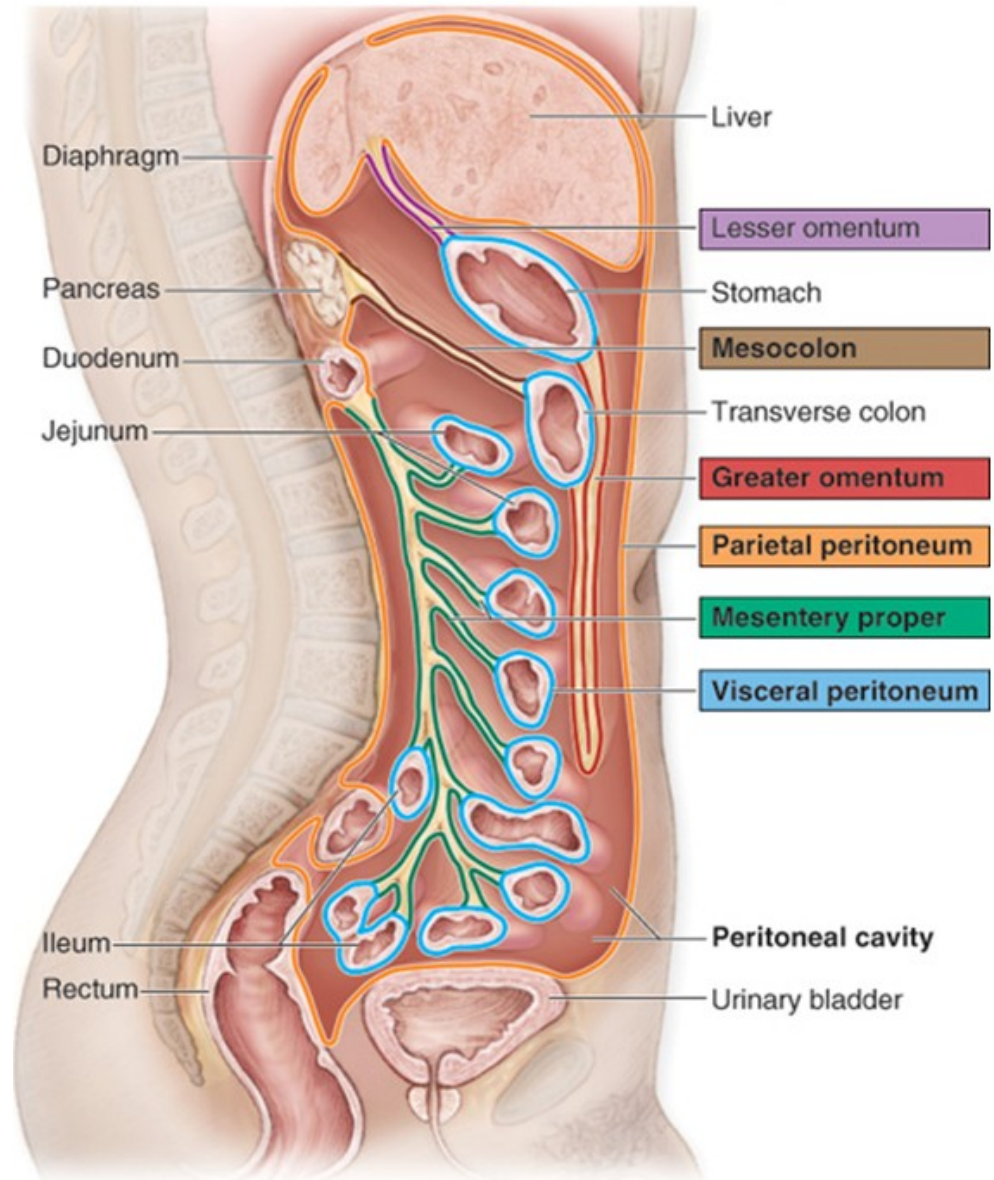
The abdominal cavity

- The abdominal cavity is:
Enclosed anterolaterally
by the dynamic musculo-
aponeurotic abdominal
walls.
- ⦿ Separated superiorly
from the thoracic cavity
by the diaphragm.
- ⦿ Continuous inferiorly
with the pelvic cavity.
- ⦿ Lined with peritoneum, a
serous membrane.

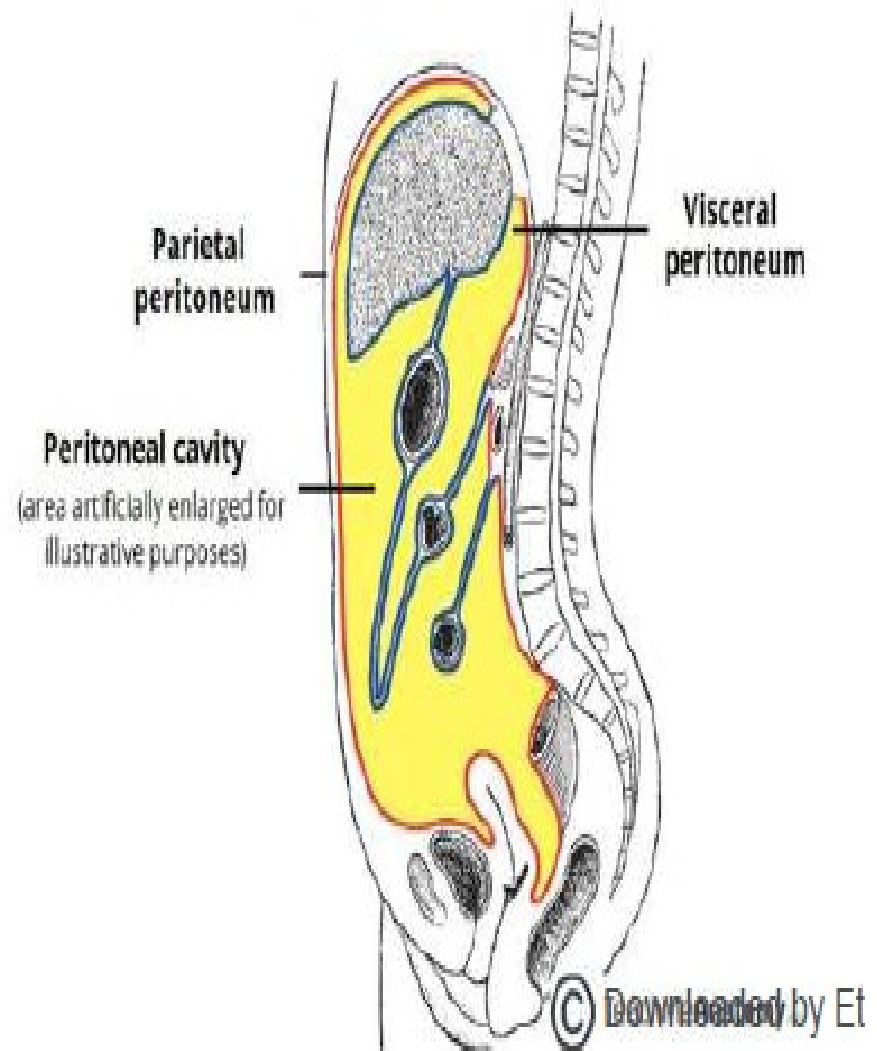


Peritoneum and Peritoneal Cavity

- **The peritoneum**, is a double layered serous membrane of the abdominal cavity,
 - **Parietal layer**, lines the abdominal wall
 - **Visceral layer**, covers the external surfaces of the abdominal organs



- **The peritoneal cavity:** is within the abdominal cavity and continues inferiorly into the pelvic cavity.
- is a potential space of capillary thinness between the parietal and visceral layers of peritoneum.
- **contains no organs** but contains a thin film of peritoneal fluid
- **Peritoneal fluid** lubricates the peritoneal surfaces, enabling the viscera to move over each other without friction
- is completely closed in **males**
- But there is a communication pathway in females to the exterior of the body through the uterine tubes, uterine cavity, and vagina.

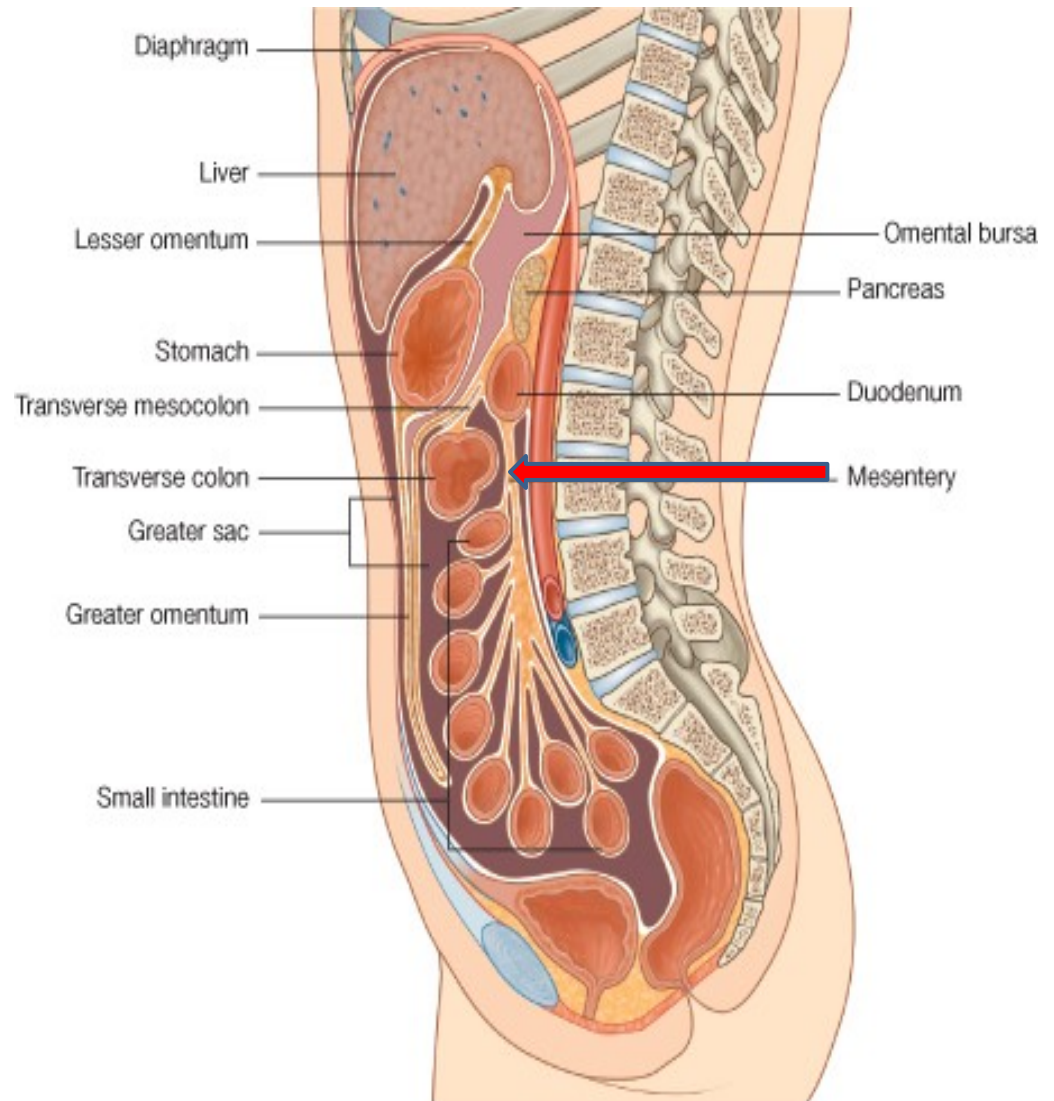


Peritoneal Reflections

- A number of terms are used to describe the folds and spaces that are part of the peritoneum.

1. Mesentery- A mesentery is double layer of visceral peritoneum. It connects an intraperitoneal organ to the posterior abdominal wall. It provides a pathway for nerves, blood vessels and lymphatics to the viscera. The mesentery of the small intestine is simply called 'the mesentery'.

- Mesentery related to other parts of the gastrointestinal system is named according to the viscera it connects to, for example the transverse and sigmoid mesocolons.

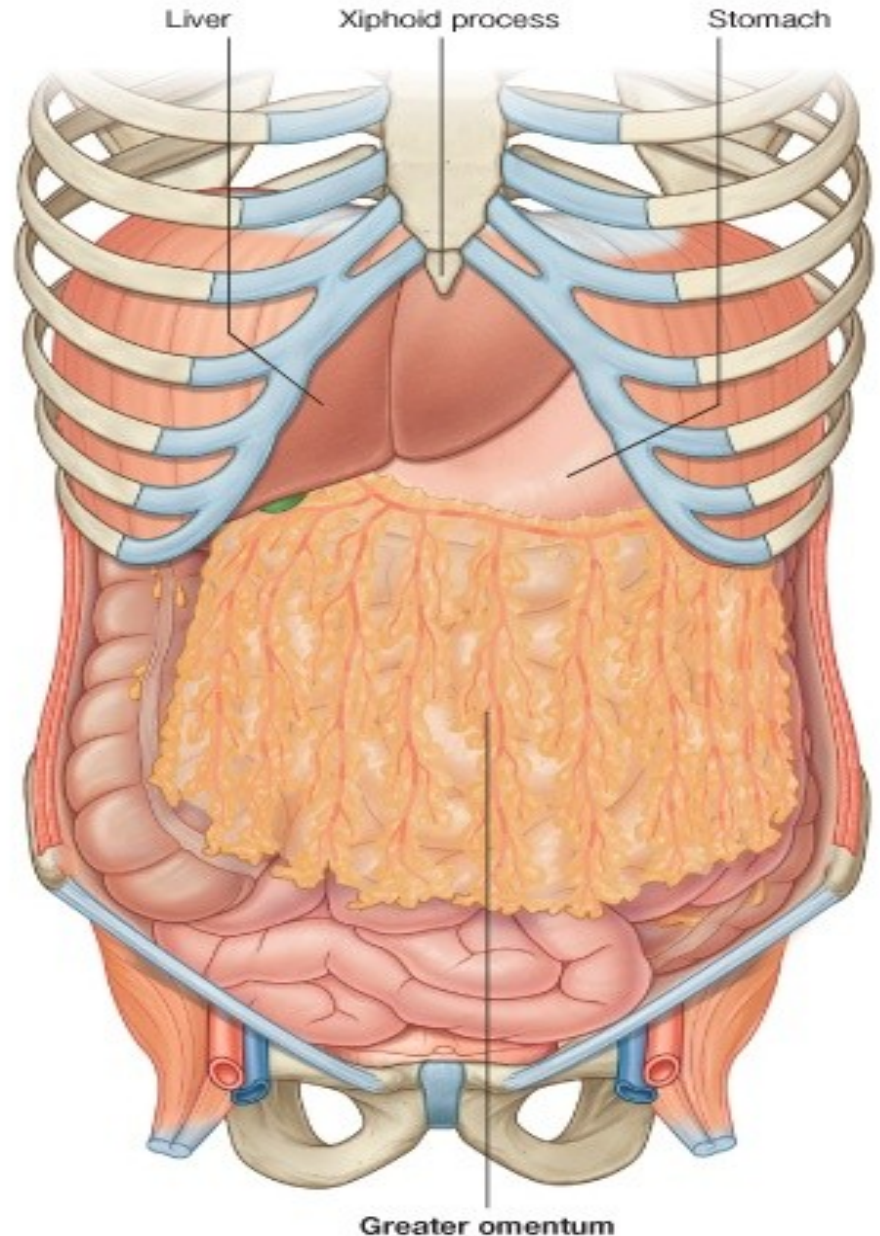


Peritoneal Reflections

2. Omentum- The omenta are sheets of visceral peritoneum that extend from the stomach and proximal part of the duodenum to other abdominal organs.

A. Greater Omentum- The greater omentum consists of four layers of visceral peritoneum.

- It descends from the greater curvature of the stomach and proximal part of the duodenum, then folds back up and attaches to the anterior surface of the transverse colon.
- It has a role in immunity and is sometimes referred to as the 'abdominal policeman' because it can migrate to infected viscera or to the site of surgical disturbance.



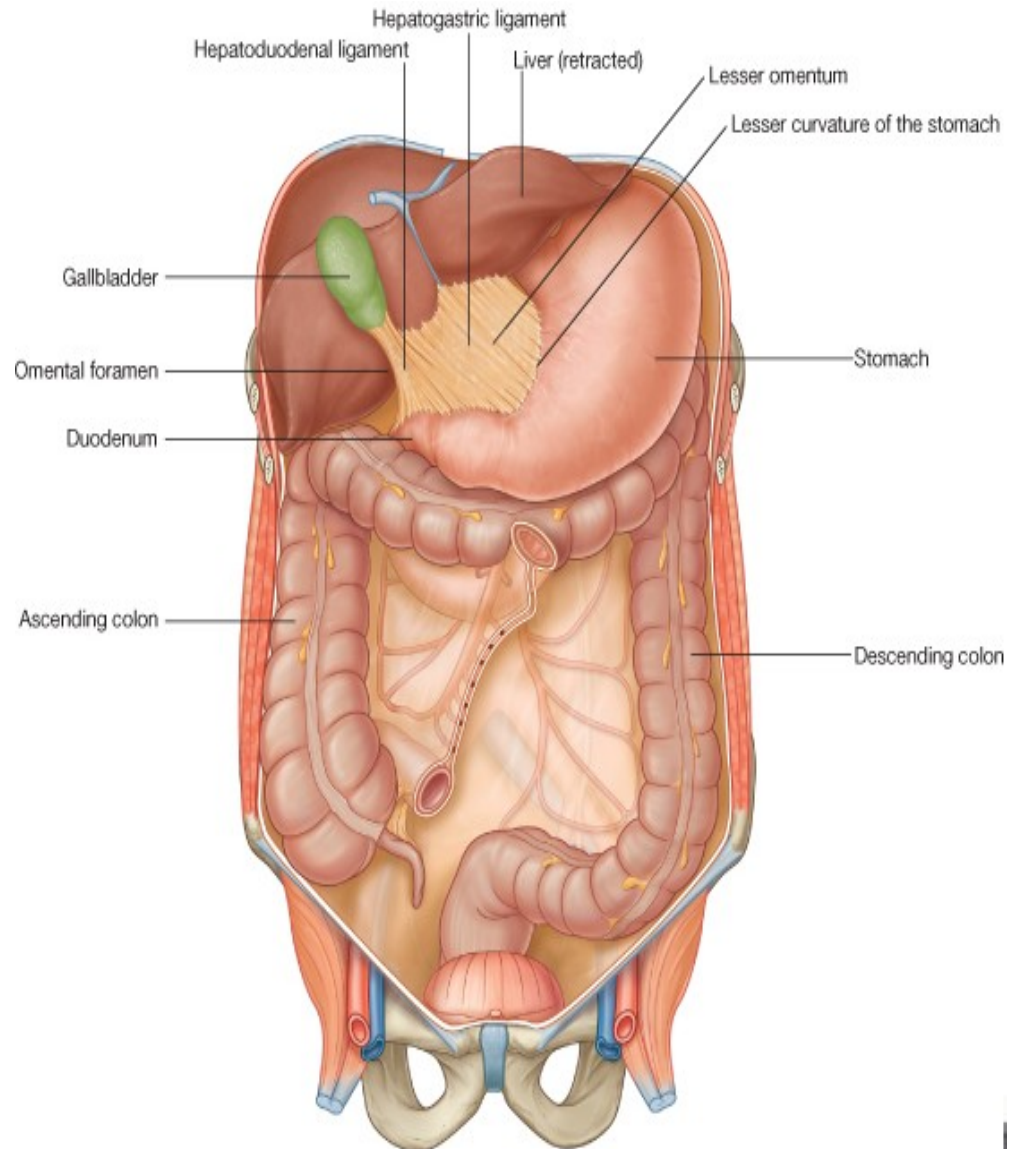
Peritoneal Reflections

B. Lesser Omentum- The lesser omentum is a double layer of visceral peritoneum, and is considerably smaller than the greater and attaches from the lesser curvature of the stomach and the proximal part of the duodenum to the liver.

- It consists of two parts: the hepatogastric ligament (the flat, broad sheet) and the hepatoduodenal ligament (the free edge, containing the portal triad).

3. Peritoneal Ligaments

- A peritoneal ligament is a double fold of peritoneum that connects viscera together or connects viscera to the abdominal wall.
- An example is the hepatogastric ligament, a portion of the lesser omentum, which connects the liver to the stomach.



Relationship between the organs and peritoneum

➤ Due to intraembryonal processes the organs have different relationship with the peritoneum.

1. **Intraperitoneal** organs are entirely covered with peritoneum. They are connected to the abdominal wall with ligaments or meso, which ensures greater mobility.

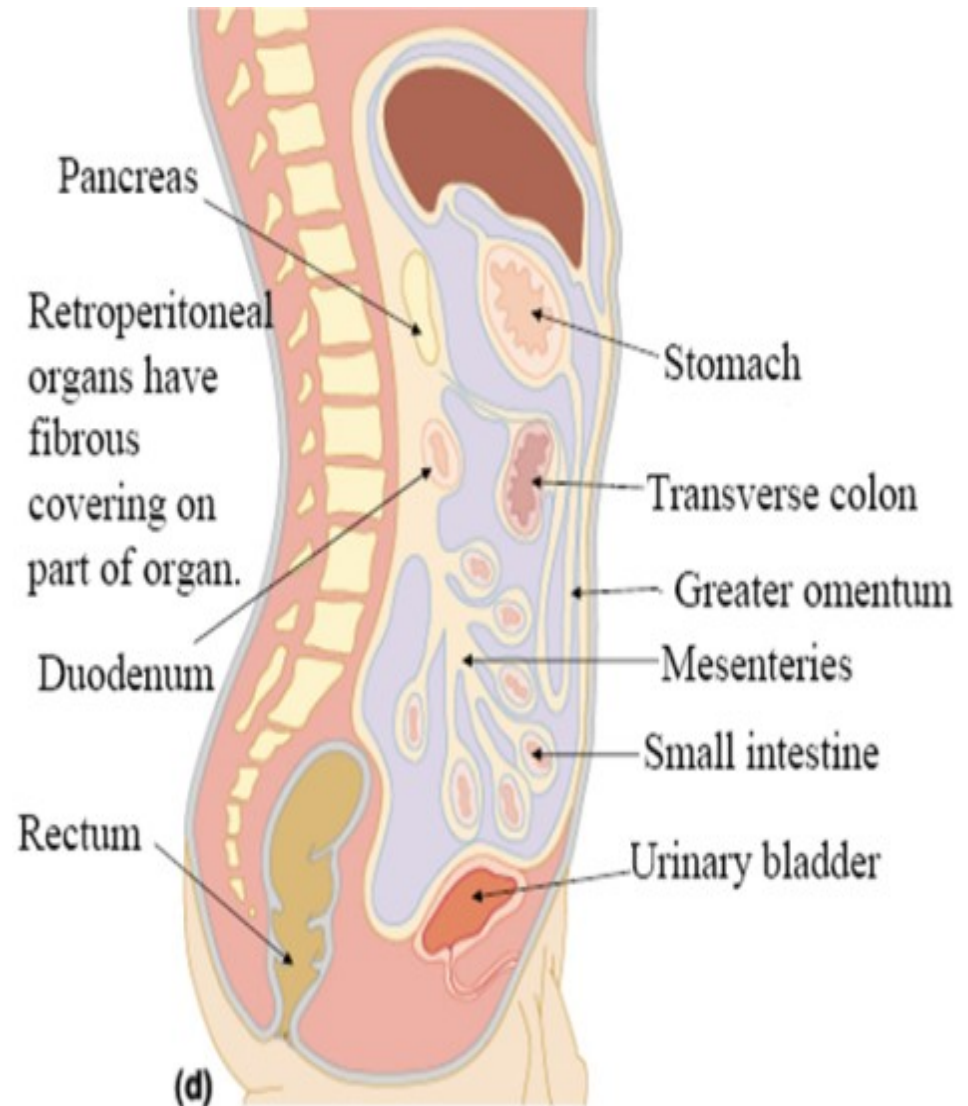
2. **Extraperitoneal** organs are partially or entirely devoid of peritoneum. They are slightly movable or immovable.

According to their position these are:

A) **Retroperitoneal** – on the posterior abdominal wall

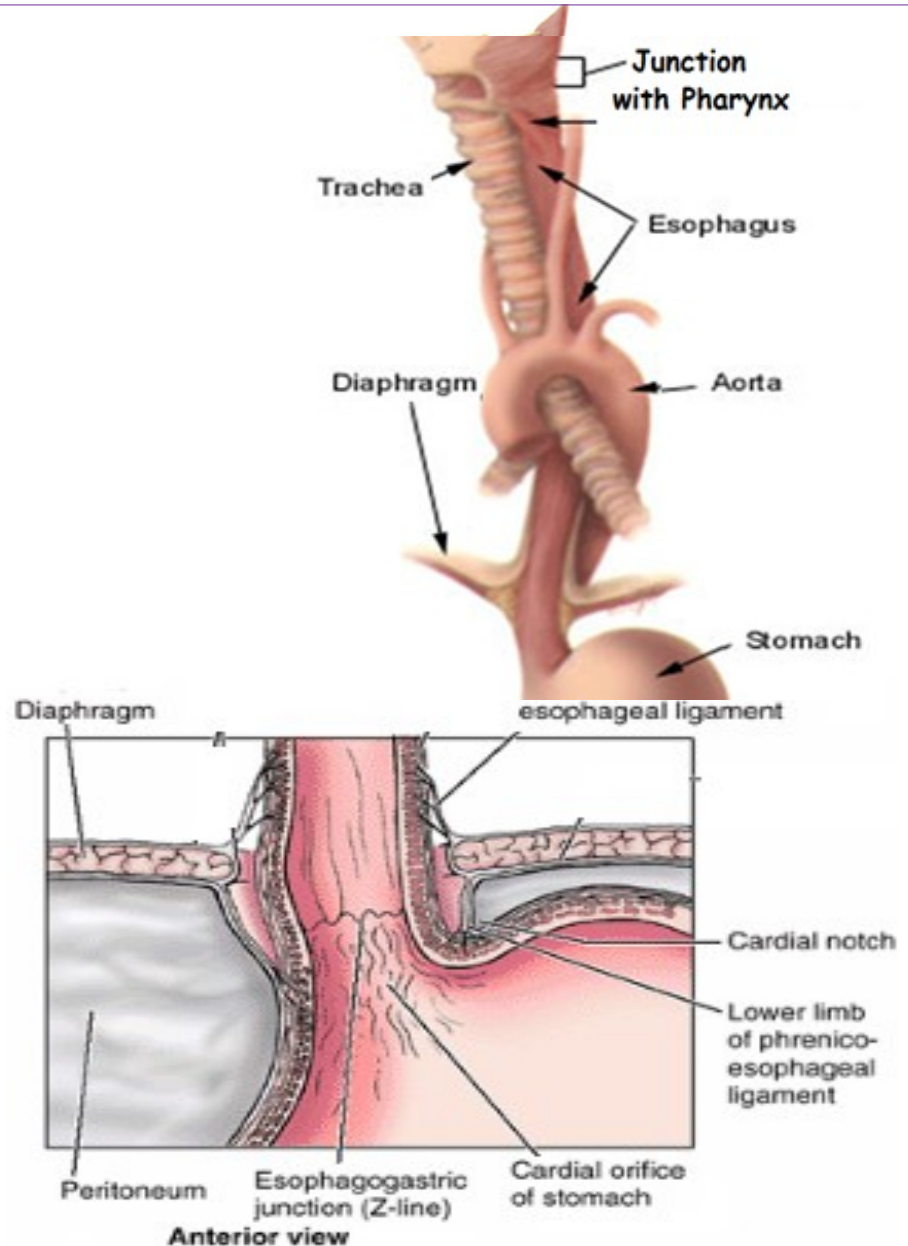
B) **Subperitoneal** – in the lesser pelvis

C) **Preperitoneal** – at the anterior abdominal wall.

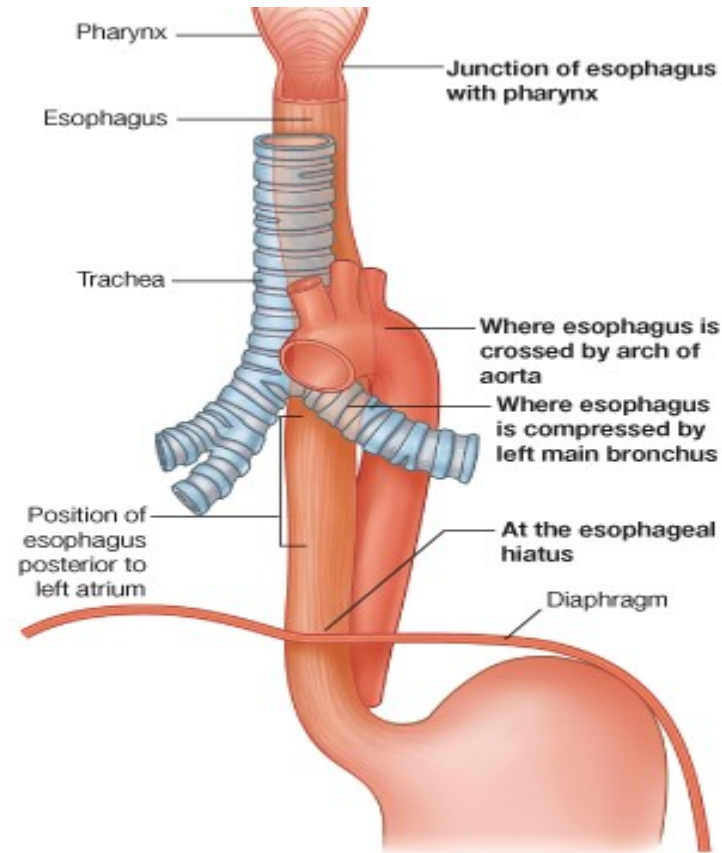


ANATOMY OF THE ABDOMINAL VISCERA

- ❑ Terminal part of the Esophagus
- ✓ Passes through the esophageal hiatus of the diaphragm
- ✓ Extends for 2-3 cm in the abdominal cavity
- ✓ Joins the stomach at gastroesophageal junction.



- ⦿ Inferior to the arch, the esophagus again inclines to the left as it approaches and passes through the esophageal hiatus in the diaphragm
- ⦿ **Has three parts:**
 - Cervical part
 - Thoracic part
 - Abdominal part
- ⦿ **The thoracic part:**
 - ⦿ Is about 16 cm long
 - ⦿ Extends from the thoracic inlet the level of T10 (oesophageal hiatus of the diaphragm)



Nerve supply is:

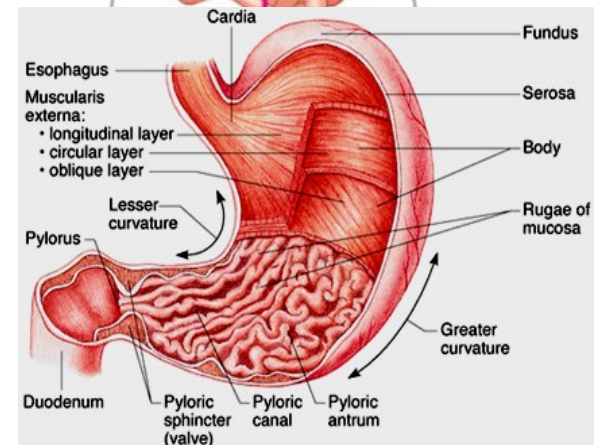
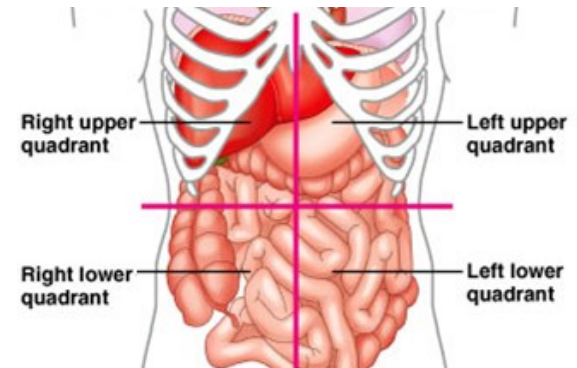
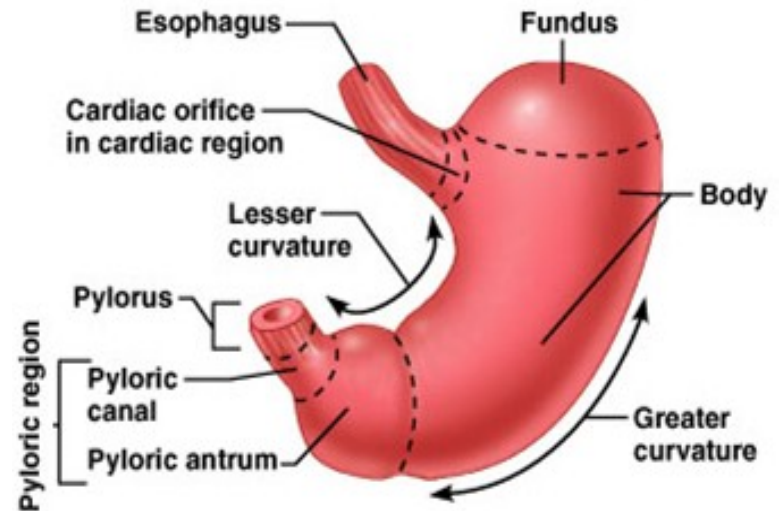
- ⊙ by vagus and the sympathetic trunk

Arterial supply:

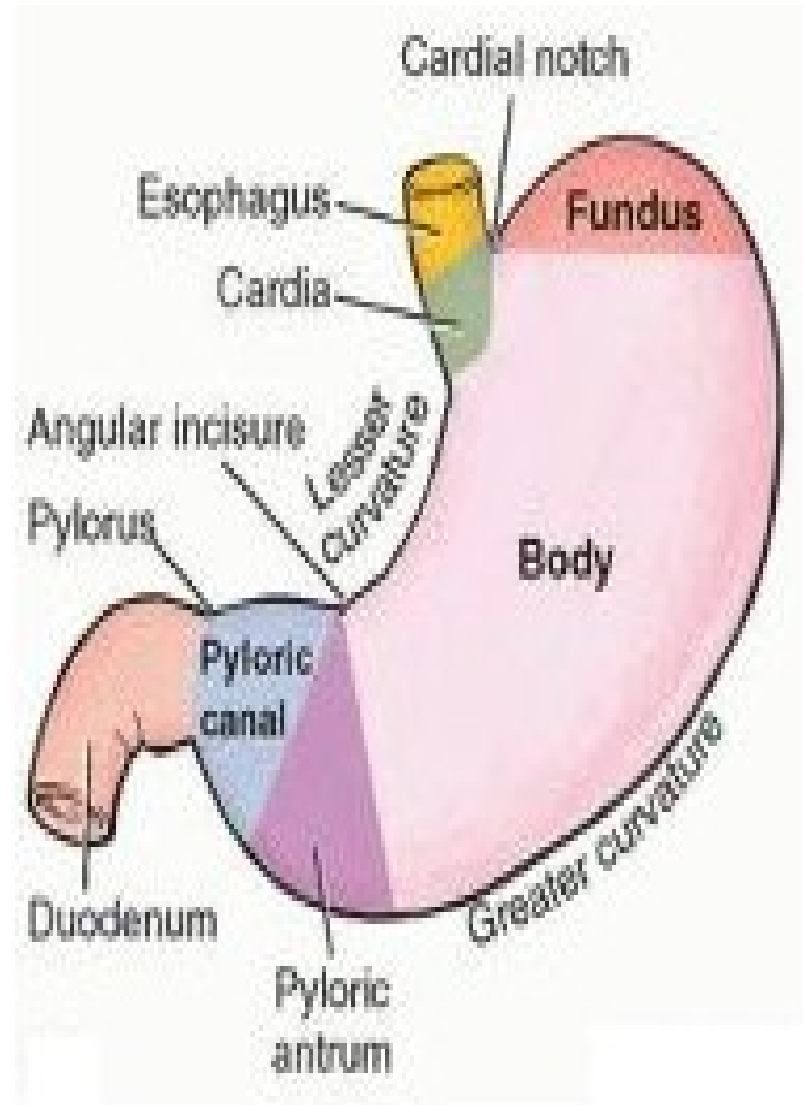
- ⊙ Cervical part: inferior thyroid artery
- ⊙ The thoracic part: oesophageal branch of the thoracic aorta
- ⊙ Abdominal part: left gastric and inferior phrenic arteries

□ The Stomach

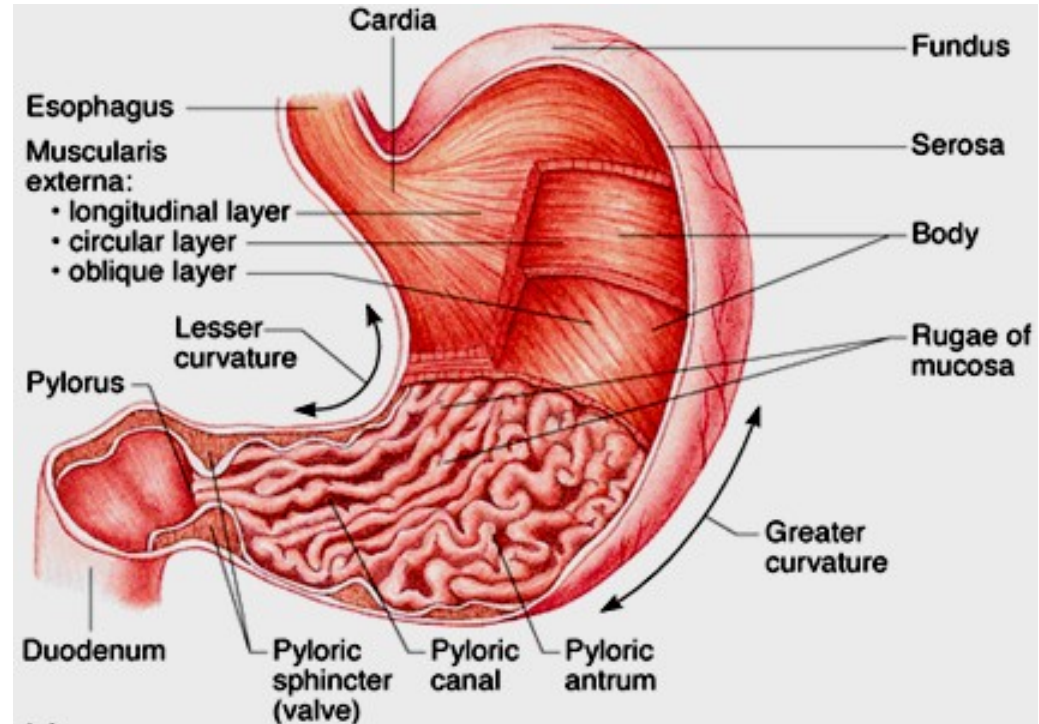
- Most dilated part of the *GIT*;
- Lies in the upper ULQ;
- Has the **cardia**, **fundus**, **body** & **pyloric** regions;
- The pyloric region has 2 parts: a pyloric **antrum** & pyloric **canal**.
- The **pyloric canal** ends at **pyloric sphincter**.



- ✓ Its lateral sides consists of:
 - A **greater curvature**
 - A **lesser curvature**
- ✓ Proximally has a **cardial notch**, an angle created when esophagus enters stomach
- ✓ Distally has an **angular incisure**, a bend on its lesser curvature.



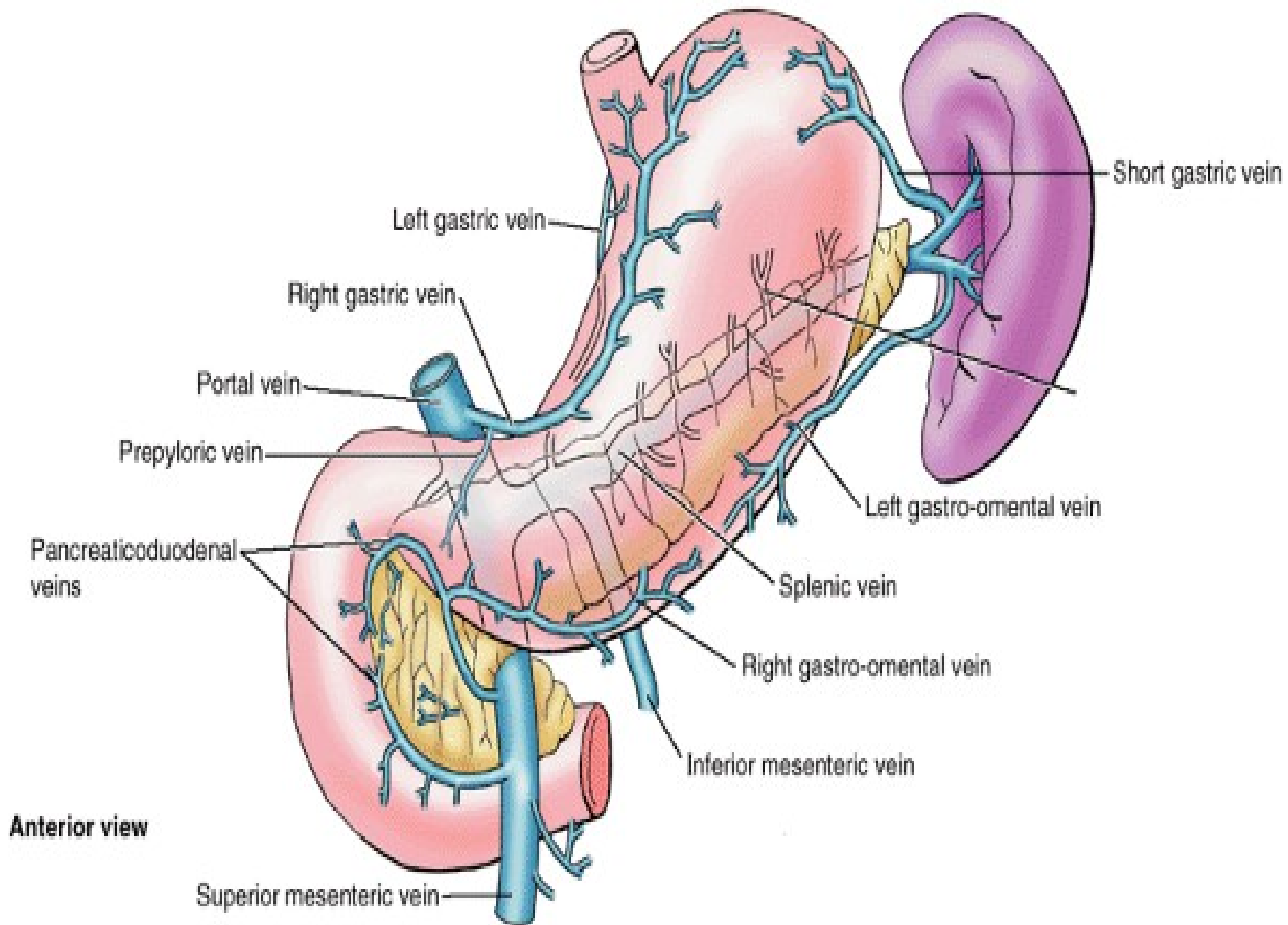
- ✓ Its **diameter & volume** varies depending on the amount of food it contains
- ✓ Can expand to hold up to **4 liters** of food; when empty, collapses inward, throwing its mucosa into large, longitudinal folds called **rugae**.



Blood Vessels

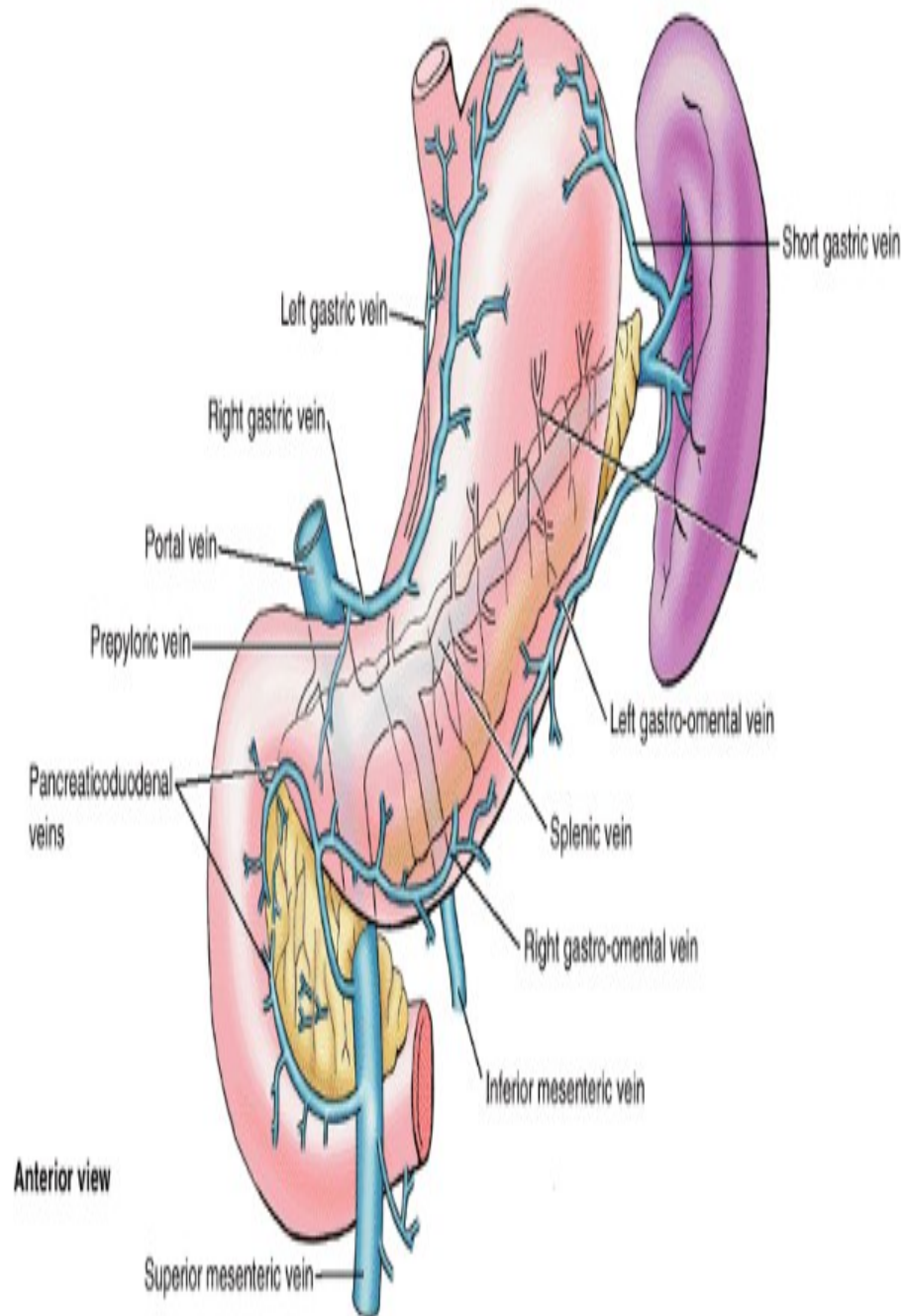
Artery of the stomach

- ⦿ The stomach has a rich arterial supply arising from the celiac trunk and its branches.
- ⦿ right gastric arteries
- ⦿ left gastric arteries
- ⦿ along the greater curvature by the right and left gastro-omental arteries.
- ⦿ The fundus and upper body receive blood from the short gastric arteries.



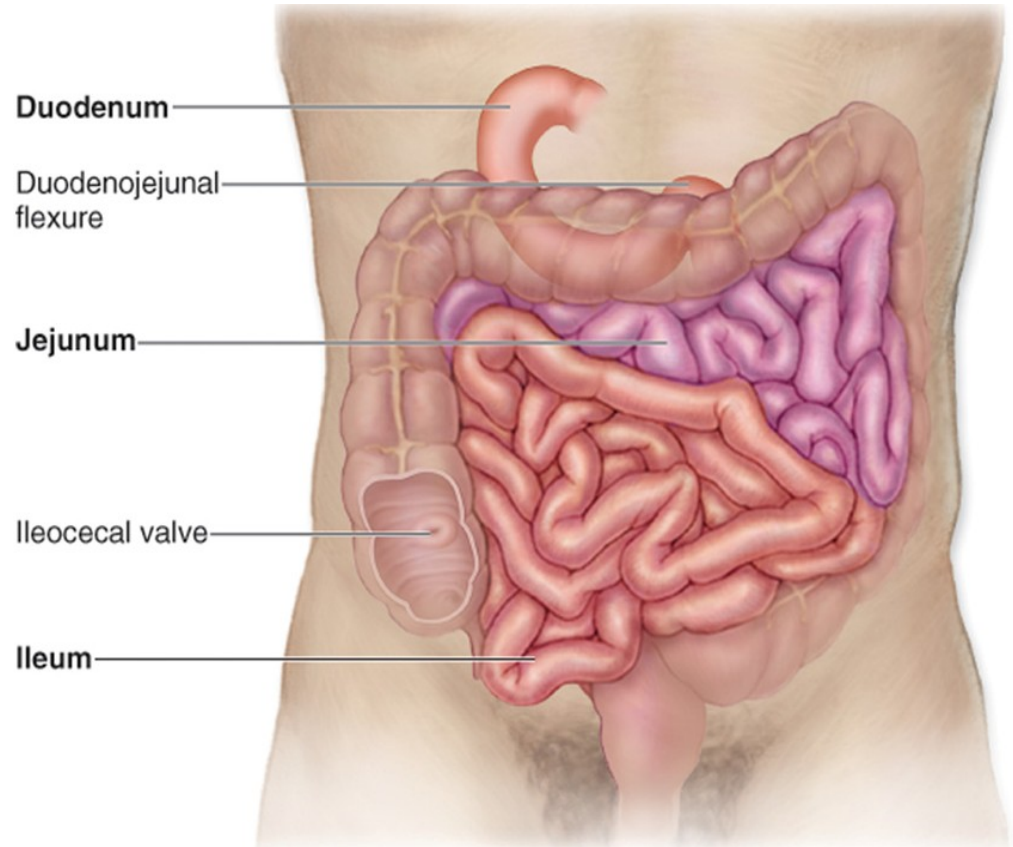
Venous drainage:

- The gastric veins parallel the arteries in position and course
- The right and left gastric veins drain into the portal vein
- the short gastric veins and left gastro-omental veins drain into the splenic vein, which joins the superior mesenteric vein (SMV) to form the portal vein.



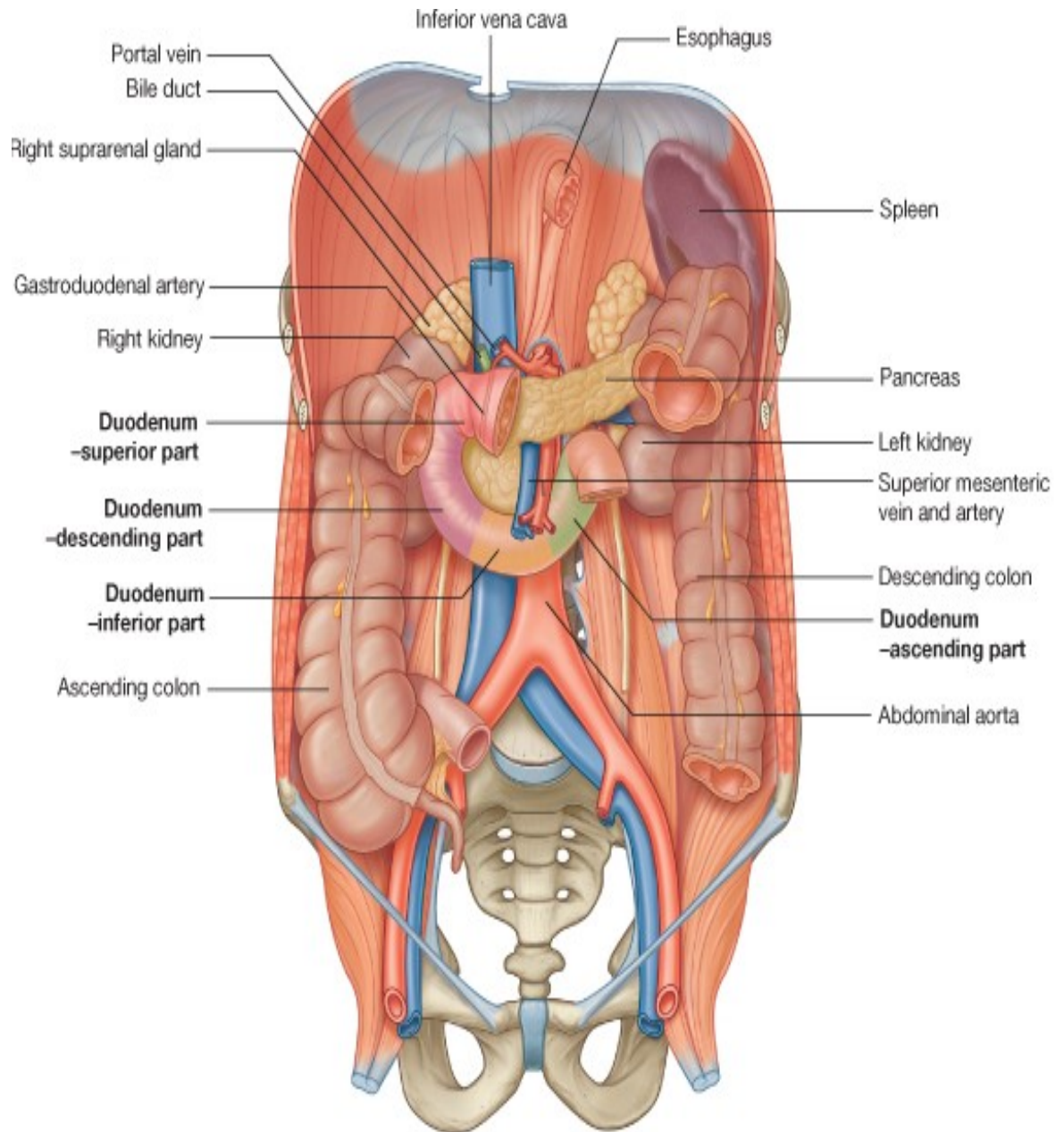
❑ The Small Intestine

- The primary site for absorption;
- The **longest**, convoluted part of the GIT;
- Extends from **pyloric sphincter** to the **ileocecal valve**;
- ✓ Has 3 parts: **duodenum**, **jejunum** & **ileum**;



The duodenum is divisible into four parts

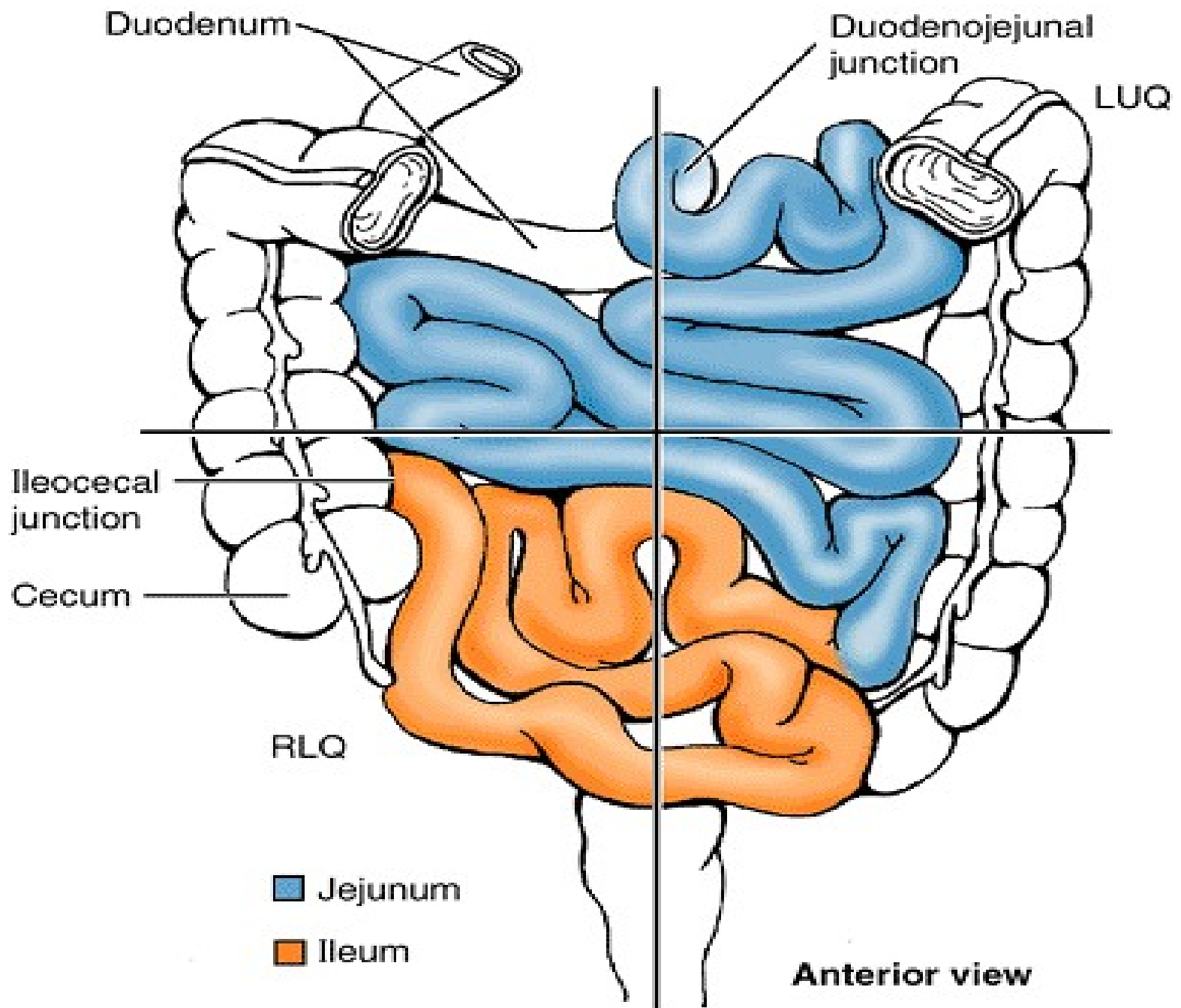
- **Superior (first) part:** short (approximately 5 cm) . lies anterolateral to the body of the L1 vertebra.
- **Descending (second) part:-** longer (7 - 10 cm) . descends along the right sides of the L1 - L3 vertebrae.
- **Horizontal (third) part:-** 6 - 8 cm long and crosses the L3 vertebra.
- **Ascending (fourth) part:** short (5 cm) and begins at the left of the L3 vertebra and rises superiorly as far as the superior border of the L2 vertebra.



Jejunum and Ileum

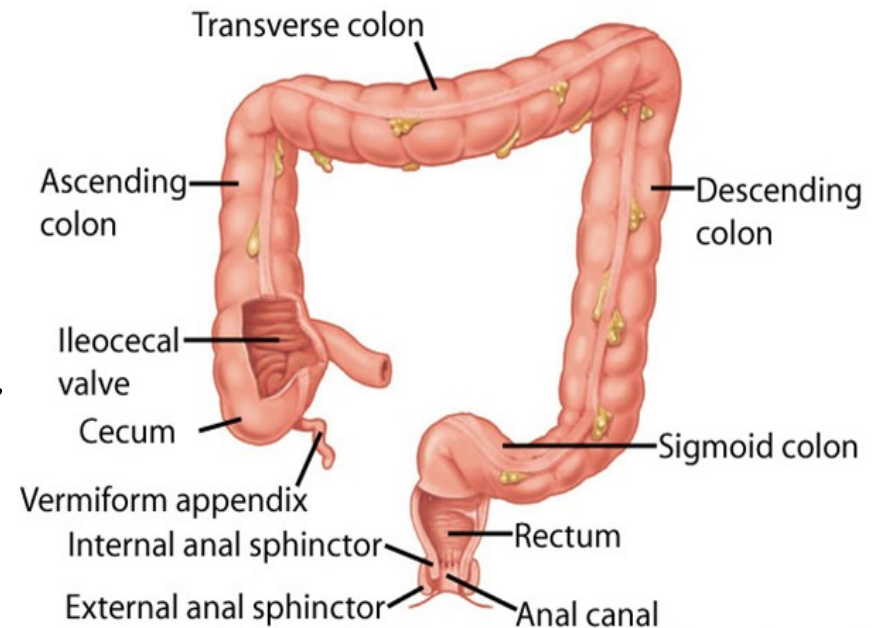
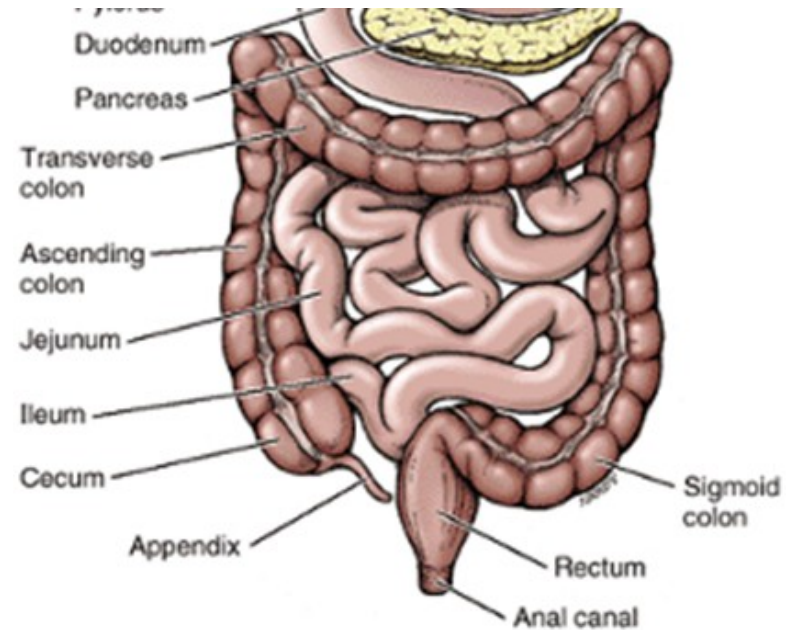
- ⦿ The second part of the small intestine, the jejunum, begins at the duodenojejunal flexure where the alimentary tract resumes an intraperitoneal course.
- ⦿ The third part of the small intestine, the ileum, ends at the ileocecal junction
- ⦿ Together, the jejunum and ileum are 6 - 7 m long, the jejunum constituting approximately **two fifths** and the ileum approximately **three fifths** of the intraperitoneal section of the small intestine.
- ⦿ Most of the jejunum lies in the left upper quadrant, whereas most of the ileum lies in the right lower quadrant.

- ④ The terminal ileum usually lies in the pelvis from which it ascends, ending in the **medial aspect of the cecum**.
- ④ Although no clear line of demarcation between the jejunum and ileum exists,
- ④ The **jejunal mesentery** is attached to the posterior abdominal wall above and to the left of the aorta, whereas the ileal mesentery is attached below and to the right of the aorta.



❑ The Large Intestine

- ✓ Involves in absorption of water, salts, fat soluble vitamins & in the production of feces;
- ✓ About 1.5 m long, which extends from the ileocecal valve to the anal canal;
- ✓ Has 5 major parts such as the cecum, appendix, colon, rectum & the anal canal.

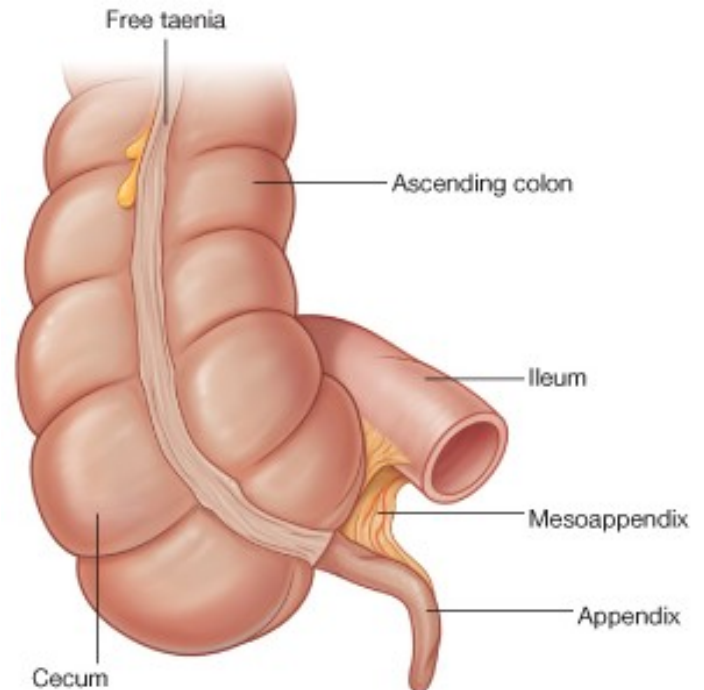
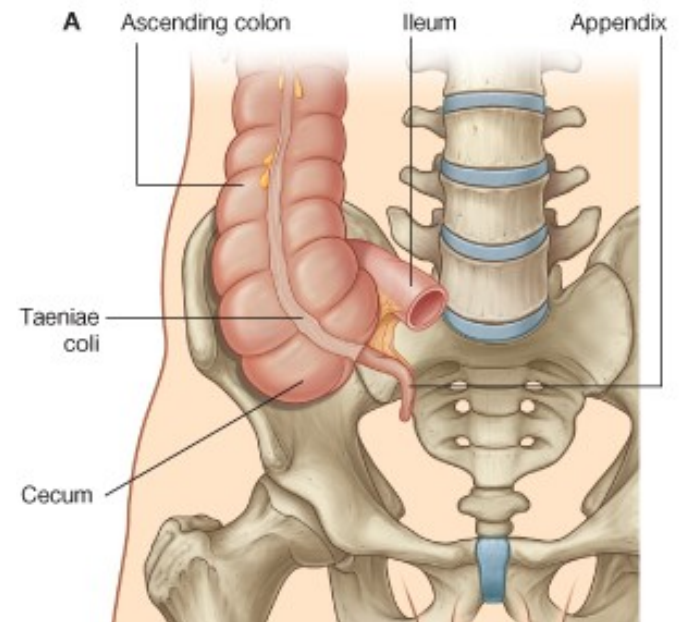


i. The Cecum

- ✓ The 1st, dilated part,
- ✓ Hanging inferiorly below the ileocecal junction;
- ✓ Continuous with the ascending colon above the ileocecal valve.

ii. The Appendix

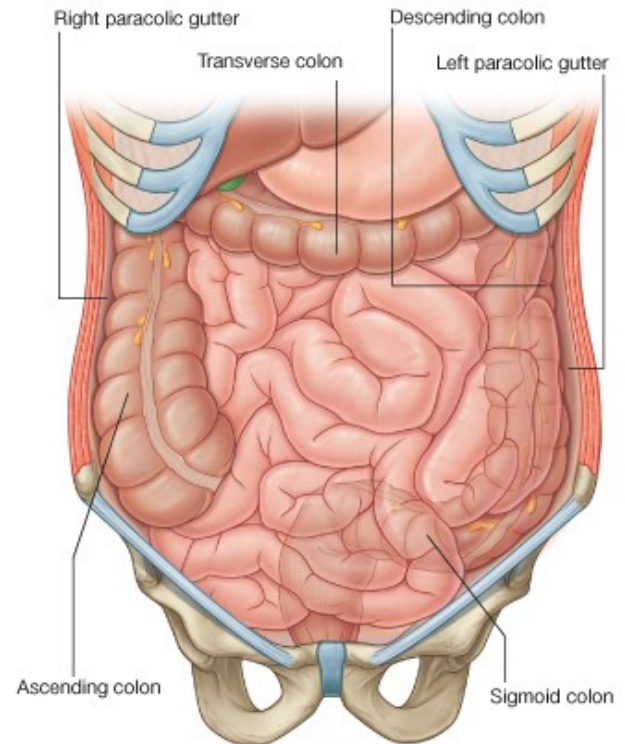
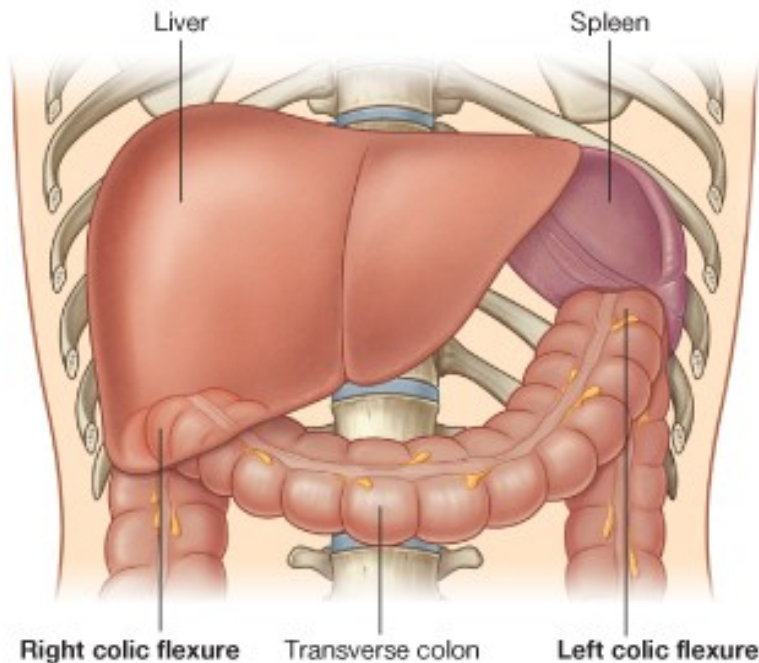
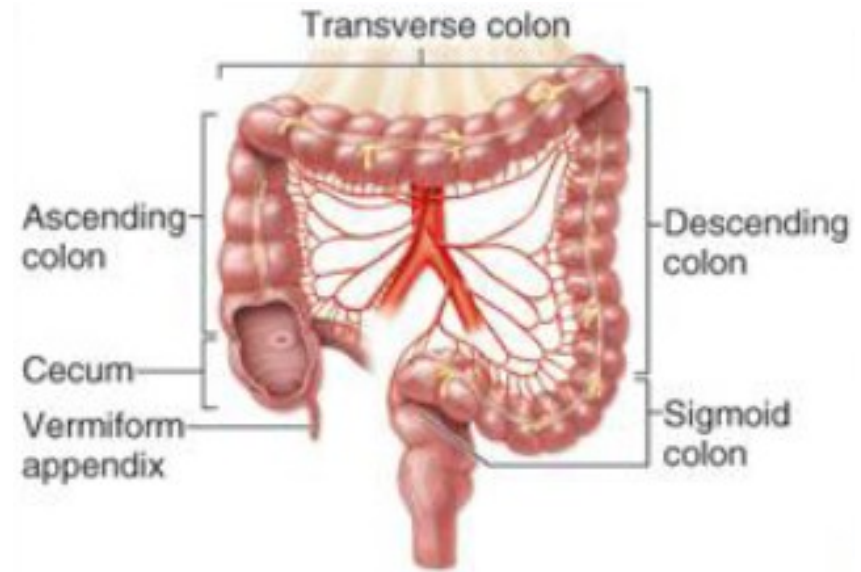
- A blind intestinal diverticulum; that contains masses of lymphoid tissue in its wall.
- Arises from **posteromedial** aspect of the cecum,
- An intraperitoneal part, has *mesoappendix*.
- suspended from the terminal ileum by the mesoappendix, which contains the appendicular vessels
- **teniae coli**



iii. The Colon:

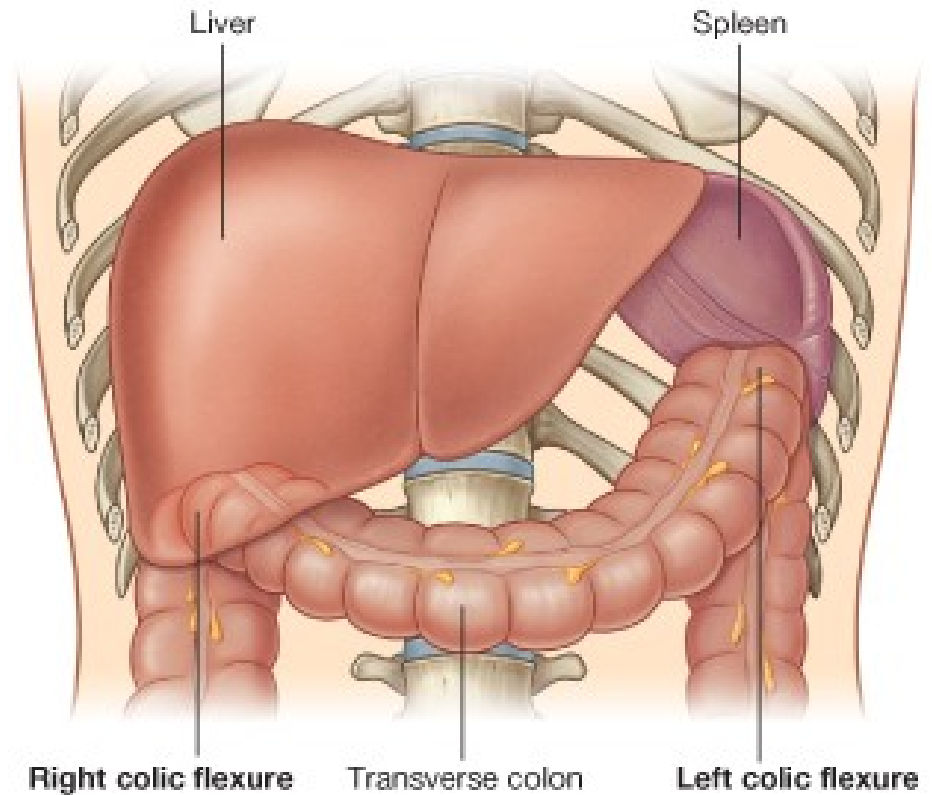
a. The ascending colon:

- ✓ About **25cm** long;
- ✓ Extends from the ileocecal valve to the **right colic flexure**.



❑ The transverse colon:

- ✓ Extends between the **right & left colic flexures**;
- ✓ The longest (50 cm long) & most mobile part, suspended from the pos. abd. wall by **the transverse mesocolon**.

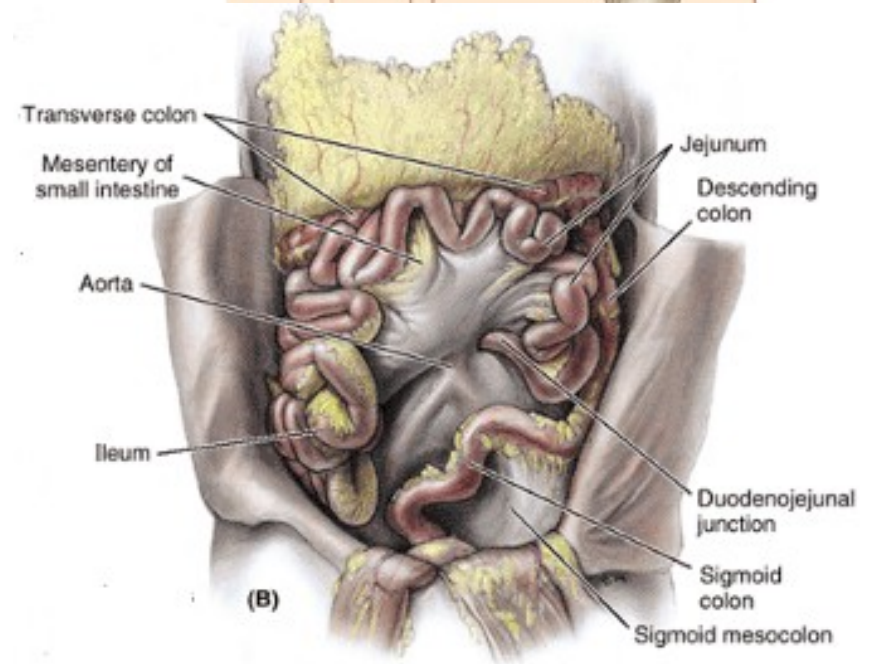
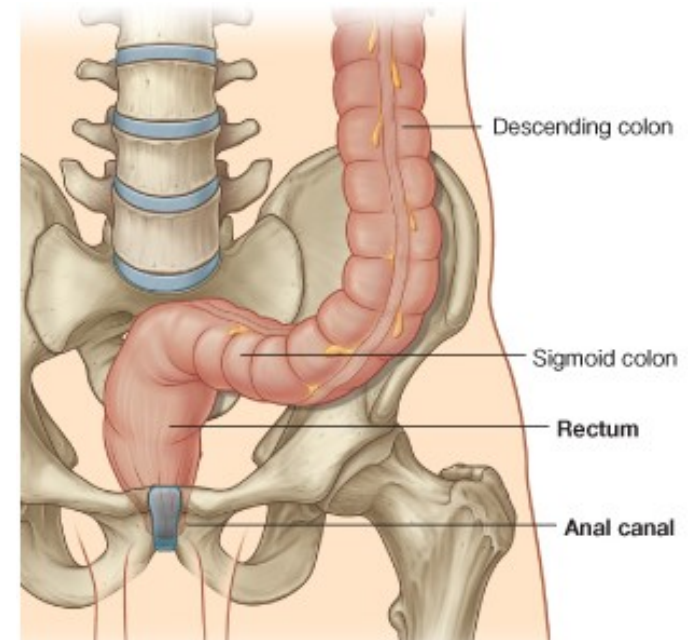


❑ The descending colon:

- ✓ Extends from the **left colic flexure** to the **pelvic brim**, then continue as the sigmoid colon.

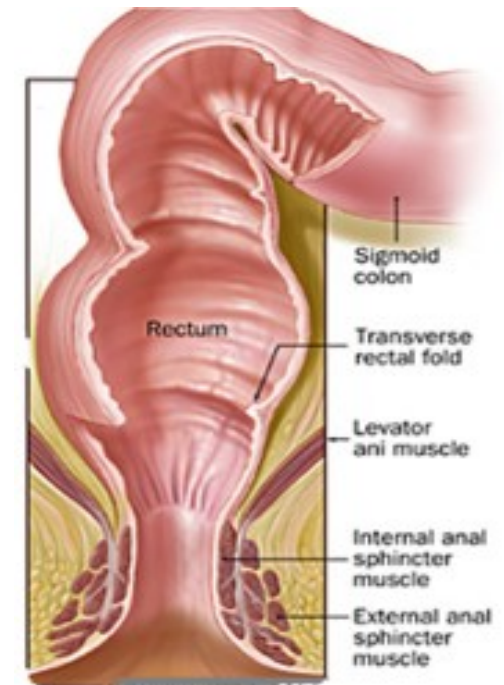
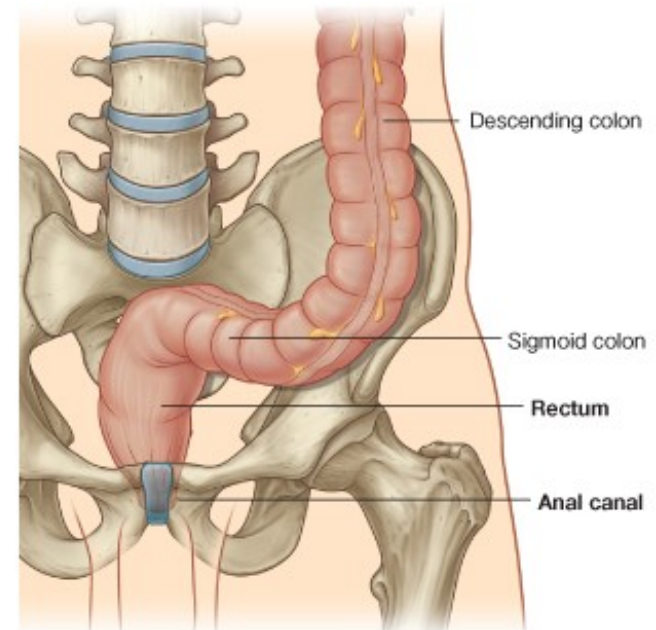
❑ The sigmoid colon:

- ✓ About **45cm** long;
- ✓ Extend to the rectum & attached to the wall by **sigmoid mesocolon**.



□ The Rectum

- About **13cm** long,
- Extends from the sigmoid colon to the pelvic diaphragm
- Fixed terminal part of the large intestine
- Inner surface contains **transverse rectal folds/valves**, that separate feces from flatus, thus allowing gas to pass.



❑ The Anal Canal

- ✓ About 3 cm long
- ✓ Continues from the rectum
- ✓ Has 2 sphincters:

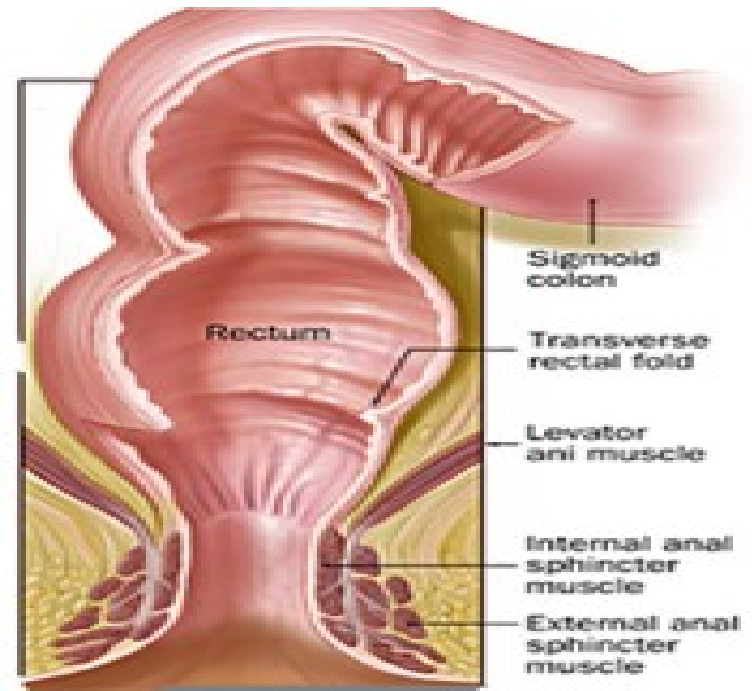
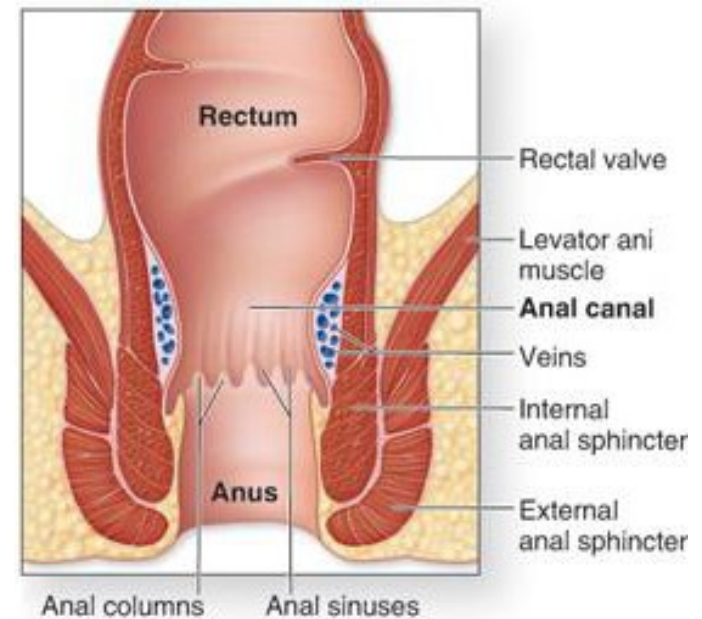
a) External anal sphincter

(composed of voluntary skeletal muscle)

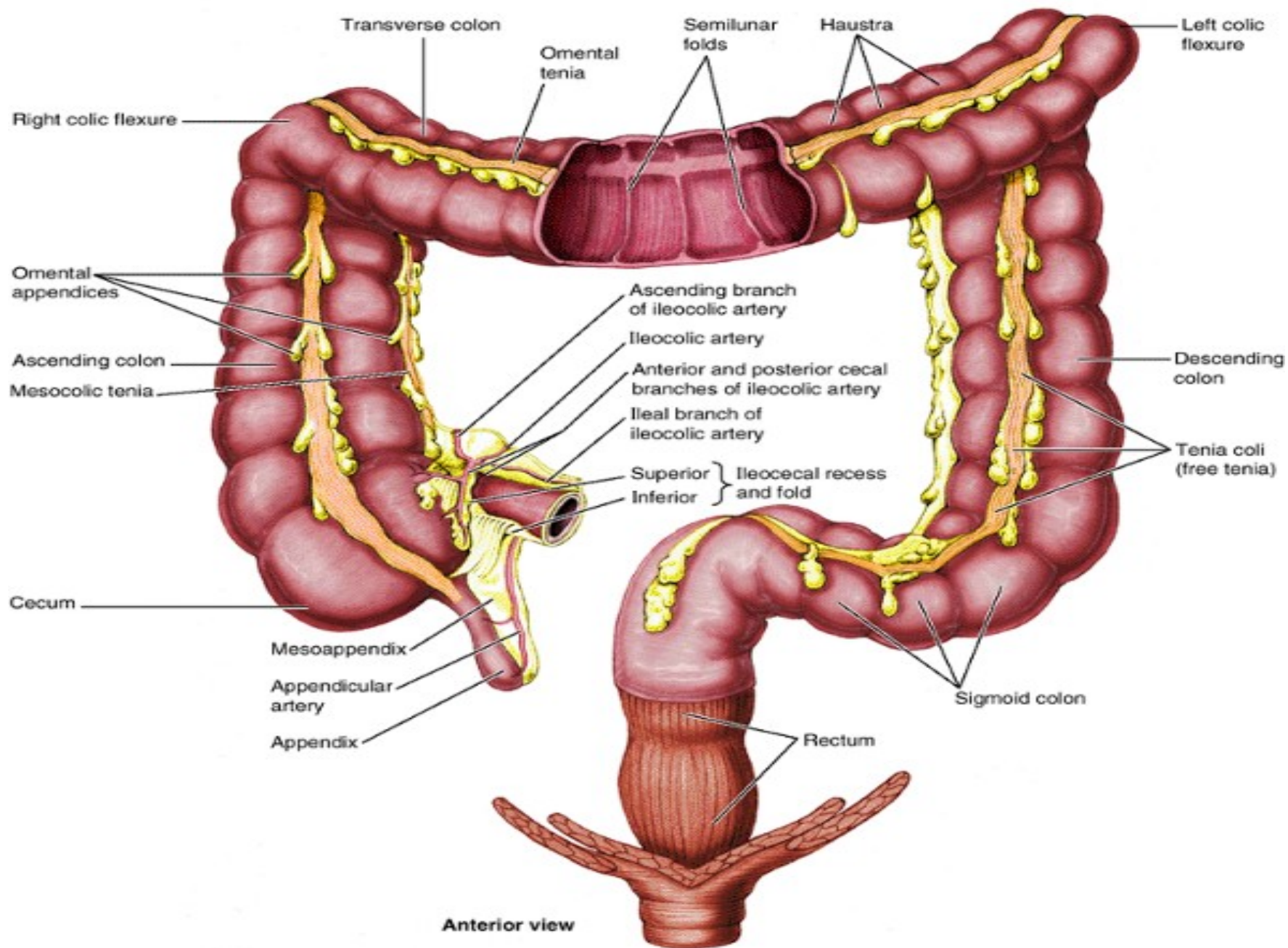
b) Internal anal sphincter

(composed of involuntary smooth muscle)

- These sphincters which act to open and close the anus,

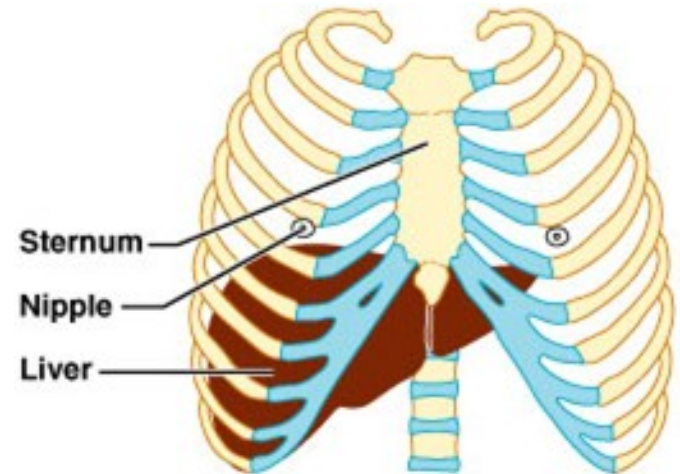
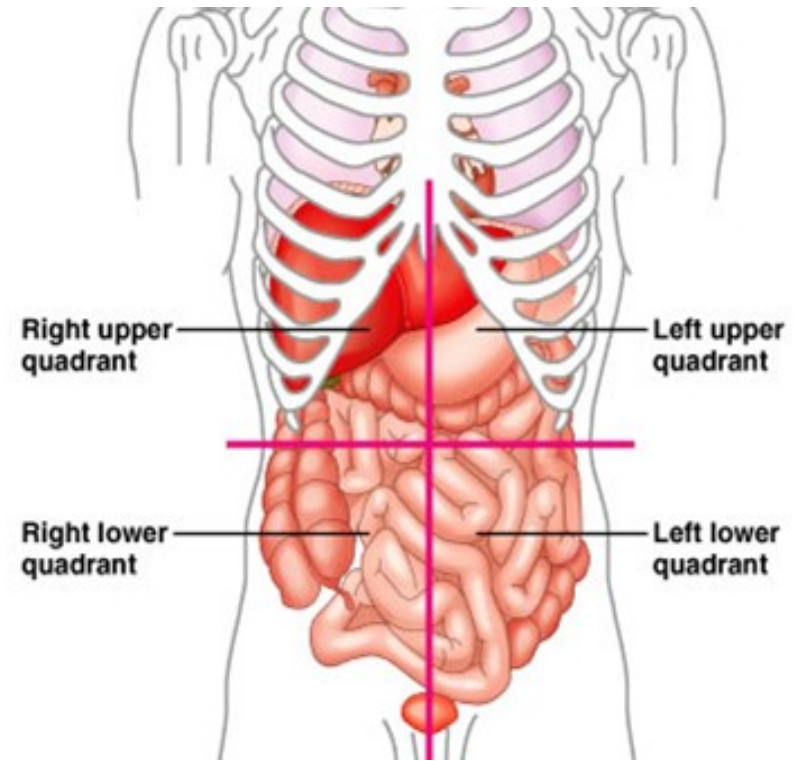


- ◎ **The large intestine can be distinguished from the small intestine by:**
- ◎ **Omental appendices:**
 - small, fatty, omentum-like projections.
- ◎ **teniae coli:**
- ◎ **Haustra:**
 - sacculations of the wall of the colon between the teniae



□ The Liver

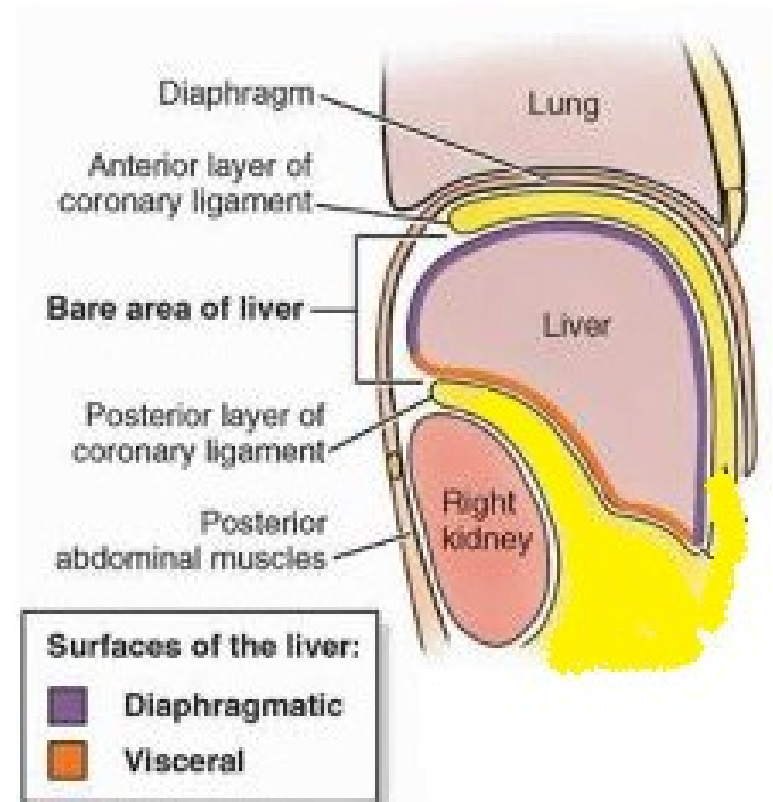
- ✓ Lies mainly in the **RUQ**
- ✓ Protected by the thoracic cage & diaphragm
 - is the largest gland in the body
 - weighs approximately **1500 g**
 - accounts for approximately **2.5%** of adult body weight.
 - In a mature fetus—when it serves as a hematopoietic organ—it is proportionately twice as large (**5%** of body weight).
 - Except for fat, all nutrients absorbed from the digestive tract are initially conveyed to the liver by the portal venous system



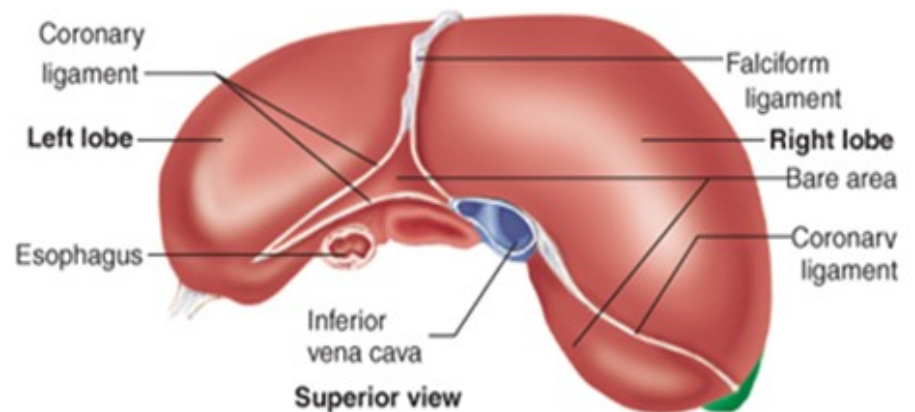
- ✓ The liver has **diaphragmatic** & **visceral** surfaces

A. The diaphragmatic surface

- Smooth & convex
- Has **anterior**, **superior** & **posterior** parts;
- Covered by **visceral peritoneum**, except posteriorly, where directly contacts the diaphragm, **bare area of the liver**.

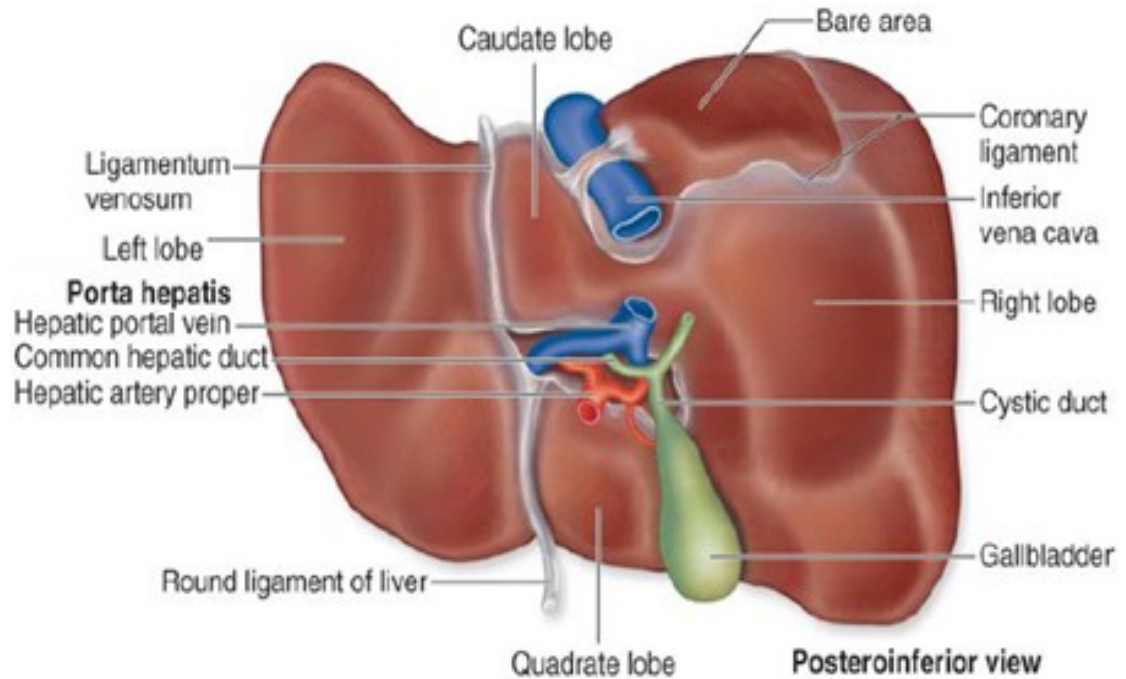


(A) Right lateral view - schematic sagittal section



B. The Visceral surface:

- Lies in the **postero-inferior** direction,
- Covered by visceral peritoneum except at the **fossa** for **gallbladder** & at the **porta hepatis**.



➤ The porta hepatis:

- A transverse fissure where the hepatic **vessels, nerve plexus, & ducts** enter & leave the liver.

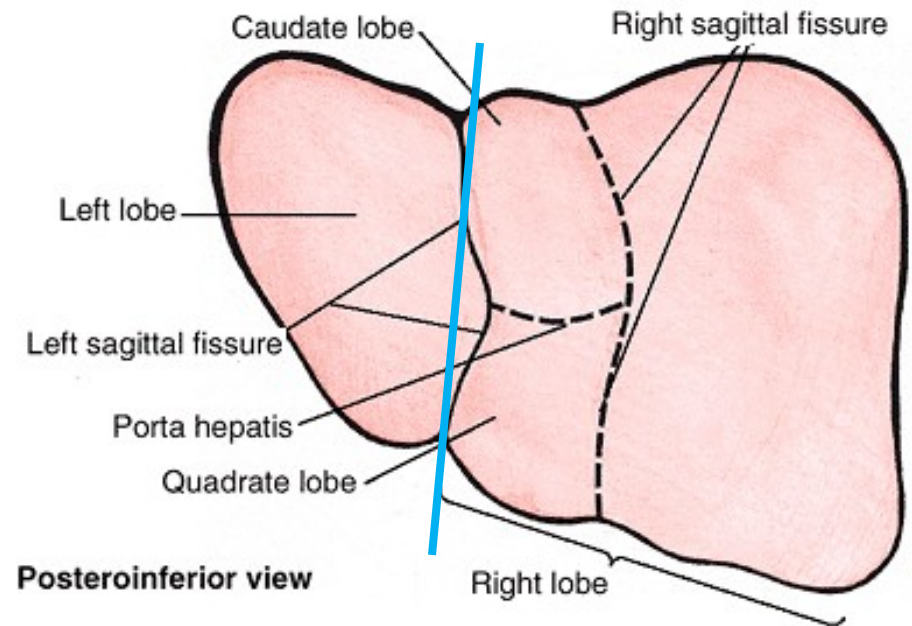
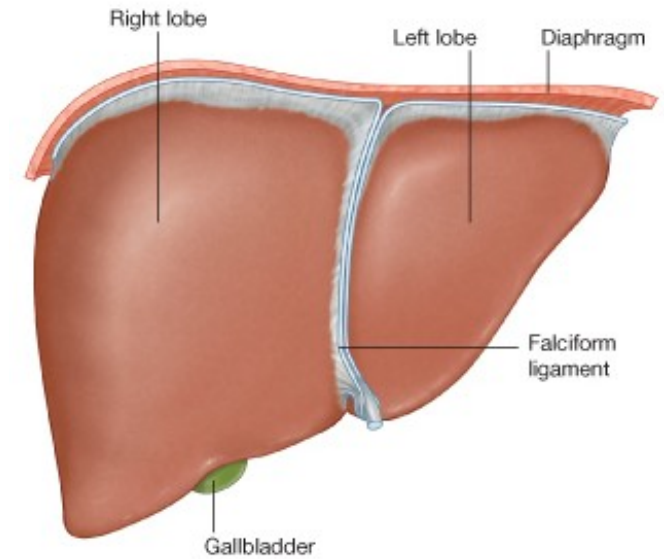
1. Anatomical subdivisions:

✓ The liver divided into Rt &

Lt lobes:

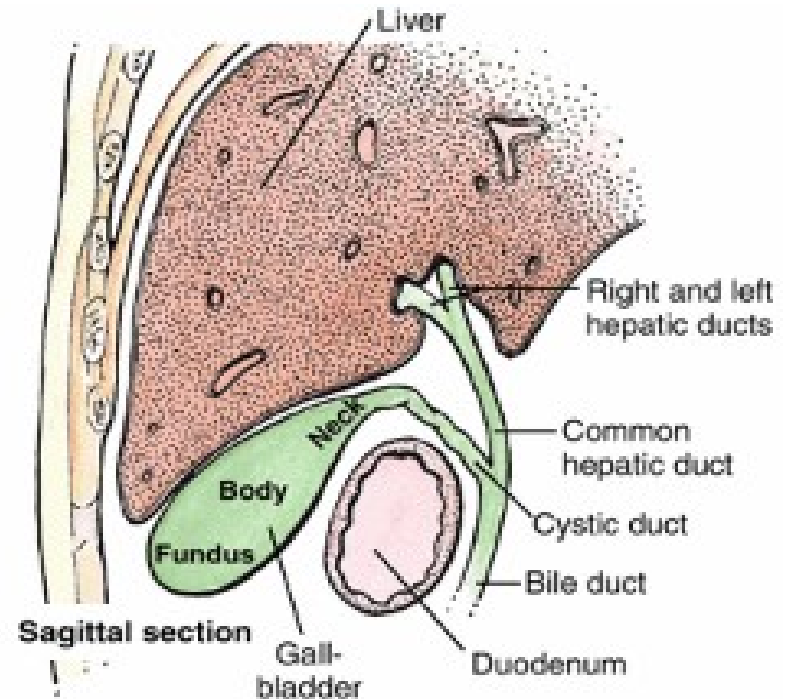
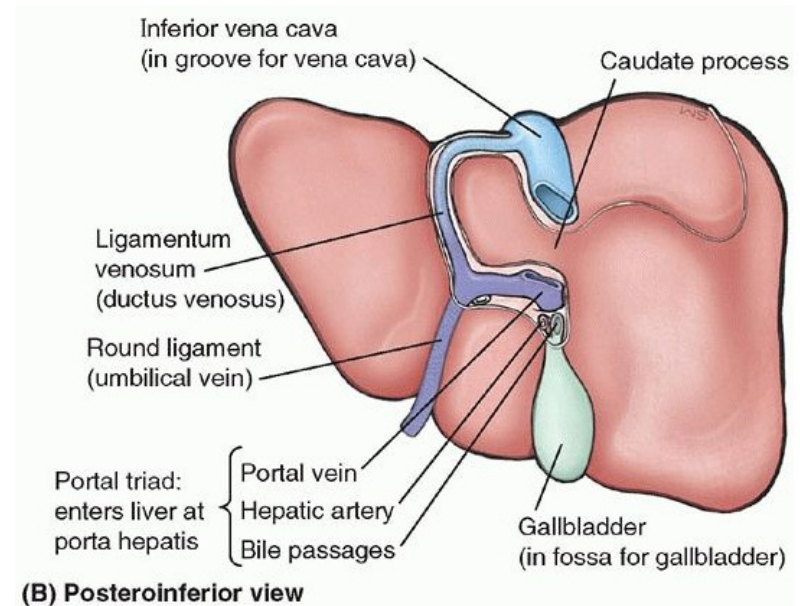
- Anteriorly by falciform ligament
- Posteriorly by Lt sagittal fissure

✓ According to this division the right lobe includes quadrate & caudate lobes.



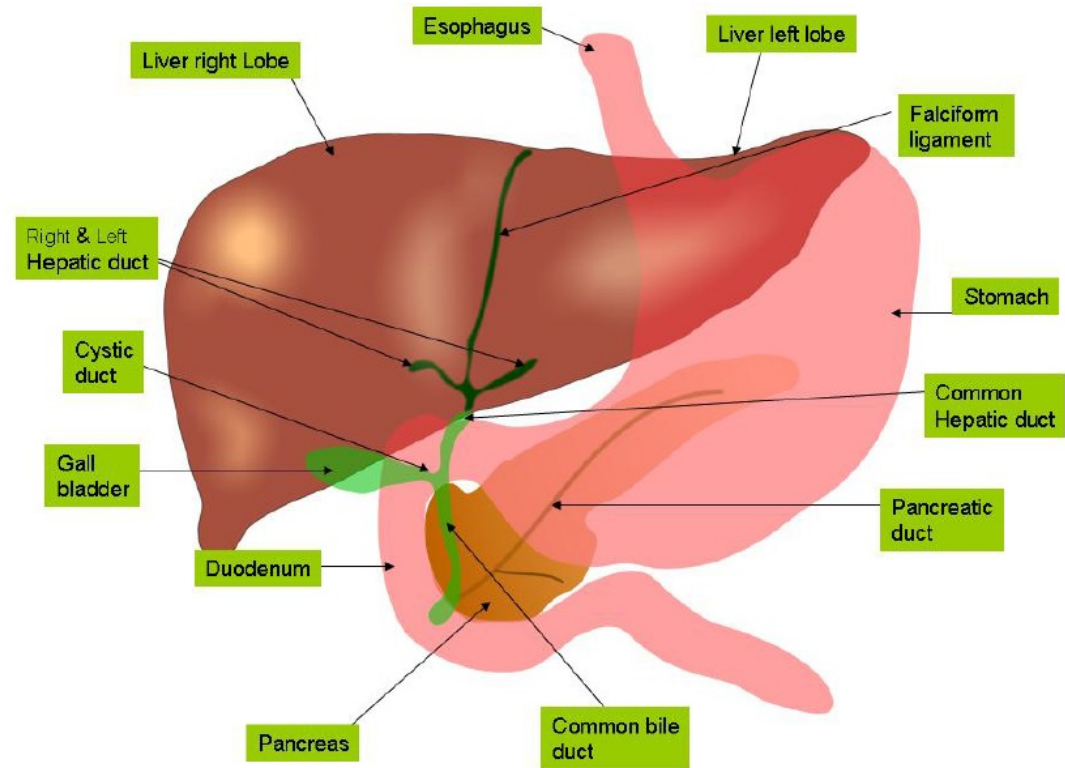
❑ The Gallbladder

- Lies at the fossa for the gallbladder
- Has three parts; fundus, body & neck
- The **cystic duct** connects the neck to common hepatic duct
- **Receives, concentrates, & stores** bile from the liver.

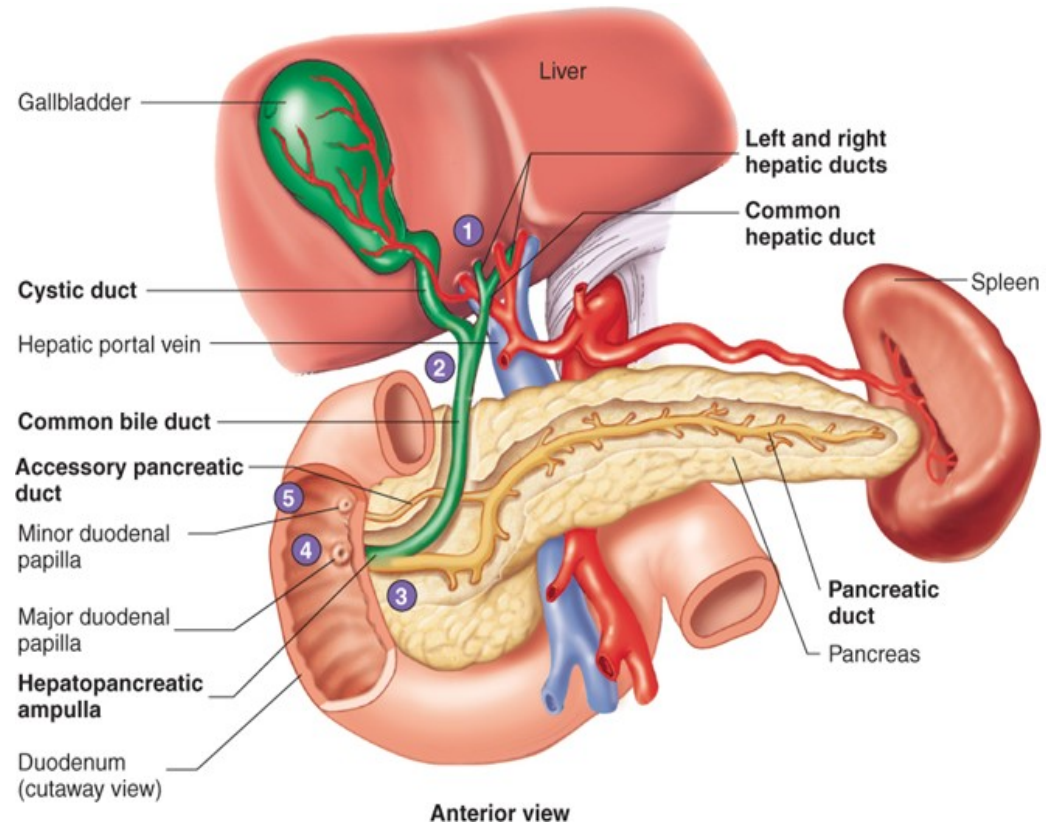


❑ The Biliary Ducts

- ✓ The Rt & Lt hepatic ducts carry bile from the liver & join to form the **common hepatic duct**;
- ✓ The common hepatic duct then joins the cystic duct to form **common bile duct**.

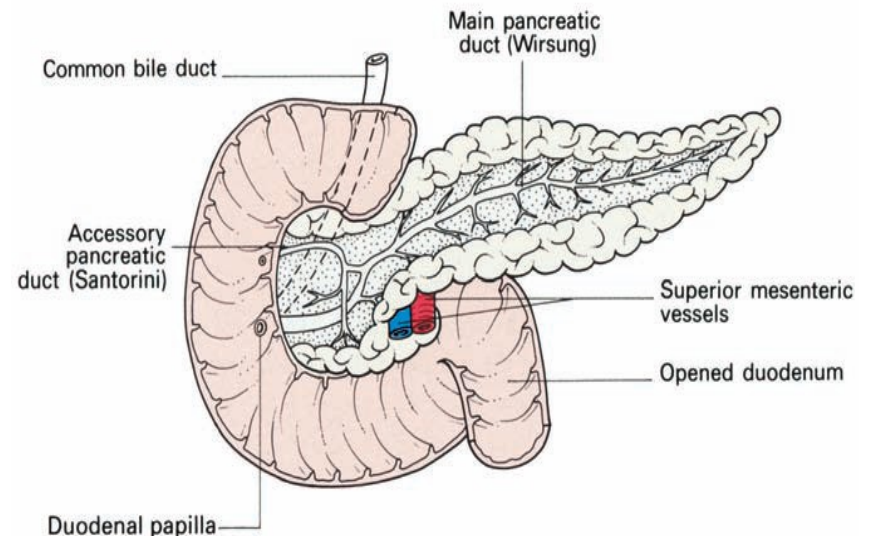
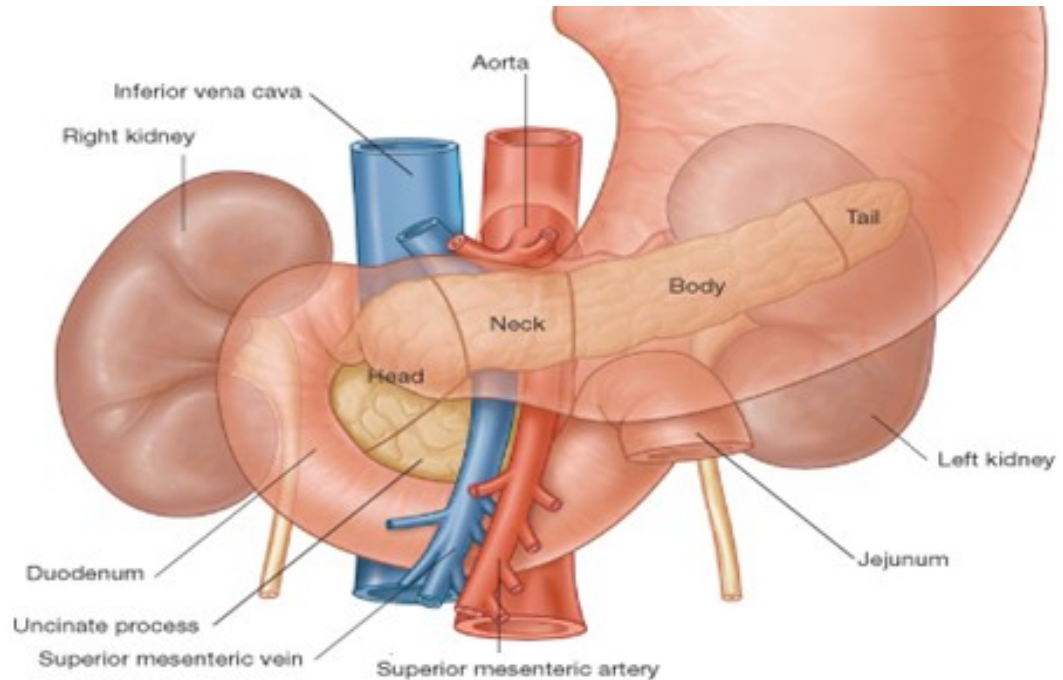


- ✓ The common bile duct joins main pancreatic duct to form hepatopancreatic ampulla, which opens at 2nd part of the duodenum through major duodenal papilla;
- ✓ An accessory pancreatic duct opens into the minor duodenal papilla.



□ The Pancreas

- Lies retroperitoneally across the PAW, except its tail;
- Has four parts: **head**, **neck**, **body**, & **tail** (also has **uncinate** process);
- Functions as an **endocrine** & an **exocrine** gland.

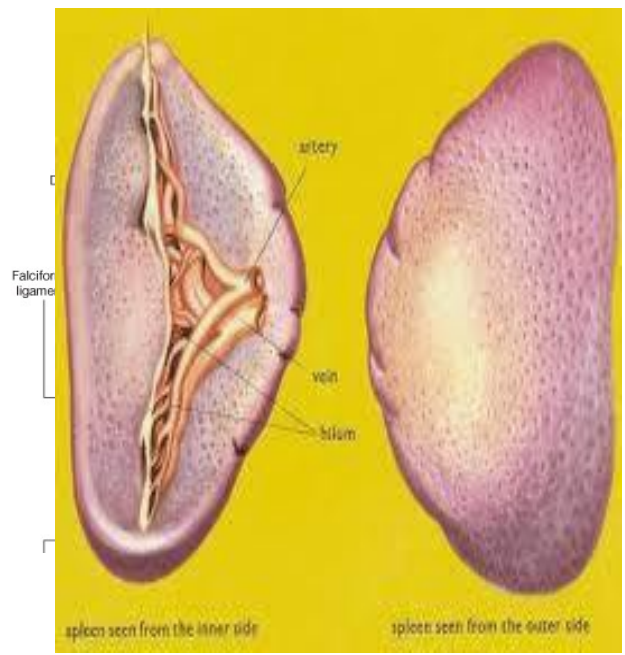
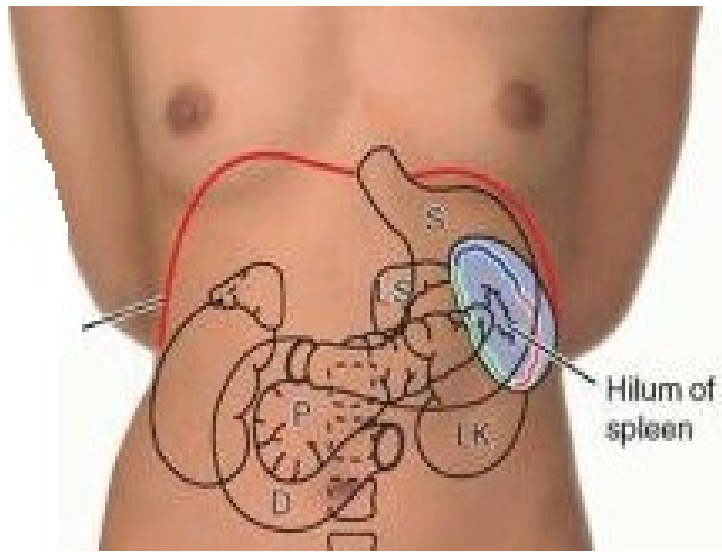


■ **The Spleen**

- Lies intraperitoneal in the left upper quadrant
- 12 cm long, 6 - 8 cm wide, 3 cm thick and weighs about 150 gm.
- Entirely surrounded by peritoneum except at the hilum (the entry point for splenic vessels).
- Related to the 9th, 10th & 11th ribs on the left side.
- A lymphatic organ
- The spleen, which is the largest endocrine (ductless) gland, appears to act to remove dead blood cells and fight infection

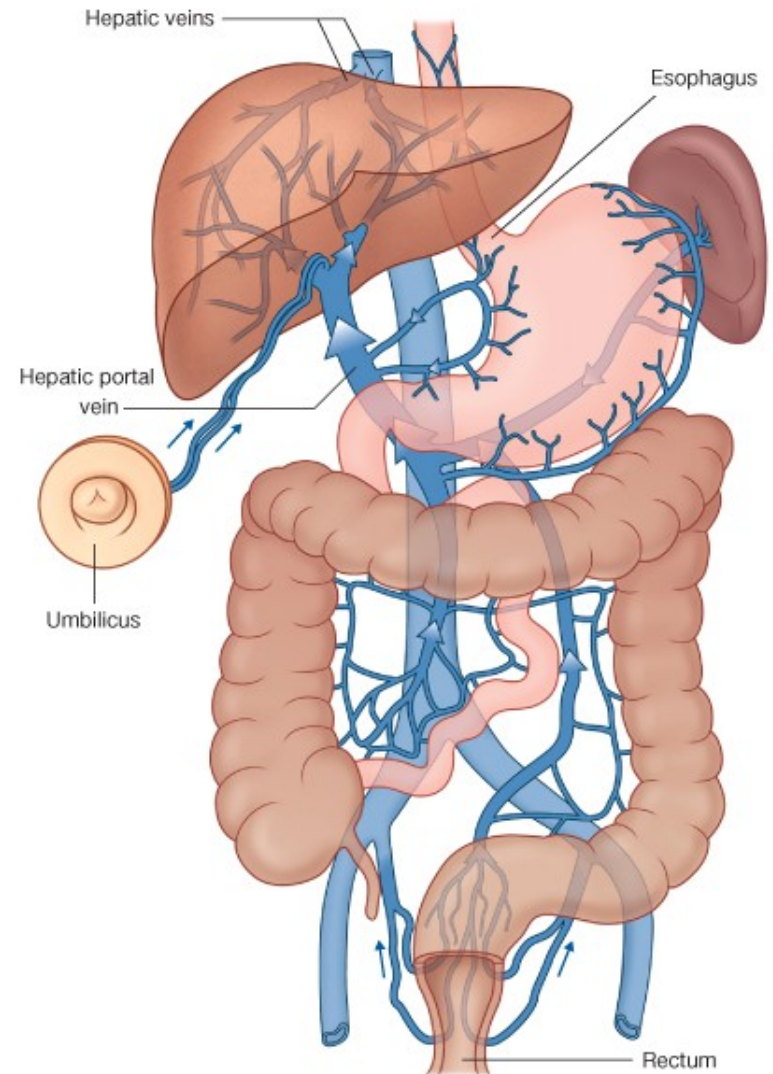
✓ **Major functions:**

- Participates in the body's defense system; removes expended RBCs & PLTs;
- Has smooth & convex **diaphragmatic** & an irregular concave **visceral surfaces**;



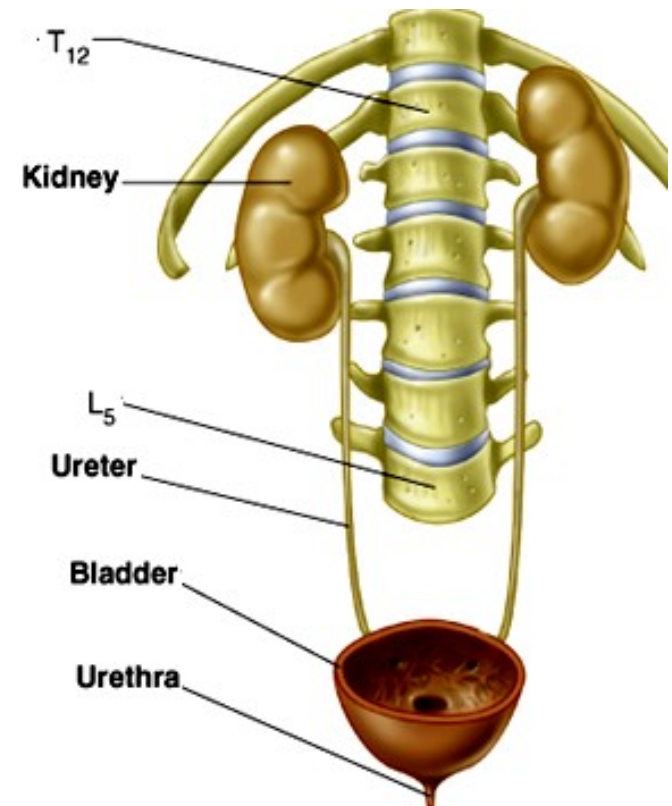
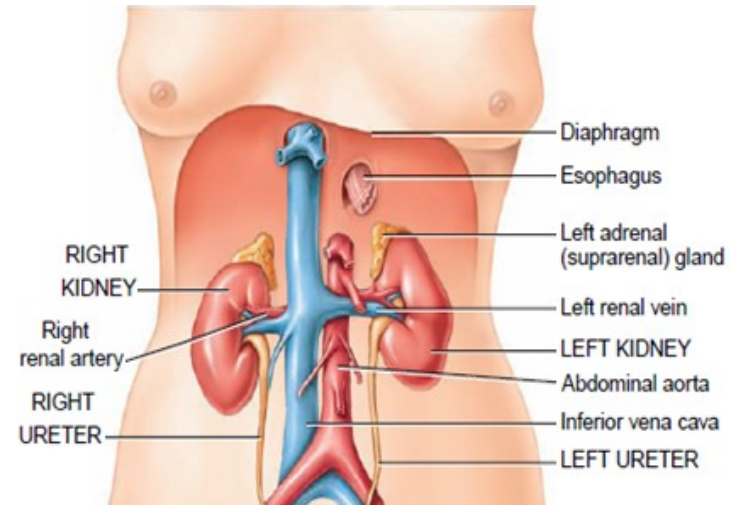
□ The Hepatic Portal System

- Drains venous blood from the digestive tract, spleen, pancreas & gallbladder, then the blood enters the liver through the hepatic portal vein;
- The hepatic portal vein is formed by union of splenic & superior mesenteric veins



❑ The Kidneys

- ✓ Paired bean-shaped organs;
- ✓ Lie retroperitoneally in the PAW;
- ✓ Extend at the level of T12-L3 vertebrae
- ✓ The Rt kidney is **slightly lower in position** than the Lt kidney
- Removes wastes of metabolism from the blood;



❖ Supporting and Protecting Structures

☐ Renal capsule:

- The innermost dense connective tissue layer,
- Adheres to the surface of the kidney,
- Protects against trauma & infection & used to maintain shapes of the kidneys.

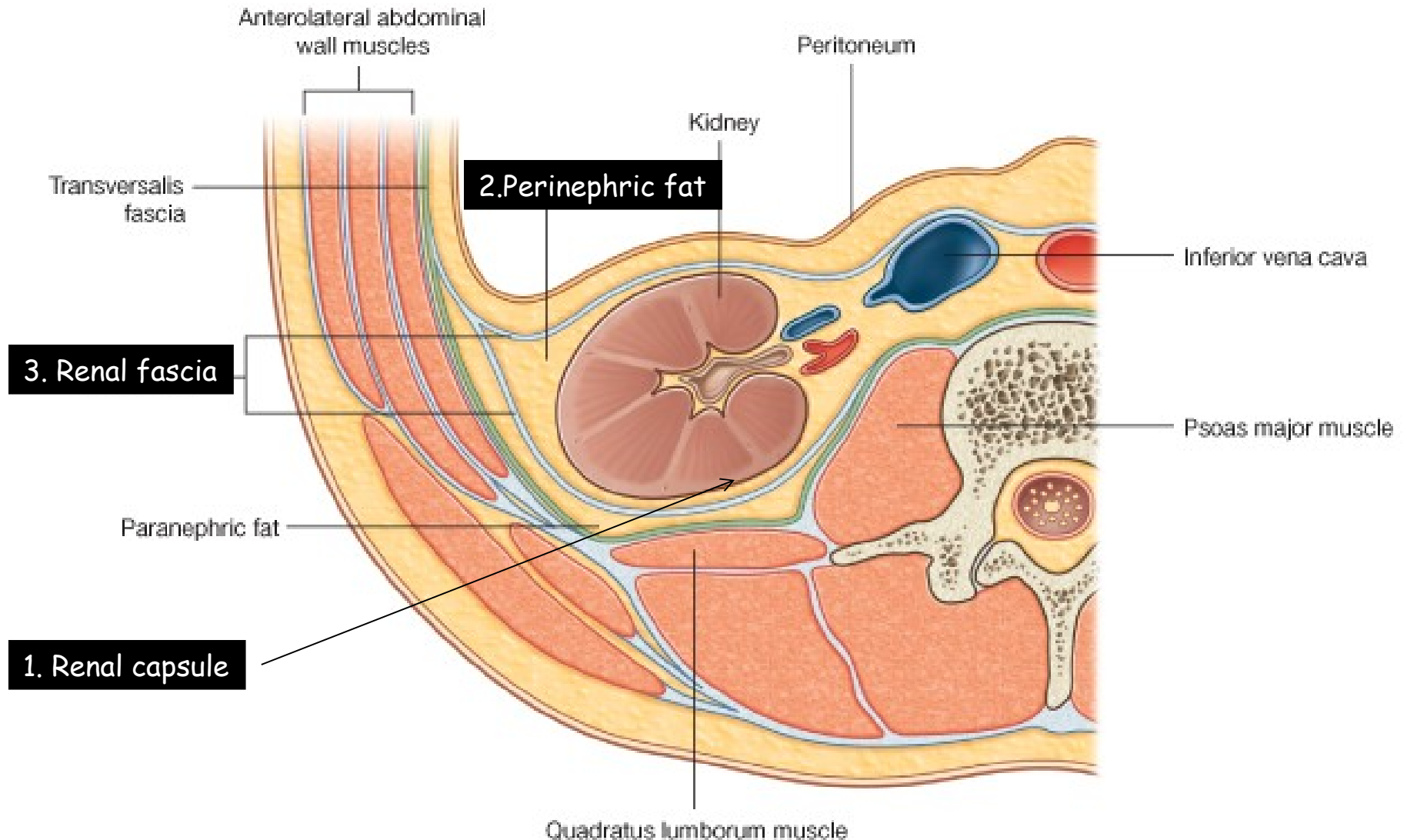
☐ Perinephric fat/adipose capsule:

- The middle fatty tissue layer,
- Protects against trauma & holds the kidney firmly in place within the posterior abdominal wall.

☐ Renal fascia:

- ✓ The outermost membranous layer
- ✓ Anchors the kidney to the surrounding structures
- ✓ Has anterior & posterior layers; medially,

Supporting and protecting structures



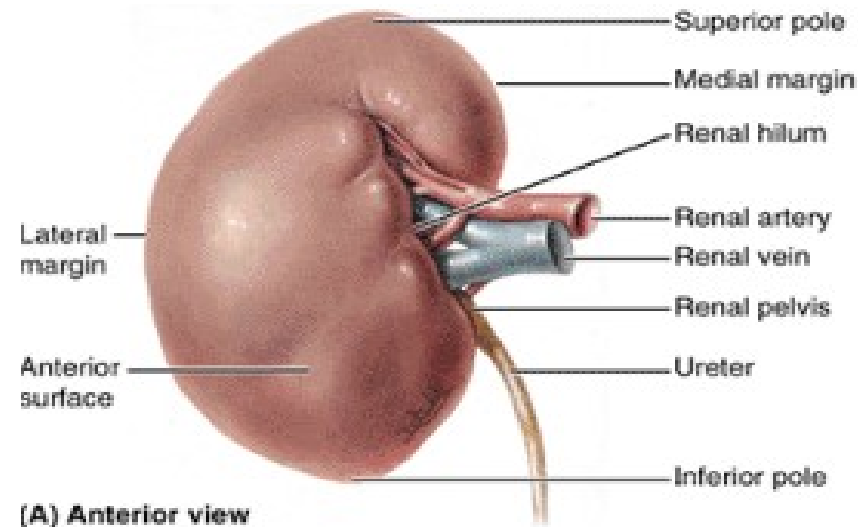
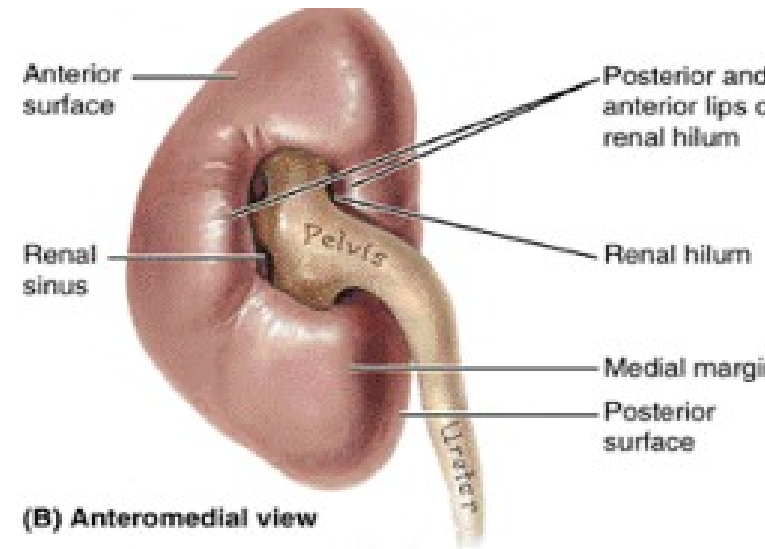
❑ External Anatomy

✓ Each kidney has:

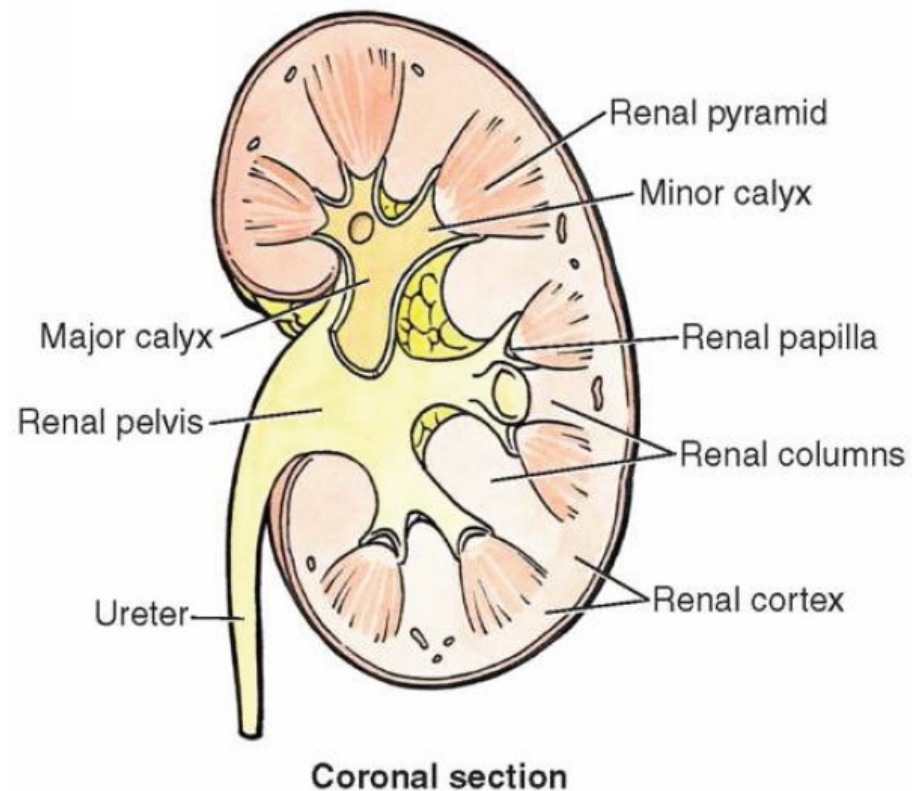
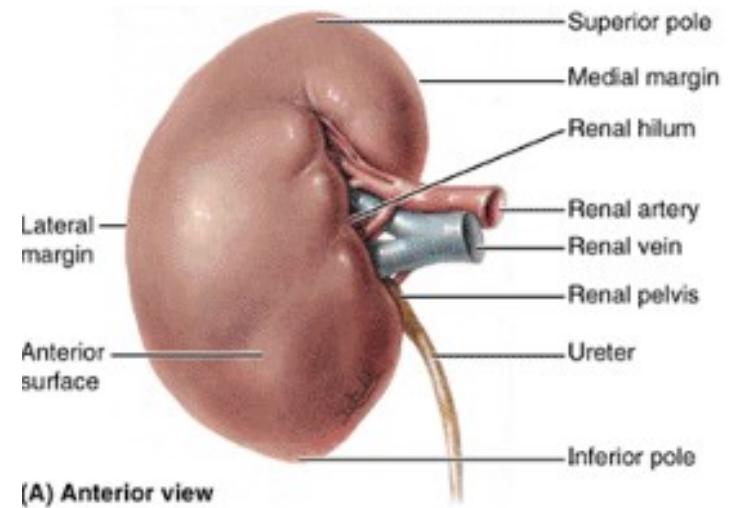
- Anterior & posterior surfaces;
- Superior & inferior poles;
- Lateral & medial borders.

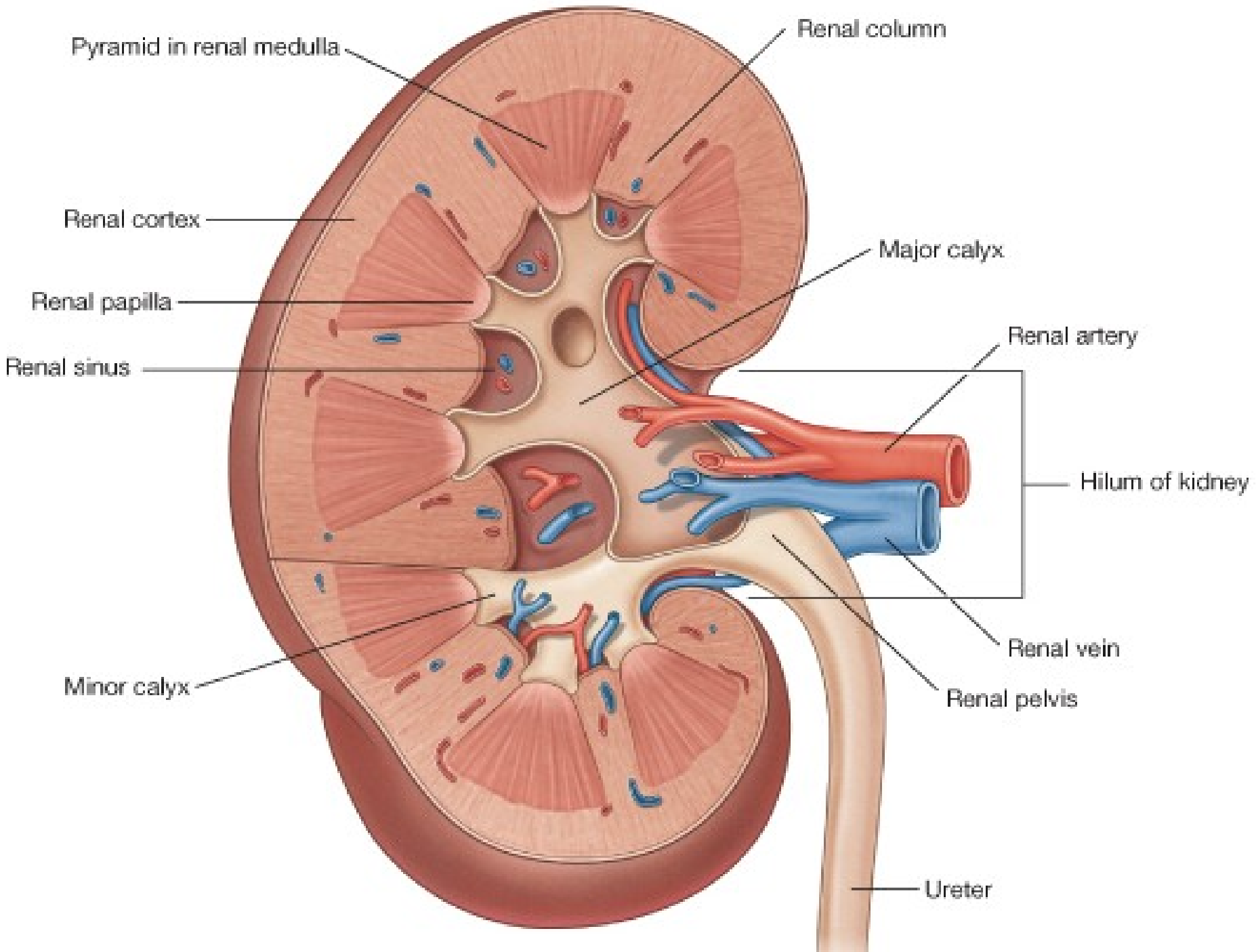
❖ At the medial border is an indentation, the **renal hilum**;

❖ At the hilum, the renal vein is anterior to the renal artery, which is anterior to the renal pelvis



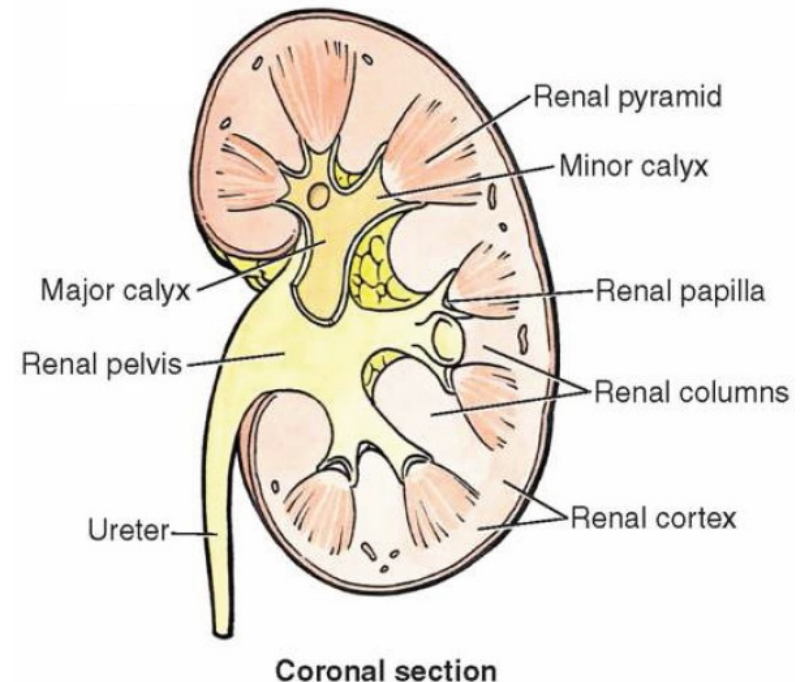
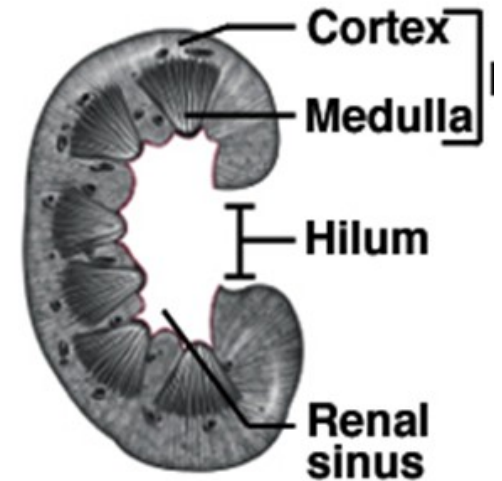
- **The renal hilum** that leads to a space within the kidney, the **renal sinus**.
- **The renal sinus**, is occupied mainly by **perinephric fat** in which the renal **pelvis, calices, vessels & nerves** are embedded.



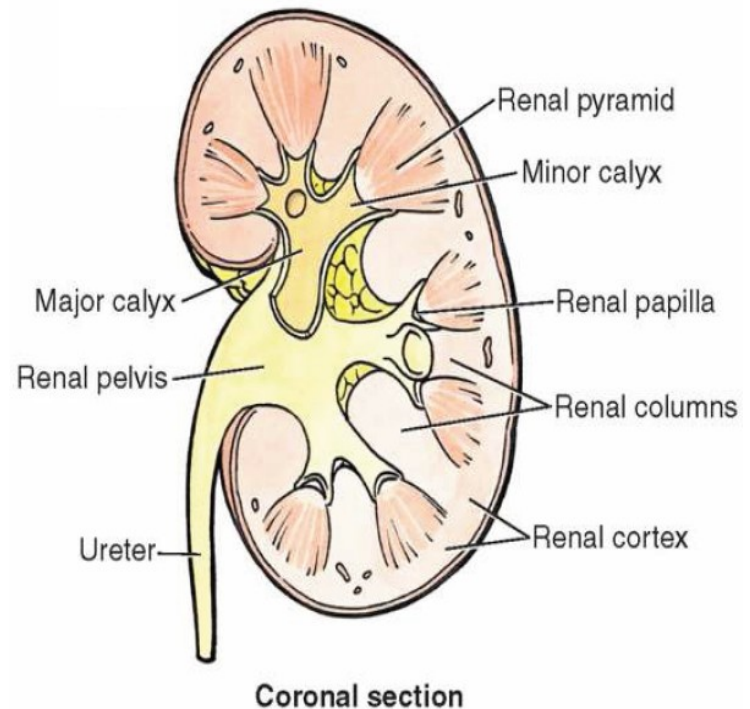


❑ Internal Anatomy:

- ✓ Each kidney has an outer **cortex** & an **inner** medulla region
- **The Cortex** is a superficial region that sends extensions into the medulla, **renal columns**
- **The Medulla** is an inner darker part that contains **renal pyramids**, which are separated by the renal columns,



- **The Renal pyramids** are triangular units in the medulla that house the nephron.
- **Renal pelvis:** funnel-shaped expansion of the superior end of the ureter;
- **Major calyces:** the branching channels of the renal pelvis that collect urine from minor calyces and empty urine into the pelvis
- **Minor calyces** the branching channels of the major calyces, receive urine from the **renal papillae**,
- **Renal Papillae:** tips of the renal pyramids that release urine into minor calices.



Renal Vasculature

➤ Renal Arteries:

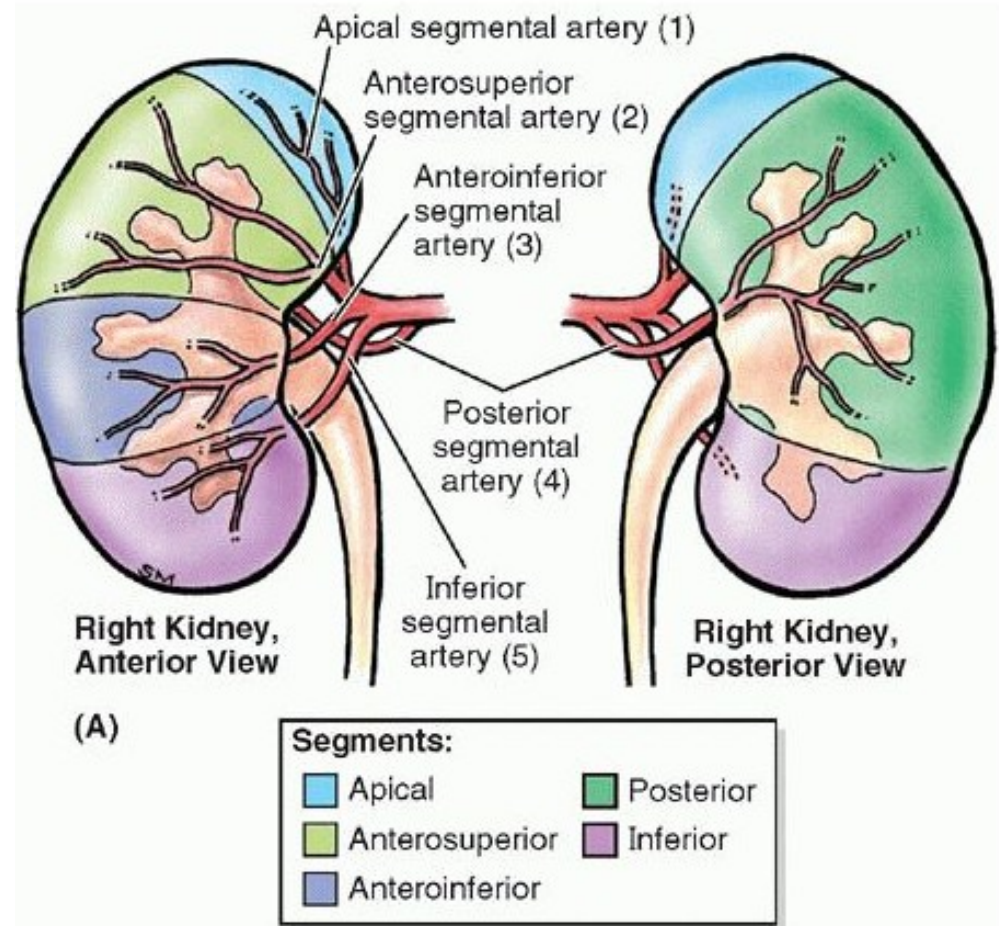
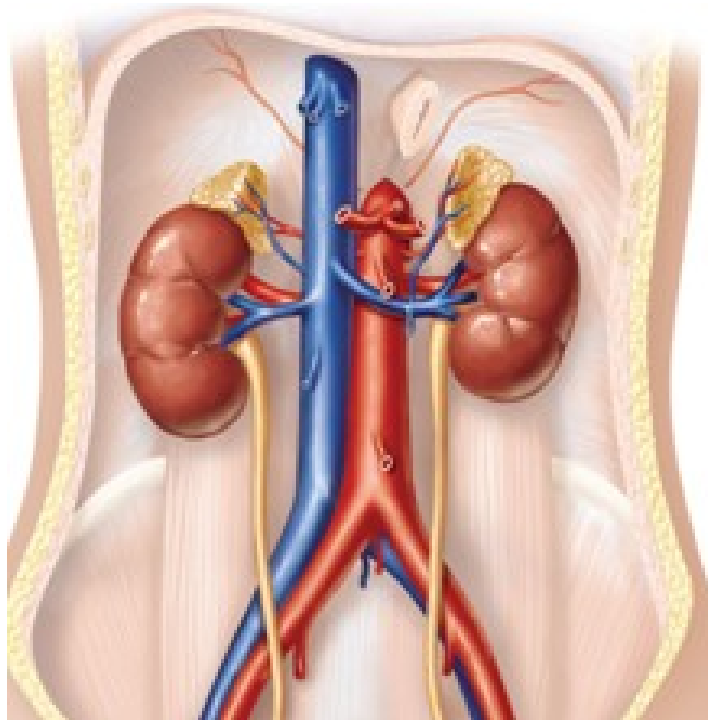
- Branches of abdominal aorta,
- Divide into anterior and posterior branches,

1. The anterior branch subdivided into:

- ✓ Superior, anterosuperior, anteroinferior & inferior segmental arteries to supply the corresponding segments.

2. The posterior branch:

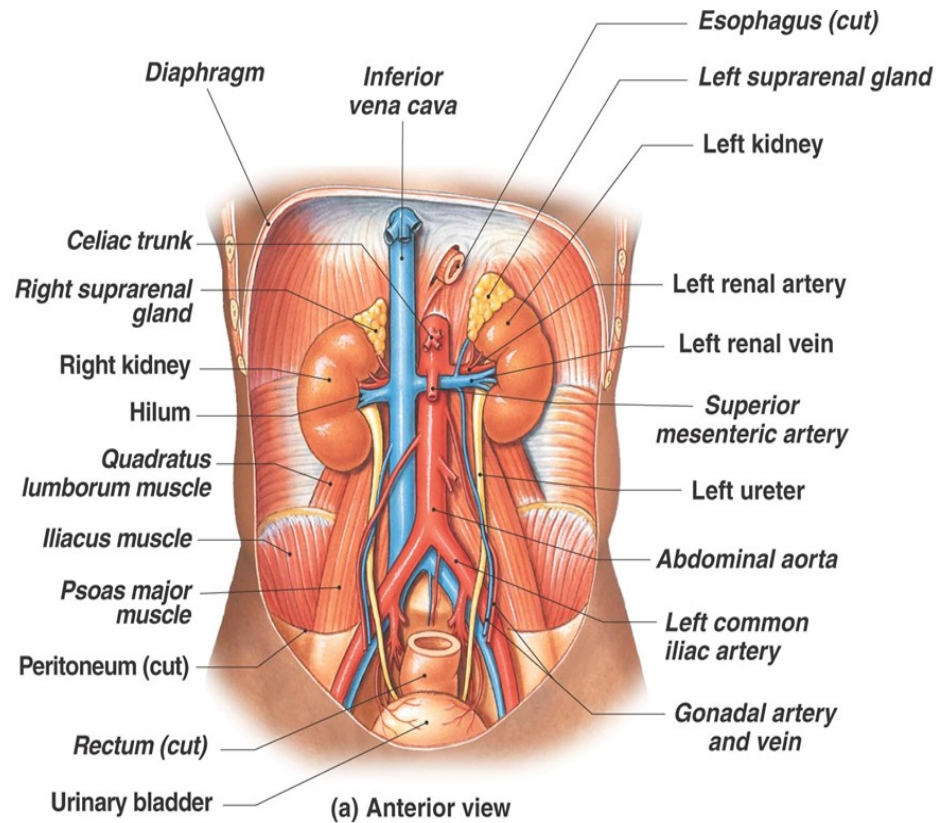
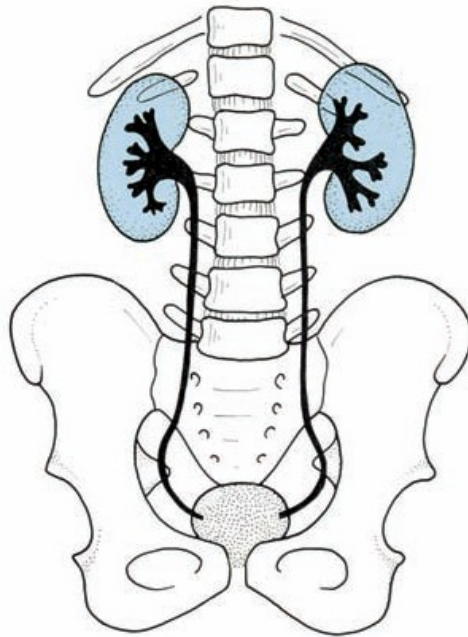
- ✓ Continue as, posterior segmental artery, to supply the posterior segment of the kidney.



➤ Each renal vein drains into the IVC.

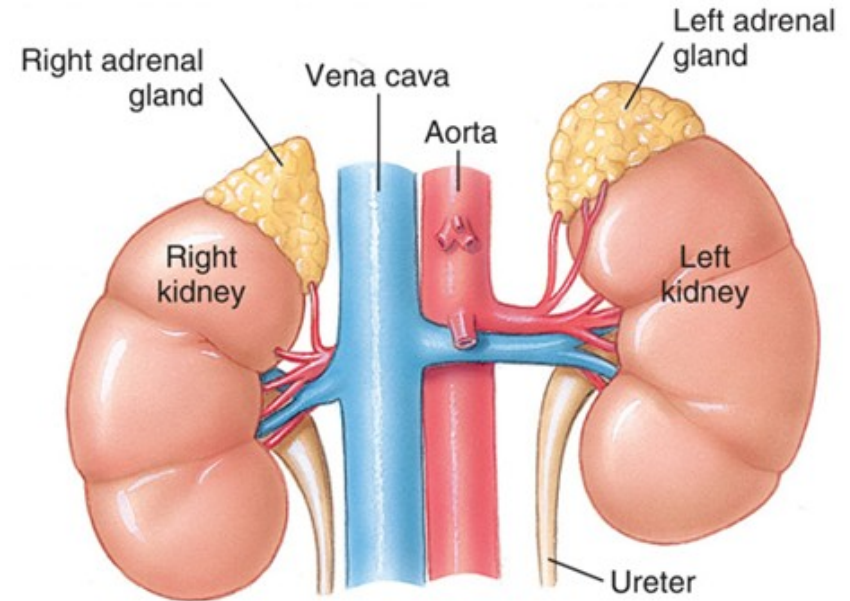
□ URETERS

- Muscular tubes, extend retroperitoneally from the kidneys to posterior surface of urinary bladder;



□ The Adrenal Glands

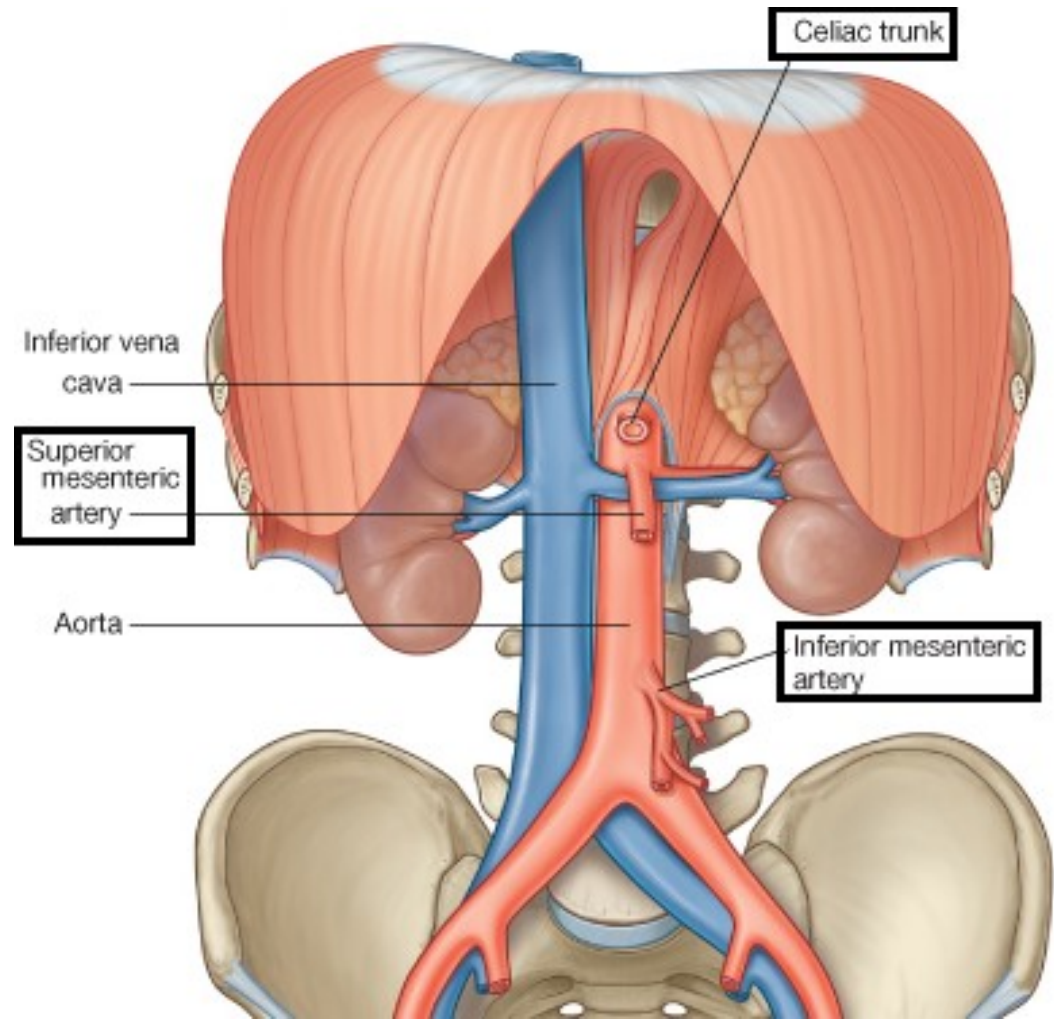
- Are paired **retroperitoneal** glands,
- Lie near the **superomedial** aspects of the kidneys;
- Are surrounded by **renal fascia** (but are separated from the kidneys by the **perirenal fat septum**).



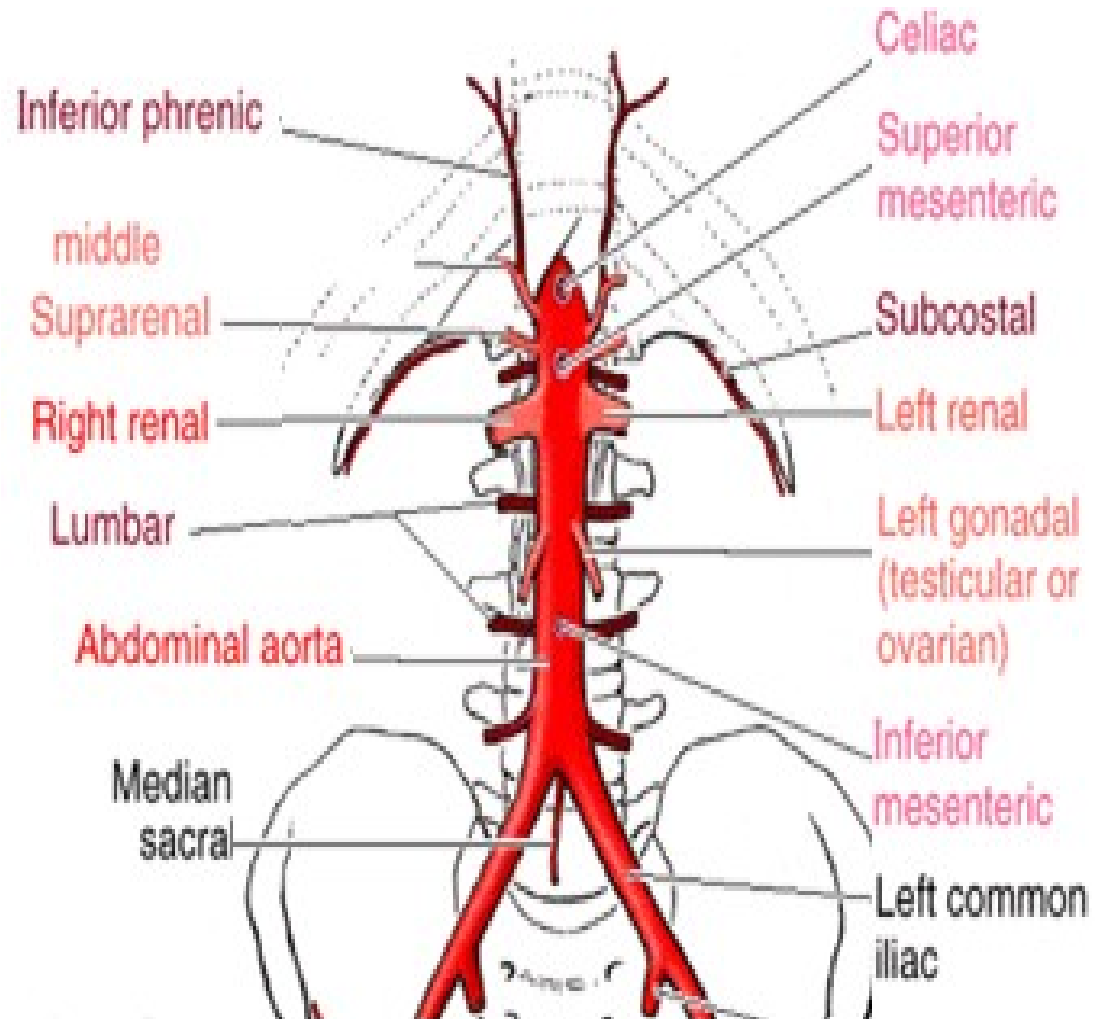
Major Neurovascular Structures of the Abdomen

□ The Abdominal Aorta

- Begins at the aortic hiatus
- Ends at the level of L4 vertebra by bifurcating into Rt & Lt **common iliac arteries**.

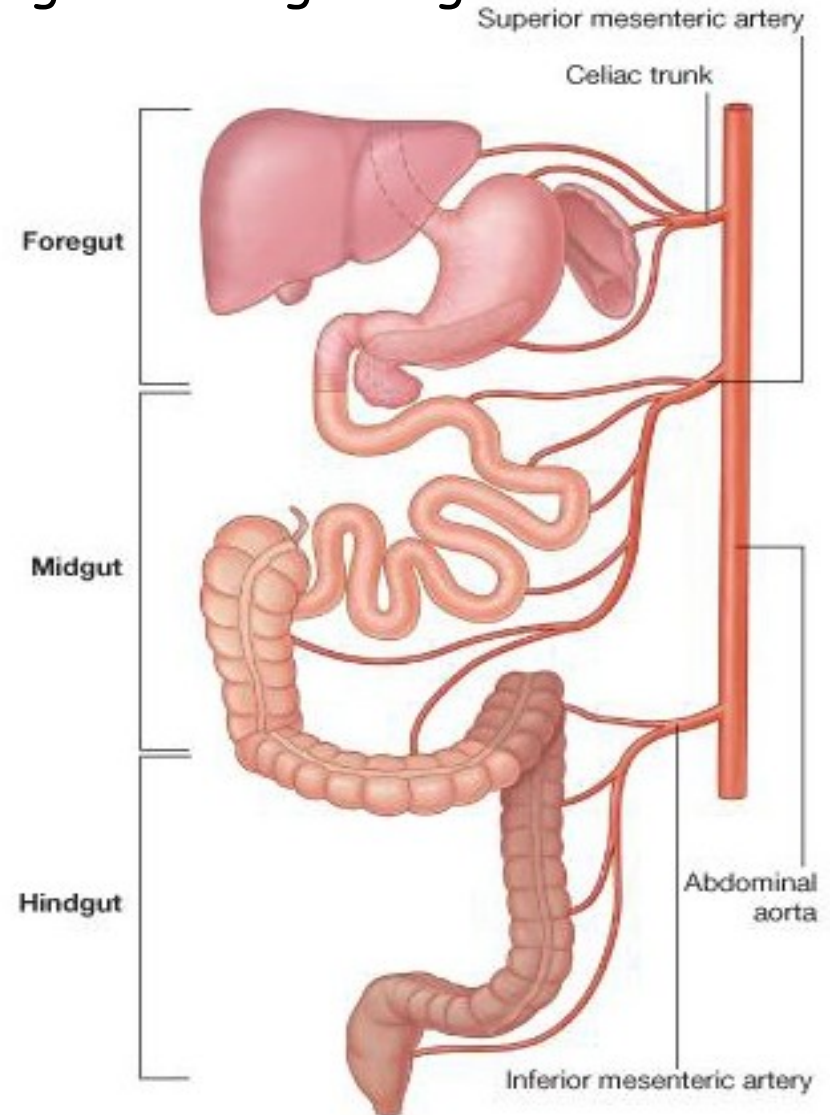


- Anterior,
- lateral,
- posterior &
- Terminal branches



- ✓ During its course the abdominal aorta, gives rise to:
- 3 large unpaired anterior branches:
 - a. The celiac trunk:
 - b. The superior mesenteric artery
 - c. The inferior mesenteric artery:

□ The **primitive gut** includes foregut, midgut & hindgut regions.



➤ The unpaired posterior branch:

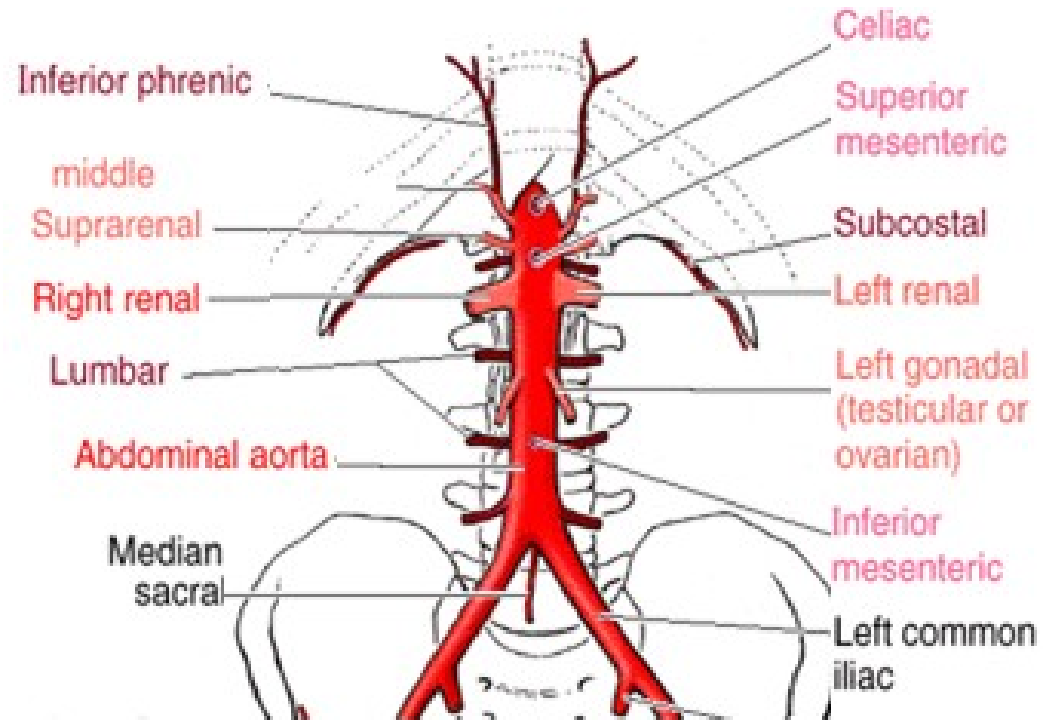
median sacral artery

➤ Paired lateral branches:

- Middle suprarenal,
- Renal,
- Gonadal (ovarian or testicular),
- Inferior phrenic,
- Lumbar

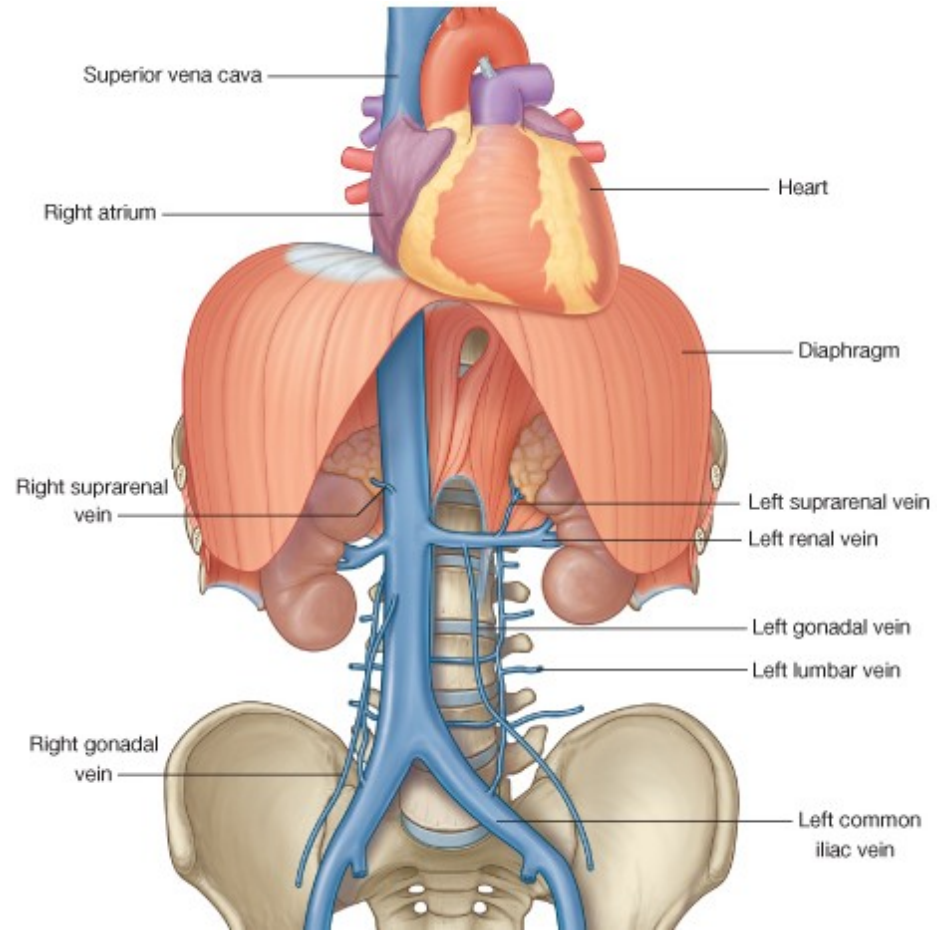
➤ Terminal branches:

- Rt & Lt common iliac arteries

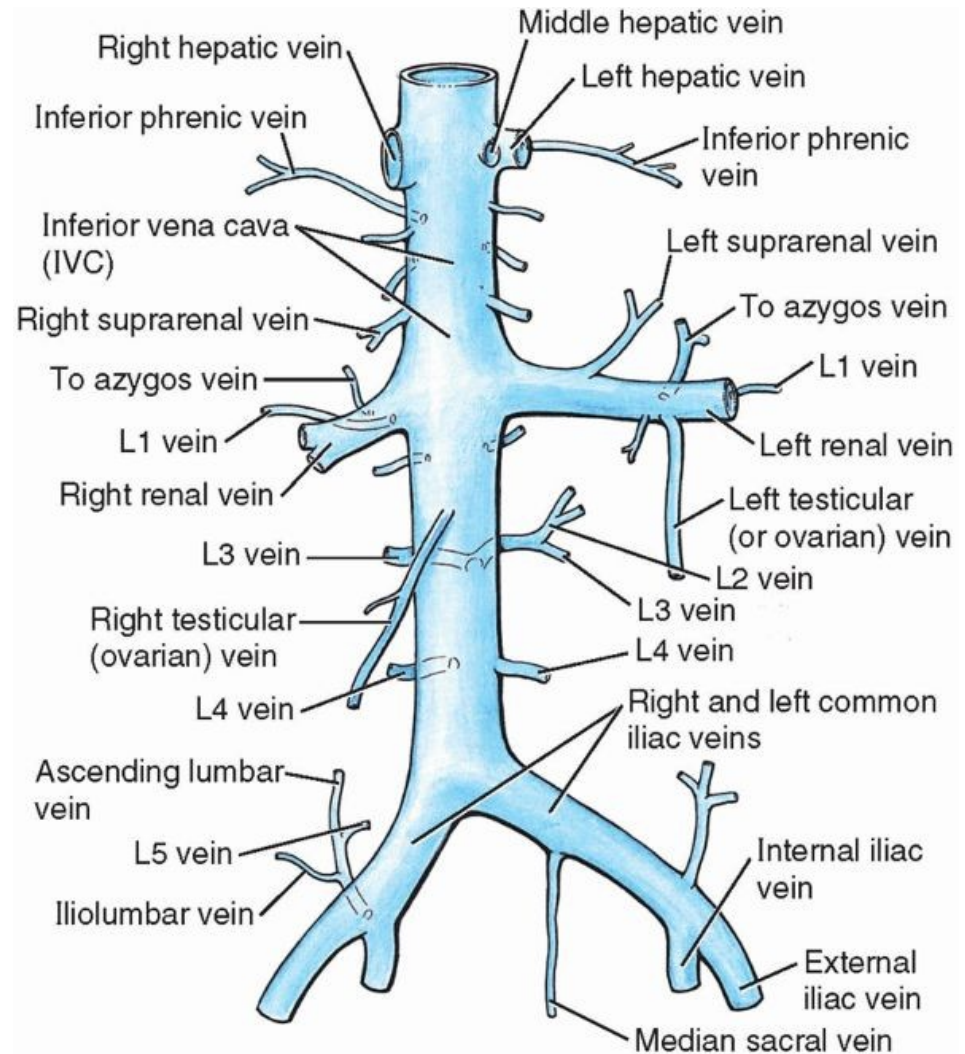


❑ The Inferior Vena Cava

- ✓ Drains the abdomen, pelvis, perineum & both lower limbs
- ✓ Formed when the two **common iliac veins** come together
- ✓ Ascends through the posterior abdominal wall
- ✓ Leaves the abdomen by piercing the central tendon of the diaphragm thru **vena caval foramen** and opens into Rt atrium of the heart.



- ✓ Tributaries of the IVC
- include:
- The common iliac veins;
 - The lumbar veins;
 - The Rt testicular or ovarian vein;
 - The renal veins;
 - The right suprarenal vein;
 - The inferior phrenic veins;
 - The hepatic veins





Galatoomaa